RMG-1: Relational Medium Gravity

Gravity as the low-energy relational geometry of a stable base-medium

Internal core draft, version 6.2
2026-05-11

Purpose

RMG formulates gravity as the low-energy relational geometry of a stable base-medium.

Core sentence:

$$\boxed{\text{gravity is the low-energy relational geometry of a stable base-medium.}}$$

The starting object is a base-medium $\mathcal{M}$ with a stable low-energy phase. The tested GR/ADM structure is required to emerge as the low-energy fixed point of $\mathcal{M}$.

The protected low-energy phase admits only operational relational observables, not a separate local medium-frame field.

A central core gate is that all physical low-energy principal operators share one kinetic metric kernel.

A known no-go result is addressed at the outset. The Weinberg–Witten theorem forbids a massless spin-2 particle from emerging as a composite in a relativistic quantum field theory carrying a Lorentz-covariant conserved stress-energy tensor on a fixed background. RMG is not such a theory: it has no fixed background spacetime and no Lorentz-covariant stress-energy tensor defined on one. Spacetime, rods, clocks and the operational metric are themselves emergent relational records of $\mathcal{M}$, in the sense of Operational Relational Covariance. The background-independence of RMG is therefore the intended route around the Weinberg–Witten obstruction. This escape is necessary but not sufficient: the positive construction of exactly two transverse-traceless tensor radiation modes, with no extra physical scalar or vector gravitational mode, remains an open requirement carried by the record-sector factorization and spin-2 emergence gates.

Status labels

Every statement in RMG-1 carries one of the following labels: *Primitive Postulate* (minimal starting property of the base-medium); *Primitive Protection Principle* (low-energy relational rule excluding operators that violate the protection requirements); *Working Assumption* (temporary bridge requiring deeper derivation); *Result* (statement derived from listed assumptions); *Low-Energy Target* (structure required by observation and adopted as the low-energy fixed point); *Open Theorem* (claim still to be proven from $\mathcal{M}$); *Failure Gate* (condition whose failure invalidates the local core).

1 Core Terms

RMG (Relational Medium Gravity) is a gravitational framework in which a single base-medium $\mathcal{M}$ gives rise to matter modes, rods, clocks, signals and operational geometry. The deeper task is to explain why the universal low-energy geometry is the ADM/GR fixed point.

$\mathcal{M}$ is the base-medium; it is not assumed to be an ordinary scalar field. $\mathcal{W}$ is the relational medium-history. X_F is a foliation or embedding description of the same history $\mathcal{W}$.

$s(x)$ is the local operational factor shared by clocks, rods and signals in the weak-sector bridge description. u is the compression/lapse potential, with $s = e^{-u}$; it enters the action as a constraint multiplier, not as an independent propagating field.

$g^{\text{eff}}_{\mu\nu}$ is the effective metric seen by low-energy matter modes. N, N^i, h_{ij} are the ADM lapse, shift and spatial metric variables in the low-energy target, and h^{TT}_{ij} is the transverse-traceless tensor/shear radiation sector. $u^\mu_{\mathcal{M}}$ denotes a putative independent local medium-frame field; it must not appear as an independent physical field in the tested RMG core.

$Q^{\mu\nu}_{AB}$ is the highest-derivative low-energy kinetic response matrix for perturbations of matter and geometry modes. $G^{\mu\nu}$ is the common principal metric kernel, identified with $g^{\mu\nu}_{\text{eff}}$ if the factorization gate succeeds. K_{AB} is an internal/species kinetic matrix: it carries allowed differences between particle families, tail shapes, polarizations, phases and internal geometries. $B^{\mu\nu}_{AB}$ is the non-factorizing failure matrix; it must vanish on physical low-energy principal modes, up to gauge, internal or suppressed correction terms.

$\Delta_{\mathcal{M}}$ is the residual medium contribution to the low-energy Hamiltonian/constraint algebra after the ADM/EH target terms are separated; the tested core requires $\Delta_{\mathcal{M}} = 0$, or suppression outside all tested regimes.

$\mathcal{R}$ is the record sector: low-energy modes that can participate in operational measurement as rods, clocks, detector components, memory or record components, exchanged signals or stable external matter excitations. $\mathcal{I}$ is the internal sector: confined internal modes, gauge directions, constraint variables, off-shell auxiliary modes, short-range internal responses and high-energy modes outside the tested local regime. Internal modes may contribute to mass, binding energy, charge, spin, colour, tail shape, effective couplings and total energy, but must not define a separate external spacetime cone in the tested local core.

2 Primitive Postulates

Primitive Postulate 1 (One base-medium)**.** *All low-energy matter modes, rods, clocks, signals and geometric excitations are collective modes of one base-medium $\mathcal{M}$.*

Remark (mathematical status of $\mathcal{M}$). This postulate fixes only the ontological commitment of RMG-1: one substrate whose low-energy observable structures are collective or coarse-grained modes. It does not assume that $\mathcal{M}$ is an ordinary scalar field, nor does it by itself specify a state space, action, symplectic structure, internal redundancy group or record observables. Those data belong to the later base-medium input package and candidate order-parameter construction. The postulate therefore asserts existence and exhaustiveness; the microscopic content remains an open construction target.

Primitive Postulate 2 (Stable low-energy phase)**.** *The base-medium has a stable low-energy phase in which matter modes, rods, clocks, signals and operational geometry are well-defined.*

Primitive Postulate 3 (Minimal operational node scale)**.** *The base-medium as one ontological object has no ordinary microscopic radius. A length becomes physical only when the base-medium is distinguishable as recordable nodes. The first distinguishable node/record scale is denoted*

$$\ell_*.$$

Below ℓ_*, two nodes are not yet separate operational objects: there is no independent distance, delay, clock comparison or mass comparison. Above ℓ_*, medium dilution also rescales rods, clocks *and the visible node scale. Thus ℓ_* is an operational closure scale, not the radius of a mechanical particle.*

In the present RMG-1 core, ℓ_* is identified with the observed Planck-order minimal record scale. This is not a second medium and not a species-dependent parameter; it is the universal first readable separation of the one base-medium.

Primitive Postulate 4 (Operational Relational Covariance)**.** *Local low-energy observables are invariant under re-embedding of the same relational medium-history:*

$$\mathcal{W} \sim X_F(\mathbb{R} \times \Sigma) \sim X_{F'}(\mathbb{R} \times \Sigma).$$

Thus coordinates, slicings and internal clock choices are descriptions, not extra physical structures. In particular, no independent local preferred-frame field $u^\mu_{\mathcal{M}}$ appears in the tested core.

Remark (operational content of ORC). The re-embedding statement above has three explicit operational consequences that load the closure of the gravitational core:

1. Substrate-internal variables that do not enter operational record comparisons are gauge / bookkeeping, not physical observables. In particular, no hidden substrate clock, no preferred rest frame, and no additional substrate-internal label distinguish records.

2. All stable record-forming matter reads the same local operational geometry. No substrate-internal mode produces a distinguishable second operational time, second operational length, or second principal cone.

3. Background uniform substrate states (e.g. the uniform phase advance $\langle\theta\rangle = \nu_0 t$ on the stable phase $\mathcal{M}_0$) are operationally hidden; only the fluctuations on top of such backgrounds are physical. Only fluctuations relative to the background carry operational content; this is the relational form of Galilean and Lorentz invariance.

ORC is the principal axiom that closes the three local-gravity roots (Root A: one carrier; Root B: ADM/EH closure; Root C: total-energy source) into derived results. ORC itself is taken here as a primitive postulate; its derivation from a deeper micro principle remains a long-term research target.

Primitive Postulate 5 (Positive physical energy)**.** *Physical excitations of $\mathcal{M}$ have Hamiltonian bounded below:*

$$H - H_{\text{vac}} \geq 0.$$

All physical degrees of freedom carry positive kinetic energy. The trace/volume direction of the DeWitt supermetric belongs to the constrained sector and does not propagate as an independent mode.

Primitive Postulate 6 (Local effective hierarchy)**.** *The low-energy theory is local and organized by a derivative or scale expansion:*

$$\mathcal{L}_{\text{eff}} = \mathcal{L}_{\text{core}} + \ell^2\mathcal{L}_4 + \ell^4\mathcal{L}_6 + \cdots .$$

Corrections may exist, but they must be suppressed or forbidden in tested local regimes.

3 Operational Bridge

This section gives the minimal bridge needed to recover the weak gravitational sector; the microscopic derivation is treated separately.

The universal factor $s(x)$ is a bridge variable, not a primitive object. Its deeper origin is the universal principal symbol of all low-energy matter modes.

Working Assumption 1 (Universal operational factor)**.** *In the slowly varying low-energy regime, all stable matter modes share one local operational factor:*

$$s = s(x).$$

Equivalently, all matter modes see one effective metric $g_{\mu\nu}^{\text{eff}}$*.*

Working Assumption 2 (Reciprocal clock/rod response)**.** *Local clocks and rods respond as:*

$$d\tau = sdt, \qquad d\ell = s^{-1}dx.$$

Result 1 (Square signal law)**.** *If the local physical signal speed is* $c = d\ell/d\tau$*, then:*

$$c = \frac{s^{-1}dx}{sdt} = s^{-2}\frac{dx}{dt}.$$

Therefore:

$$\boxed{\frac{c_{\text{coord}}}{c} = s^2.}$$

Result 2 (Coherent endpoint-counting route to the square law)**.** *The reciprocal clock/rod bridge is not produced by a longer courier path. A longer path at fixed carrier speed gives only a linear delay. A square response is obtained when the stable record sector is controlled by one coherent base-medium amplitude*

$$\boxed{\rho := |\Phi_{\mathcal{M}}| = \sqrt{n}}$$

and the local record coefficients scale as

$$\boxed{C_t : C_0 : C_x = 1 : \rho : \rho^2.}$$

Here C_t *is the common-rhythm time coefficient,* C_0 *is the local stable gap coefficient and* C_x *is the spatial link-stiffness coefficient. Define*

$$\boxed{s := \rho^{1/2} = n^{1/4}.}$$

Then

$$C_t : C_0 : C_x = 1 : s^2 : s^4.$$

For a stable quadratic record mode,

$$\omega_0 \propto \sqrt{\frac{C_0}{C_t}} = s, \qquad \ell_{\text{coord}} \propto \sqrt{\frac{C_x}{C_0}} = s, \qquad c_{\text{coord}} \propto \sqrt{\frac{C_x}{C_t}} = s^2 c.$$

Thus the coherent endpoint-counting completion derives

$$\boxed{d\tau = sdt, \qquad d\ell = s^{-1}dx, \qquad \frac{c_{\text{coord}}}{c} = s^2.}$$

The exponent origin is:

$$\boxed{\textit{site/local gap} \sim \rho, \qquad \textit{link/spatial stiffness} \sim \rho_a\rho_b \to \rho^2.}$$

Working Assumption 3 (Additive compression and multiplicative response). *Weak compression potentials add:*

$$u_{12} = u_1 + u_2.$$

Process factors multiply:

$$s_{12} = s_1 s_2.$$

Result 3 (Exponential layering). *The continuous solution with weak-field normalization is:*

$$\boxed{s = e^{-u}.}$$

Result 4 (ADM lapse identification). *For a normal ADM observer with zero shift displacement,*

$$d\tau = N dt.$$

The RMG bridge clock response gives:

$$d\tau = s dt.$$

Therefore, inside the operational bridge,

$$\boxed{N = s = e^{-u}.}$$

The remaining task is to derive the universal factor s and the exponential layering from $\mathcal{M}$.

Result 5 (Weak operational metric). *The bridge metric is:*

$$\boxed{ds^2 = -e^{-2u} c^2 dt^2 + e^{2u} \delta_{ij} dx^i dx^j.}$$

4 One-Kernel Factorization Gate

The universal matter metric must derive from a common highest-derivative kinetic kernel. For low-energy perturbations $\delta\Phi^A$, denote by $Q^{\mu\nu}_{AB}$ the principal kinetic response. For a covector k_μ, define the principal symbol:

$$\boxed{P_{AB}(k) = Q^{\mu\nu}_{AB} k_\mu k_\nu.}$$

Let $\Pi_{\mathcal{R}}$ project onto the physical record subspace after gauge and constraint directions are removed.

The labels A, B range over every stable record-forming excitation relevant to the local core: elementary medium excitations, effective matter modes, photons/signals, composite records such as atoms, rods and clocks, and detector components. Modes outside this class may be internal or confined, but they must not act as external spacetime carriers.

The principal symbol is extracted in the relevant low-energy effective regime, on the physical on-shell branch after gauge and constrained directions are removed. It is not a blind UV limit outside the record theory:

$$\boxed{k \ll \Lambda_{\mathrm{UV}}, \qquad \delta\Phi \in \Pi_{\mathcal{R}}, \qquad \text{dispersion evaluated on a physical record branch.}}$$

Open Theorem 1 (Factorization gate). *Derive from the protected base-medium phase:*

$$\boxed{Q^{\mu\nu}_{AB} = G^{\mu\nu} K_{AB}.}$$

Then all physical low-energy modes share one principal metric:

$$\boxed{G^{\mu\nu} = g^{\mu\nu}_{\mathrm{eff}}.}$$

Common characteristic cones do not suffice: the physical principal symbol must also be free of unsuppressed species or polarization birefringence, and a single base medium does not by itself prevent internal or confined modes from defining separate cones. The RMG statement is therefore restricted:

$$\boxed{\text{one stable record-forming low-energy phase} \Rightarrow \text{one operational spacetime cone for records.}}$$

Modes outside the record sector may have different internal dispersions, but they must not define separate external rods, clocks, signals or spacetime metrics. The pass condition reads:

$$\boxed{\Pi_{\mathcal{R}} P(k) \Pi_{\mathcal{R}} = (G^{\mu\nu} k_\mu k_\nu)\, K_{\mathcal{R}}.}$$

Allowed remainders are gauge directions, constraint variables, internal algebra, lower-derivative terms or suppressed corrections. The local core fails if the physical principal symbol contains:

$$\boxed{P^{\text{phys}}_{AB}(k) = (G^{\mu\nu} k_\mu k_\nu)\, K_{AB} + B_{AB}(k), \qquad B_{AB}(k) \neq 0,}$$

where B_{AB} produces an observable species-private cone, polarization splitting, preferred-frame tensor or clock-dependent redshift.

Why this is non-trivial in RMG. In GR, one metric for all matter is usually imposed by minimal coupling. Root A is not allowed to assume that result. It must derive why all stable record-forming modes read one external cone from the base-medium spectrum and record projection. If this fails, later ADM closure or source universality cannot repair the local core.

Open Theorem 2 (Record-sector no-birefringence principle)**.** *All stable modes that can form local records, rods, clocks and exchanged signals must belong to one operational record sector. In that sector:*

$$\boxed{B^{\text{phys}}_{AB}(k) = 0}$$

for unsuppressed principal spacetime response. Internal differences may remain in K_{AB}, masses, charges, spin, color, topology, phase and tail geometry; they must not appear as separate external spacetime cones. This is a protection principle in RMG-1 and still requires derivation from $\mathcal{M}$.

Open Theorem 3 (Record-phase route to factorization)**.** *Derive the chain:*

$$\boxed{\textit{stable record phase} \Rightarrow \textit{common operational records} \Rightarrow \textit{no observable species-private spacetime cone} \Rightarrow \textit{no physical}}$$

This chain remains to be derived from $\mathcal{M}$.

Open Theorem 4 (One-tensor factorization lemma)**.** *For any candidate common record tensor $G^{\mu\nu}$, decompose the principal kinetic matrix as:*

$$\boxed{Q^{\mu\nu}_{AB} = G^{\mu\nu} K_{AB} + B^{\mu\nu}_{AB}.}$$

If the stable record phase contains only one observable rank-two spacetime principal tensor, then the non-factorizing part $B^{\mu\nu}_{AB}$ has no unsuppressed physical projection on external record-forming modes:

$$\boxed{\Pi_{\mathcal{R}} B^{\mu\nu}_{AB} \Pi_{\mathcal{R}} = 0}$$

up to gauge, confined, gapped, damped, unstable or high-energy suppressed pieces. Otherwise $B^{\mu\nu}_{AB}$ would define a second observable cone, anisotropy, preferred-frame tensor or species-private clock response. This is an open lemma: RMG must still derive the one-observable-tensor premise from the base medium.

Gate 1 (Observable tensor basis gate). *In the tested local record core, the physical principal symbol must contain no unsuppressed observable rank-two tensor other than the common record tensor* $G^{\mu\nu}$*. Equivalently, any decomposition*

$$\boxed{Q_{AB}^{\mu\nu} = G^{\mu\nu} K_{AB} + B_{AB}^{\mu\nu}}$$

must satisfy

$$\boxed{\Pi_{\mathcal{R}} B_{AB}^{\mu\nu} \Pi_{\mathcal{R}} = 0}$$

at leading principal order, except for gauge, constraint, internal/confined, gapped, damped, unstable, high-energy suppressed or explicitly large-scale correction sectors. Forbidden local-core pieces include unsuppressed

$$\boxed{U^\mu U^\nu, \quad v^\mu v^\nu, \quad \sigma^{\mu\nu}, \quad \omega^\mu{}_\alpha \omega^{\alpha\nu}, \quad \nabla^\mu u \nabla^\nu u, \quad S_A^{\mu\nu}.}$$

If any such tensor survives in physical record propagation, RMG predicts a preferred frame, anisotropic cone, species-private metric or birefringence. This gate is open from $\mathcal{M}$*.*

Open Theorem 5 (Master gate: no-birefringent record carrier). *The master local-core gate is:*

$$\boxed{\textit{all low-energy record-forming tails project onto one common carrier}}$$

so that the physical record-sector principal symbol has no unsuppressed species or polarization birefringence:

$$\boxed{P_{AB}^{\text{phys}}(k) = (G^{\mu\nu} k_\mu k_\nu)\, K_{AB}.}$$

If this gate closes, then one operational cone, one universal clock factor $s(x)$*, the ADM lapse bridge* $N = s = e^{-u}$*, the total-energy source gate and the compact internal-energy/no-external-cone split become linked consequences of the same record-sector protection.*

Here "one common carrier" means one shared base-medium vibration channel, not one elementary scalar field and not one tail substance per particle. Different sources may imprint different local patterns on the same medium; the tested local core allows only one external record carrier/cone.

Open Theorem 6 (Stable-tail common-carrier route). *For every stable record-forming oscillon or matter mode* A*, decompose its long tail as:*

$$\boxed{T_A = \Pi_{\text{common}} T_A + \Pi_{\text{internal}} T_A.}$$

The theorem target is:

$$\boxed{\text{principal}(\Pi_{\text{common}} T_A) = G^{\mu\nu} k_\mu k_\nu \quad \textit{for all record-forming } A,}$$

while $\Pi_{\text{internal}} T_A$ *may carry source-specific frequency, phase, spin, charge, topology, multipoles and short-range structure, but must not define a separate external spacetime cone.* T_A *is a source imprint in the same base-medium vibration, not a separate long-range field assigned to species* A*. Modes whose tails cannot project onto the common carrier are internal, confined, gapped, constrained or non-record-forming.*

Open Theorem 7 (Universal operational factor from one cone). *For a clock family* A*, write:*

$$d\tau_A = s_A(x) dt.$$

If the one-kernel factorization gate holds, then all record-forming clock families read the same spacetime kernel $G^{\mu\nu}$. Hence:

$$\boxed{s_A(x) = C_A\, s(x)\, [1 + \epsilon_A(x)].}$$

Here C_A is a constant internal clock calibration, while $\epsilon_A(x)$ is the species-dependent environmental response that would violate the universality requirement. The tested core requires:

$$\boxed{\epsilon_A(x) = 0}$$

or suppression outside tested regimes. After constant calibration,

$$\boxed{s_A(x) = s_B(x) = s(x).}$$

Equivalently,

$$\boxed{\nabla_\mu \ln \frac{s_A}{s_B} = 0}$$

up to suppressed corrections. If this fails, the failure is the clock-level version of a nonzero $B^{\mu\nu}_{AB}$.

The exact target is $\epsilon_A = 0$. If a candidate leaves a residual ϵ_A, it must be bounded against clock-comparison, redshift, composition-dependence and species-dependent signal-cone tests; it cannot remain an unconstrained symbolic remainder.

Different oscillons may excite the same base-medium with different local frequencies, phases, polarizations and internal geometries. The factorization requirement constrains only the external record carrier of their long-range tails, which must be the same base-medium vibration channel:

$$\boxed{\text{diverse local imprints, one base-medium, one operational principal cone.}}$$

The possible failure is:

$$\boxed{Q^{\mu\nu}_{AB} = G^{\mu\nu} K_{AB} + B^{\mu\nu}_{AB}, \qquad B^{\mu\nu}_{AB} \neq 0.}$$

In the tested core, $B^{\mu\nu}_{AB}$ must not contain an independent preferred-frame tensor, species-private metric, or geometry-sector cone different from the matter cone.

5 Tail Superposition And Projection Gate

Different oscillons may produce different local tail imprints in the same base-medium. Gravity does not read the full microscopic imprint. It reads the common pressure/refraction projection:

$$\boxed{\Pi_{\text{grav}}(T_A) = q_A \mathcal{G}.}$$

Here T_A is the source imprint associated with particle family A, $\mathcal{G}$ is the common base-medium pressure carrier, and q_A is the pressure-deficit strength. T_A must not be read as a separate long-range field or separate spacetime channel.

Open Theorem 8 (Universal gravitational projection)**.** *For all stable low-energy record-forming matter modes, derive:*

$$\boxed{q_A \propto E_A/c^2.}$$

Here E_A means total physical energy of the source mode or composite system:

$$\boxed{E_A = E_{\text{rest}} + E_{\text{kinetic}} + E_{\text{binding}} + E_{\text{internal}}.}$$

Confined/internal modes do not need their own free long gravitational tails; they contribute through the total energy of the composite external tail.

Then many different tails source one pressure profile:

$$\boxed{u(\mathbf{x}) = \sum_i u_i(\mathbf{x}), \qquad \nabla^2 u \sim -\rho_E}$$

in the weak-field regime, and the same u determines the effective geometry.

Equivalently, the far long-range pressure tail must have an energy monopole:

$$\boxed{u_A(r,\Omega) = \frac{Q_A^{(0)}}{r} + O(r^{-2}), \qquad Q_A^{(0)} = \eta\, E_A/c^2.}$$

The theorem target is that the common pressure source density equals total physical energy density in mass units:

$$\boxed{\sigma_{\text{grav}} = \alpha_E\, \rho_E/c^2.}$$

This is expected because the long common pressure channel should couple to the conserved scalar generator of stable medium evolution, namely total energy.

A more specific source theorem target is:

$$\boxed{S_u(x) := \frac{1}{N\sqrt{h}} \frac{\delta S_{\text{m}}}{\delta u(x)} = \rho_E(x) + \text{suppressed corrections.}}$$

Here S_u is the source seen by the compression/lapse variable u. The reason is that u changes the local operational process rate, and the generator of local process-rate change is the Hamiltonian density. Thus the common pressure channel should read $\rho_E = \mathcal{H}_{\text{phys}}$, not a species-private tail detail.

Species-specific tail structure may remain in K_{AB} or in internal sectors (spin, charge, colour, phase, polarization, topology), but must not create a second long-range spacetime carrier. If an unsuppressed species-private source term remains in S_u, the equivalence-principle source gate fails.

Open Theorem 9 (Matter-coupled closure and source universality)**.** *Beyond vacuum ADM closure, the matter-coupled constraints must close as well:*

$$\boxed{\mathcal{H}_{\text{tot}} = \mathcal{H}_{\text{grav}} + \mathcal{H}_{\text{matter}} \approx 0, \qquad \mathcal{H}_{i,\text{tot}} = \mathcal{H}_{i,\text{grav}} + \mathcal{H}_{i,\text{matter}} \approx 0.}$$

The closure target is:

$$\boxed{\{H_{\text{tot}}[N], H_{\text{tot}}[M]\} = D_{\text{tot}}\left[h^{ij}(N\partial_j M - M\partial_j N)\right].}$$

This requires stable matter families to use the same record metric and the same total-energy source:

$$\boxed{S_{\text{matter}} = S_{\text{matter}}[g,\Psi], \qquad S_u = \rho_E.}$$

No unsuppressed species-private gravitational source may remain:

$$\boxed{\rho_{\text{grav}} \neq \rho_E + \sigma_{\text{species}} \quad \textit{with } \sigma_{\text{species}} \textit{ physical and unsuppressed.}}$$

Internal/confined modes, including binding and gluon-like internal energy, must contribute to E_{total} while not defining an independent external spacetime carrier. This gate is open from $\mathcal{M}$.

For the ADM matter action

$$\boxed{S_{\mathrm{m}} = \int dt\, d^3x\, \left(\pi_A \dot{\psi}_A - N\mathcal{H}_{\mathrm{m}} - N^i \mathcal{H}_i\right), \qquad N = e^{-u},}$$

variation at fixed h_{ij} gives

$$\boxed{\delta_u S_{\mathrm{m}} = \int dt\, d^3x\, N\mathcal{H}_{\mathrm{m}}\, \delta u, \qquad \frac{1}{N\sqrt{h}} \frac{\delta S_{\mathrm{m}}}{\delta u} = \rho_E.}$$

Thus the low-energy ADM bridge already identifies the u-source with the Hamiltonian density. It remains to derive $N = e^{-u}$ and the universal ADM bridge from the base-medium.

6 Delta-M Candidate Theorem

Open Theorem 10 (Conditional Delta-M theorem)**.** *If the protected low-energy phase has:*

$$\boxed{\textit{one stable operational record phase with a common carrier/principal cone,}}$$

$$\boxed{\textit{exact Operational Relational Covariance,}}$$

$$\boxed{Q^{\mu\nu}_{AB} = G^{\mu\nu} K_{AB},}$$

and a local two-derivative low-energy hierarchy, then the tested core permits no unsuppressed independent medium residue:

$$\boxed{\Delta_{\mathcal{M}} = 0}$$

up to suppressed high-energy or large-scale correction windows.

7 Static Weak-Field Sector

Working Assumption 4 (Hamiltonian source)**.** *In the weak static regime, the compression/lapse constraint is sourced by total physical energy density ρ_E:*

$$\rho_E(\mathbf{x}) \geq 0.$$

The inertial mass of a localized source is:

$$M = \frac{1}{c^2} \int d^3x\, \rho_E(\mathbf{x}).$$

Working Assumption 5 (Base-node calibration of G)**.** *The source coefficient in the weak static sector is fixed by the minimal operational node scale:*

$$\boxed{G_{\mathrm{RMG}} = \frac{c^3 \ell_*^2}{\hbar}.}$$

Equivalently, the gravitational stiffness is

$$\boxed{\frac{c^4}{4\pi G_{\mathrm{RMG}}} = \frac{\hbar c}{4\pi \ell_*^2}.}$$

When the universal operational node scale is identified with the observed Planck-order length, G_{RMG} equals the measured Newton constant. The remaining deeper task is not to choose a smaller length, but to derive ℓ_* itself from the stable base-medium record dynamics.

Below, G denotes this calibrated G_{RMG} unless explicitly stated.
Use the static energy functional:

$$\mathcal{E}[u] = \frac{c^4}{8\pi G_{\mathrm{RMG}}} \int d^3x \, |\nabla u|^2 - \int d^3x \, \rho_E u.$$

Result 6 (Static source equation)**.** *Varying u gives:*

$$\boxed{\nabla^2 u = -\frac{4\pi G_{\mathrm{RMG}}}{c^4} \rho_E.}$$

With $u(\infty) = 0$:

$$u(\mathbf{x}) = \frac{G_{\mathrm{RMG}}}{c^4} \int d^3x' \, \frac{\rho_E(\mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|}.$$

Far from the source:

$$\boxed{u(r) = \frac{G_{\mathrm{RMG}} M}{rc^2} + O(r^{-2}).}$$

Thus the gravitational charge is inertial mass because the source is total energy.

8 PPN Weak-Field Check

Using:

$$U = c^2 u, \qquad U(r) = \frac{G_{\mathrm{RMG}} M}{r},$$

the metric gives:

$$g_{00} = -e^{-2u} = -1 + 2u - 2u^2 + O(u^3),$$
$$g_{ij} = e^{2u} \delta_{ij} = (1 + 2u + O(u^2))\delta_{ij}.$$

Therefore:

$$\boxed{\gamma = 1, \qquad \beta = 1.}$$

9 Constraint And Radiation Targets

The low-energy dynamic description must be expressible in ADM variables:

$$ds^2 = -N^2 c^2 dt^2 + h_{ij}(dx^i + N^i dt)(dx^j + N^j dt).$$

Weak split:

$$N = 1 - u + O(2),$$
$$h_{ij} = (1 + 2u)\delta_{ij} + h_{ij}^{TT} + O(2),$$
$$N_i = \mathcal{A}_i + O(2).$$

Low-Energy Target 1 (Scalar and vector sectors)**.** *The scalar/compression variable and the vector/flow variable must be constraint or near-zone variables at low energy, not independent vacuum radiation modes.*

The absence of scalar/vector radiation is protected only if lapse and shift remain non-propagating constraint variables.

Target form:

$$\nabla^2 u = -4\pi G\rho/c^2,$$
$$\nabla^2 \mathcal{A}_i = -\kappa_V G\mathcal{P}_i^T/c^3.$$

Low-Energy Target 2 (Tensor/shear radiation)**.** *The only propagating vacuum gravitational modes at low energy must be:*

$$h_+, \qquad h_\times.$$

They must have positive energy and propagate on the single record cone:

$$c_{\mathrm{GW}} = c.$$

Here c is not one of several speeds tuned to agree; it is the unique local signal speed defined by the common record cone after Root A closes.

Target wave equation:

$$\left(\frac{1}{c^2}\partial_t^2 - \nabla^2\right) h_{ij}^{TT} = \kappa_T G T_{ij}^{TT}/c^4.$$

Vacuum:

$$\left(\frac{1}{c^2}\partial_t^2 - \nabla^2\right) h_{ij}^{TT} = 0.$$

The propagation target also excludes massive vector modes, massive spin-2 leakage, a second metric tensor mode, and any extra shear channel in the tested local sector. Propagation alone is not sufficient: compact-binary emission must use the same protected tensor channel, with no leading scalar-dipole or scalar-monopole excess in binary-pulsar or waveform tests.

Open Theorem 11 (GW emission and radiation-reaction audit)**.** *Propagation and emission are separate gates. The local TT channel must not only propagate on the record cone; it must also carry the outgoing energy flux emitted by compact binaries. The sourced radiative target is*

$$\boxed{\left(\frac{1}{c^2}\partial_t^2 - \nabla^2\right) h_{ij}^{TT} = \Pi_{TT}\,\mathcal{S}_{ij}[g_{\mathrm{rec}}, T_{\mu\nu}] + \textit{suppressed corrections},}$$

where $T_{\mu\nu}$ comes from $S_{\mathrm{m}}[g_{\mathrm{rec}}, \Psi]$ and $\rho_E = T_{\mu\nu}n^\mu n^\nu$. The static and radiative constants must be the same:

$$\boxed{G_{\mathrm{Newton}} = G_{\mathrm{radiation}} = G_{\mathrm{record}}, \qquad c_{\mathrm{matter}} = c_{\mathrm{GW}} = c_{\mathrm{record}}.}$$

A mismatch between the Newtonian coupling, the conservative near-zone coupling and the radiative flux normalization is a GW-emission failure.

The emission audit proceeds in the local asymptotically flat binary regime:

$$\boxed{r \gg R_{\mathrm{source}}, \qquad r > \lambda_{\mathrm{GW}}, \qquad \lambda_{\mathrm{GW}} = c/f_{\mathrm{GW}}.}$$

The weak-field near zone must be matched to the wave zone; leading tensor radiation reaction is the first dissipative correction, while full high-PN waveform templates, merger and ringdown are deferred.

The leading multipole hierarchy must be derived rather than inserted:

$$\boxed{\textit{monopole conserved}, \qquad \textit{dipole removed by center-of-energy/sensitivity universality}, \qquad \textit{quadrupole first radiati}}$$

The dipole-zero condition inherits Root C:

$$\boxed{\alpha_A = \frac{q_A}{E_A/c^2}, \qquad \alpha_A = \alpha_B}$$

or a bound/suppression on $\alpha_A - \alpha_B$. Without this, scalar-like dipole radiation is not excluded.

The outgoing tensor flux must be positive and must equal the loss of near-zone mechanical energy:

$$\boxed{P_{TT} = \int r^2 d\Omega\, \mathcal{F}_{TT}[\dot{h}^{TT}], \qquad \frac{dE_{\rm orbit}}{dt} = -P_{TT}.}$$

For eccentric binaries and timing systems the angular-momentum balance must also hold:

$$\boxed{\frac{dL_{\rm orbit}}{dt} = -\mathcal{F}_{TT}^{\rm ang}.}$$

The leading chirp mass is built from Root C source masses

$$\boxed{m_A = E_A/c^2, \qquad \mathcal{M}_c = \frac{(m_1 m_2)^{3/5}}{(m_1 + m_2)^{1/5}}.}$$

Non-tensor channels – scalar monopole, scalar dipole, vector dipole, massive spin-2 leakage, second metric flux or pure breathing radiation – must be constraint, gauge, near-zone, short-range, gapped, damped, high-energy suppressed or below the relevant binary-pulsar and waveform bounds. If such a classification is not supplied, the channel is an emission anomaly.

Quantitative emission MVP. The first concrete emission target is a binary-pulsar timing comparison, with PSR B1913+16 as a canonical benchmark. The table to be filled by the calculation is:

quantity	RMG MVP target	failure mode
$\dot{P}_b$	from $dE_{\rm orbit}/dt = -P_{TT}$	wrong damping
m_A	E_A/c^2 from Root C	source mass mismatch
$\alpha_A - \alpha_B$	0 or below timing bounds	scalar dipole loss
eccentricity effects	$dL_{\rm orbit}/dt = -\mathcal{F}_{TT}^{\rm ang}$	wrong timing evolution

The benchmark observational scale is the Hulse–Taylor orbital decay, $\dot{P}_b \sim -2.4 \times 10^{-12}$ s/s, after the standard timing and environmental corrections. RMG must reproduce the tensor energy-loss channel without adding a private scalar/vector flux. For GW speed, the multimessenger benchmark remains

$$\boxed{|c_{\rm GW}/c - 1| \lesssim 10^{-15}}$$

as a propagation gate that emission must not disturb.

Gate 2 (GW detection: common-mode cancellation versus shear signal)**.** *A uniform or adiabatic compression of the local record system rescales the apparatus, clocks, rulers and light standards together; the scalar/compression channel u therefore couples to an interferometer signal only through differential terms. The local Michelson-like difference channel projects out the common-mode part:*

$$\boxed{\Pi_{\rm det}\, \delta g_{\rm common} = 0.}$$

The detector-visible gravitational-wave channel must instead be a differential record deformation:

$$\boxed{\Pi_{\rm det}\,\delta g_{\rm rec} = \Pi_{TT}\,\delta g_{\rm rec} + \Pi_{\rm shear/defect}\,\delta u_{\rm noncommon}.}$$

The first term is the protected tensor/shear sector h_{ij}^{TT}. The second term is allowed only as a controlled conversion of breathing/compression into detector-visible shear, tidal or symmetry-defect response. It must not survive as an unsuppressed local scalar radiation mode.

Here "defect projection" means a detector-visible differential record deformation which has a TT/shear component after projection onto the detector response. It is a placeholder for a later detector-response calculation, not a license to re-label a pure scalar breathing mode as a tensor wave.

A local apparatus cannot register a uniform rescaling of the medium from which its rods, clocks and light standards are built; it can register only anisotropic or differential record deformation. Observed gravitational-wave signals must therefore be carried by the protected TT/shear response or by a suppressed noncommon conversion channel, not by a pure common-mode scalar breathing.

Open Theorem 12 (Constraint protection route)**.** *Derive from the base-medium that the local low-energy variables enter in ADM-like form:*

$$\boxed{S_{\rm local} = \int dt\, d^3x \left(\pi^{ij}\dot{h}_{ij} - N\mathcal{H} - N^i\mathcal{H}_i\right)}$$

with no independent low-energy kinetic terms

$$\boxed{\dot{N}^2, \qquad \dot{N}_i\dot{N}^i, \qquad \dot{u}^2, \qquad \dot{\mathcal{A}}_i\dot{\mathcal{A}}^i}$$

for physical scalar/vector gravitational variables. If N and N^i impose first-class Hamiltonian and momentum constraints which close under time evolution, then u and local flow/shift variables are constraint/gauge or near-zone variables, not independent radiation. The vacuum gravitational record sector then has only

$$\boxed{h_+, \qquad h_\times}$$

as physical propagating modes. This is the local mechanism behind flow decoupling and the preferred-frame null target.

10 ADM/EH Fixed Point

Low-Energy Target 3 (Nonlinear low-energy completion)**.** *The nonlinear low-energy fixed point is:*

$$\boxed{S_{\rm low} = \frac{c^4}{16\pi G}\int d^4x\sqrt{-g_{\rm rec}}\,(R[g_{\rm rec}] - 2\Lambda) + S_{\rm matter}[g_{\rm rec},\Psi] + S_{\rm suppressed}.}$$

Coefficient status. In the ADM filter below, α is a temporary curvature coefficient. The Einstein–Hilbert target fixes $\alpha = 1$ after the normalization of the measured Newton constant G is chosen. The cosmological term Λ is kept as a local-core background parameter in the MVP, but in the full program it must be derived, regulated, or absorbed into the cosmological-background closure. It may not be used to repair a local ADM/EH or source-universality failure.

ADM form:

$$S_{\rm ADM} = \int dt\, d^3x \left(\pi^{ij}\dot{h}_{ij} - N\mathcal{H} - N^i\mathcal{H}_i\right).$$

Hamiltonian and momentum constraints:

$$\mathcal{H} \approx 0, \qquad \mathcal{H}_i \approx 0.$$

The target constraint algebra includes:

$$\{H[N], H[M]\} = D\big[h^{ij}(N\partial_j M - M\partial_j N)\big] .$$

Operational Relational Covariance makes ADM closure plausible, but it does not replace the proof.

Open Theorem 13 (Main closure theorem)**.** *Derive from the base-medium:*

$$\boxed{\mathcal{M} \Rightarrow \textit{universal principal symbol} \Rightarrow \textit{ADM constraint closure} \Rightarrow S_{\mathrm{EH}} + S_{\mathrm{matter}}.}$$

Open Theorem 14 (Hypersurface-deformation closure filter)**.** *The local Hamiltonian constraint must pass the ADM deformation-algebra filter. For the two-derivative ansatz*

$$\boxed{\mathcal{H} = \frac{1}{\sqrt{h}}\left(\pi^{ij}\pi_{ij} - \lambda\pi^2\right) - \sqrt{h}(\alpha R - 2\Lambda) + \Delta_{\mathcal{M}},}$$

the target bracket is:

$$\boxed{\{H[N], H[M]\} = D\big[h^{ij}(N\partial_j M - M\partial_j N)\big] .}$$

This filter requires, in the protected local core:

$$\boxed{\lambda = \frac{1}{2}, \qquad \Delta_{\mathcal{M}} = 0 \quad \textit{or constrained/gauge/suppressed},}$$

and no unsuppressed higher-derivative curvature terms at the tested two-derivative fixed point. If this closure filter is derived from $\mathcal{M}$, the trace/volume direction of the DeWitt supermetric belongs to the constrained sector and does not propagate as an independent scalar mode, and the local two-derivative fixed point is $S_{\mathrm{EH}} + S_{\mathrm{matter}}$ up to suppressed corrections.

Open Theorem 15 (Constraint preservation and matter-coupled Dirac closure)**.** *ADM form is not enough. The Hamiltonian and momentum constraints must be preserved by the evolution they generate:*

$$\boxed{\dot{\mathcal{H}} \approx 0, \qquad \dot{\mathcal{H}}_i \approx 0.}$$

The matter-coupled Dirac analysis must show that primary constraints lead to the Hamiltonian and momentum constraints, that secondary constraints close without making lapse or shift physical, and that any second-class remnants are solved or remove only nonphysical variables. If matter coupling turns the first-class ADM system into a non-closing algebra, the local target action fails even if the vacuum bracket has the desired form.

11 Local Gravity Closure Package

The many open local-core gates reduce to three root theorems.

Open Theorem 16 (Root A: one external record carrier)**.** *Derive from the base-medium stable record phase:*

$$\boxed{\textit{stable external record-forming mode} \Rightarrow \textit{one common external long-tail carrier channel.}}$$

All non-common branches must be internal, confined, gapped, damped, constrained, gauge or unstable. If Root A closes, it gives one operational principal cone, no low-energy birefringence and the factorization

$$\boxed{Q^{\mu\nu}_{AB} = G^{\mu\nu} K_{AB}}$$

on physical record modes.

Open Theorem 17 (Root A spectral/resonance closure)**.** *Root A must be proven by a spectral statement around the stable record phase $\mathcal{M}_0$. Linearize the base-medium dynamics:*

$$\boxed{\delta\mathcal{E}[\mathcal{M}_0] = \mathcal{L}_{\mathcal{M}_0}\delta\Phi + \cdots .}$$

The required split is:

$$\boxed{\mathrm{Spec}(\mathcal{L}_{\mathcal{M}_0}) = \mathcal{B}_{\mathrm{rec}} \cup \mathcal{B}_{\mathrm{int}}, \qquad N_{\mathrm{cone}}(\mathcal{B}^{\mathrm{gapless}}_{\mathrm{rec}}) = 1}$$

as one external record carrier channel. This is a one-common-carrier condition, not a one-scalar-mode condition. The common channel may contain shared polarization, shear or tensor patterns, but it must not split into species-private external carriers. Stable record-forming tail imprints must phase-lock to the base-medium rhythm ν_0 or its allowed harmonic channel:

$$\boxed{\Pi_{\mathrm{rec}} T^{\mathrm{long}}_A = Z_A \mathcal{C} + \textit{internal/lower-order details}, \qquad \mathrm{principal}(\mathcal{C}) = G^{\mu\nu} k_\mu k_\nu .}$$

All other branches must be internal, confined, gapped, damped, constrained, gauge, unstable, high-energy suppressed or large-scale-only. This is the microscopic calculation still needed for Root A.

Suppression catalogue. *These escape categories are not free passes. Each requires a structural or quantitative reason why it is absent from local record observables:*

- *high-energy suppressed: corrections scale with E/Λ_{UV} or k/Λ_{UV} and are below laboratory, Solar-System and GW bounds;*
- *large-scale-only: the term activates only outside the local tested regime and does not affect Solar-System, binary-pulsar or local GW tests;*
- *confined or internal: the mode cannot propagate as an external rod, clock, signal or detector record;*
- *gapped or damped: the mode has no coherent long-range low-energy propagation in the tested regime;*
- *gauge or constrained: the mode is removed by $\Omega_{\mathcal{M}}$, a first-class generator or an elliptic/near-zone constraint;*
- *unstable: the branch is excluded from the stable phase or decays before record formation.*

If no such suppression or removal test is available, the branch remains a Root A fail-risk rather than an allowed remainder.

Root A–Root B compatibility. Root A must not assume ADM closure, but its record/internal projectors and the gauge/constraint reduction induced by $\Omega_{\mathcal{M}}$ must remain compatible with the canonical analysis needed in Root B. A one-cone spectrum obtained by a projection that destroys the later ADM phase space is not a valid Root A pass.

Remark (Root A closure status under ORC). Conditional on the Operational Relational Covariance postulate of Section "Primitive Postulates", the Root A open theorem closes as a derived Result. The argument proceeds in three steps: (i) Goldstone–Bogoliubov spectral counting on the stable phase $\mathcal{M}_0$ with broken $U(1)_\nu$ rhythm yields one type-I scalar Goldstone (the phonon) plus two transverse-traceless tensor modes; (ii) the "one operational geometry" clause of ORC forces $c_{\text{phase}} = c_{\text{graviton}}$, so all gapless record modes share the single principal symbol $G^{\mu\nu}k_\mu k_\nu$; (iii) stable record-forming oscillons project on this single channel with species-dependent amplitudes Z_A but a common cone. Without ORC the closure remains open; the microscopic derivation of ORC itself is left as the deeper open program.

Open Theorem 18 (Root B: record-embedding ADM closure)**.** *Derive that changes of slicing, labels and record parametrization are gauge redundancies of the same physical record history. These redundancies must generate first-class constraints:*

$$\boxed{\mathcal{H} \approx 0, \qquad \mathcal{H}_i \approx 0,}$$

with the ADM hypersurface-deformation algebra. If Root B closes, local scalar/vector gravitational sectors are constrained, flow is shift-like in the tested core, only $h_+, h_\times$ propagate, $\lambda = 1/2$ is selected and the two-derivative fixed point is S_{EH}.

Open Theorem 19 (Root B record-action construction bridge)**.** *Root B may audit a projected record action, but it cannot treat that action as a magic input. The medium package must supply the chain*

$$\boxed{\mathcal{D}_{\mathcal{M}} \longrightarrow S^{\text{rec}}_{\text{eff}} \longrightarrow S_{\mathcal{R}}[g_{\text{rec}}, \Psi_{\text{rec}}] + \textit{controlled remainders.}}$$

The object delivered to Root B must be local up to controlled nonlocal tails, have a well-defined two-derivative truncation, match the projected record spectrum $(\mathcal{L}_{\mathcal{M}_0}, \Pi_{\mathcal{R}}, \Pi_{\mathcal{I}})$, give declared tensor or density type to every record variable, remove visible preferred-frame leaks through $\mathcal{G}_{\text{int}}$, and introduce matter through record observables rather than a private lapse, shift, source or metric. If this bridge is missing, Root B remains a target audit rather than a medium-derived ADM closure proof.

Open Theorem 20 (Root B embedding/symplectic closure)**.** *Root B requires showing that the physical object is a relational record history $W_{\mathcal{R}}$, while a foliation/embedding X_F is only a description of that history:*

$$\boxed{X_F \sim X'_F, \qquad \Gamma_{\text{phys}} = \Gamma_{\mathcal{R}}/\mathcal{G}_{\text{emb}}.}$$

This equivalence must be a true gauge degeneracy of the record symplectic structure:

$$\boxed{\iota_{\delta_\xi}\Omega_{\mathcal{R}} \approx 0.}$$

Then the embedding generators must take the ADM form

$$\boxed{G[\xi] = D[N^i] + H[N], \qquad D[N^i] = \int d^3x\, N^i \mathcal{H}_i, \qquad H[N] = \int d^3x\, N\mathcal{H},}$$

with

$$\boxed{\mathcal{H}_i \approx 0, \qquad \mathcal{H} \approx 0.}$$

The composition of two embedding changes must be the hypersurface-deformation algebra:

$$\boxed{\{D[N^i], D[M^i]\} = D[[N, M]^i],}$$

$$\boxed{\{H[N], D[M^i]\} = H[\mathcal{L}_M N],}$$

$$\boxed{\{H[N], H[M]\} = D\big[h^{ij}(N\partial_j M - M\partial_j N)\big]\,,}$$

where the structure function h^{ij} is the same inverse spatial metric seen by the record sector. Lapse and shift are then embedding multipliers, not extra physical medium particles:

$$\boxed{S_{\rm loc} = \int dt\, d^3x\, \left(\pi^{ij}\dot{h}_{ij} - N\mathcal{H} - N^i\mathcal{H}_i\right), \qquad \dot{N}^2 = \dot{N}_i\dot{N}^i = 0.}$$

This no-kinetic lapse/shift structure is a target output, not a starting assumption. It must follow from the projected record action and its presymplectic form: the momenta conjugate to N and N^i must be primary constraints, and the Dirac algorithm must preserve them without making lapse or shift physical. It may not be imposed merely by choosing an ADM-looking gauge. If this theorem closes, the scalar and vector gravitational sectors are constraints/near-zone bookkeeping, local flow is shift-like, only two tensor modes propagate, the DeWitt value $\lambda = 1/2$ is selected and the two-derivative local fixed point is $S_{\rm EH}$.

Open Theorem 21 (Root B finite sufficiency checklist)**.** *Root B is reduced to five sufficient conditions on the low-energy record sector:*

$$\boxed{\begin{array}{ll} B1 & \textit{no observable absolute record time,} \\ B2 & \textit{one record geometry } h_{ij}, \\ B3 & \textit{embedding changes are presymplectic null directions,} \\ B4 & \textit{null directions have local generators } D[N^i], H[N], \\ B5 & \textit{generator composition is anomaly-free.} \end{array}}$$

The algebraic danger is a residual anomaly in the normal-normal bracket:

$$\boxed{\{H[N], H[M]\} = D\big[h^{ij}(N\partial_j M - M\partial_j N)\big] + A[N, M].}$$

Root B requires

$$\boxed{A[N, M] = 0}$$

on the local tested record sector. A nonzero A signals an unsuppressed preferred-frame, scalar, vector or non-record tensor structure in the local sector. Equivalently, lapse/shift or local flow has become physical rather than embedding bookkeeping. The action-level check is:

$$\boxed{\dot{N}^2, \quad \dot{N}_i\dot{N}^i, \quad \textit{unsuppressed independent scalar/vector kinetic terms} \quad \textit{are absent.}}$$

Again, this is a derived condition: lapse/shift no-kinetic status must be obtained from the presymplectic/Dirac audit, not inserted as a gauge choice. Thus Root B is not an aesthetic GR assumption. It is the finite test:

$$\boxed{B1 + B2 + B3 + B4 + B5 \Rightarrow \textit{ADM local closure.}}$$

Open Theorem 22 (Root B physical motivation status)**.** *The base-medium picture supports Root B at the following graded level:*

condition	motivation status
$B1$	*strong support: time/measurement are operational and relational*
$B2$	*partial: requires Root A one-carrier closure*
$B3$	*open: requires a record presymplectic form* $\Omega_{\mathcal{R}}$
$B4$	*open: requires local moment-map generators* D, H
$B5$	*open: requires anomaly cancellation* $A[N, M] = 0$

A single undivided base medium has no internal ruler, delay or comparison until it self-differentiates into distinguishable record-forming nodes. This strongly supports the absence of an observable absolute record time. It does not by itself prove the canonical statement

$$\boxed{\iota_{\delta_\xi}\Omega_{\mathcal{R}} \approx 0 \quad \Rightarrow \quad G[\xi] = D[N^i] + H[N].}$$

The next Root B step is to construct the low-energy record phase space $\Gamma_{\mathcal{R}}$*, its presymplectic form* $\Omega_{\mathcal{R}}$*, the embedding-variation vector fields* δ_ξ *and the local generators* D, H*. The pass condition remains*

$$\boxed{\{H[N], H[M]\} - D\big[h^{ij}(N\partial_j M - M\partial_j N)\big] = 0}$$

in the tested local record sector.

Open Theorem 23 (Record phase space and presymplectic Root B construction)**.** *The non-circular Root B construction starts from the covariant record phase space*

$$\boxed{\Gamma_{\mathcal{R}} = \{\Phi_{\mathcal{R}} \text{ solving the record equations}\}/\mathcal{G}_{\text{int}}, \qquad \Phi_{\mathcal{R}} = (g_{\text{rec}}, \Psi_{\text{rec}}),}$$

where $\mathcal{G}_{\text{int}}$ *removes internal/non-record redundancies but does not yet quotient embedding freedom. Let the low-energy record action satisfy*

$$\boxed{\delta L_{\mathcal{R}} = E_{\mathcal{R}}\delta\Phi_{\mathcal{R}} + d\theta_{\mathcal{R}}(\delta\Phi_{\mathcal{R}}).}$$

Define the covariant presymplectic current

$$\boxed{\omega_{\mathcal{R}}(\delta_1\Phi, \delta_2\Phi) = \delta_1\theta_{\mathcal{R}}(\delta_2\Phi) - \delta_2\theta_{\mathcal{R}}(\delta_1\Phi)}$$

and the slice form

$$\boxed{\Omega_{\mathcal{R},\Sigma} = \int_\Sigma \omega_{\mathcal{R}}.}$$

On shell and without boundary flux, $d\omega_{\mathcal{R}} = 0$*, so*

$$\boxed{\Omega_{\mathcal{R},\Sigma} = \Omega_{\mathcal{R},\Sigma'}.}$$

This is the formal version of one record history described by different slices. An embedding deformation acts as

$$\boxed{\delta_\xi\Phi_{\mathcal{R}} = \mathcal{L}_\xi\Phi_{\mathcal{R}} \quad \textit{up to internal gauge compensation.}}$$

This statement must be evaluated field by field:

$$\boxed{\begin{array}{rcl} \delta_\xi g_{\rm rec} & = & \mathcal{L}_\xi g_{\rm rec}, \\ \delta_\xi \varphi & = & \xi^\mu \partial_\mu \varphi, \\ \delta_\xi T & = & \mathcal{L}_\xi T \quad \textit{for its declared tensor type}, \\ \delta_\xi \rho & = & \mathcal{L}_\xi \rho \quad \textit{including density weight.} \end{array}}$$

If a record variable has no declared tensor or density type, the embedding audit is not yet computable. Root B requires this direction to be presymplectically null:

$$\boxed{\Omega_{\mathcal{R}}(\delta\Phi, \delta_\xi\Phi) \approx 0.}$$

Equivalently, the Noether current

$$\boxed{J_\xi = \theta_{\mathcal{R}}(\mathcal{L}_\xi \Phi_{\mathcal{R}}) - \xi \cdot L_{\mathcal{R}}}$$

must split as

$$\boxed{J_\xi = C_\xi + dQ_\xi, \qquad \int_\Sigma C_\xi = H[N] + D[N^i], \qquad \xi = Nn + N^i e_i.}$$

The pass condition is therefore

$$\boxed{\Omega_{\mathcal{R}}(\delta\Phi, \mathcal{L}_\xi\Phi) = \delta\big(H[N] + D[N^i]\big)}$$

with $H \approx 0$ and $D \approx 0$ in the tested local record sector. Any bulk remainder that is not a constraint or boundary charge is the Root B preferred-frame/scalar/vector anomaly.

The brackets must be computed from $\Omega_{\mathcal{R}}$, not copied from the target algebra. Canonical variables and momenta are read from $\theta_{\mathcal{R}}$; second-class constraints require the Dirac bracket; auxiliary fields must be eliminated before the physical mode count; and boundary terms must be separated before any bulk anomaly $A[N, M]$ is declared.

Open Theorem 24 (Noether split and background-free record-action gate)**.** *The Noether split required by Root B follows if the local tested record action is background-free:*

$$\boxed{L_{\mathcal{R}}^{\rm local} = L_{\mathcal{R}}[g_{\rm rec}, \Psi_{\rm rec}]}$$

up to constraints, gauge choices, boundary terms, integrated-out internal contributions, high-energy suppressed corrections and large-scale-only flow/MOND terms. In that case an embedding deformation is a record diffeomorphism:

$$\boxed{\delta_\xi L_{\mathcal{R}} = \mathcal{L}_\xi L_{\mathcal{R}} = d(\xi \cdot L_{\mathcal{R}}),}$$

and the Noether current can be written as

$$\boxed{J_\xi = C_\xi + dQ_\xi, \qquad C_\xi = N\mathcal{H} + N^i\mathcal{H}_i.}$$

The forbidden local failure is a non-record background structure B:

$$\boxed{L_{\mathcal{R}}^{\rm local} = L_{\mathcal{R}}[\Phi_{\mathcal{R}}; B].}$$

If B is not a gauge, constraint, boundary, integrated-out, suppressed or large-scale-only object, then the embedding variation leaves a bulk anomaly:

$$\boxed{\mathcal{A}_\xi \sim \frac{\delta L_{\mathcal{R}}}{\delta B}\mathcal{L}_\xi B.}$$

The local-core forbidden B's are:

$$\boxed{T_{\mathcal{M}}, \quad U^\mu_{\mathcal{M}}, \quad v^\mu v^\nu, \quad \textit{species-private metric/clock}, \quad \dot{N}^2, \quad \dot{N}_i\dot{N}^i, \quad \textit{extra record cone.}}$$

Every bulk remainder in J_ξ must therefore be classified as exactly one of:

$$\boxed{\textit{constraint/gauge/boundary/internal/gapped/damped/suppressed/large-scale-only/anomaly.}}$$

Root B requires the final class to vanish in the local tested sector:

$$\boxed{\mathcal{A}^{\mathrm{local}}_\xi = 0 \quad\Longleftrightarrow\quad A[N,M] = 0.}$$

The allowed classes require a structural or scale test. High-energy remainders must scale below the relevant observation row, for example as $(E_{\mathrm{test}}/\Lambda_{\mathrm{UV}})^2$ or $(k_{\mathrm{test}}/\Lambda_{\mathrm{UV}})^2$; local PPN-sensitive terms need at least the corresponding weak-field bound. Large-scale-only terms must activate only outside Solar-System, binary-pulsar and local GW regimes; gapped or damped terms must lack coherent long-range propagation through the tested baseline.

Open Theorem 25 (Background-free two-derivative record-action fixed point)**.** *If Root A supplies one record metric g_{rec}, Root B removes all local non-record background structures, and the tested local regime admits a two-derivative effective hierarchy, then the local record action has the form*

$$\boxed{L^{\mathrm{local}}_{\mathcal{R}} = \sqrt{-g_{\mathrm{rec}}}\,[\alpha R[g_{\mathrm{rec}}] - 2\Lambda + L_{\mathrm{m}}(g_{\mathrm{rec}}, \Psi_{\mathrm{rec}})] + dB + O(\partial^4/M_*^2).}$$

At the two-derivative fixed point this is

$$\boxed{S_{\mathrm{EH}} + S_{\mathrm{matter}}}$$

up to normalization, cosmological term, boundary terms and suppressed higher-derivative corrections. The two-derivative ansatz is itself part of the audit: higher-derivative operators such as R^2, $R_{\mu\nu}R^{\mu\nu}$, Weyl-squared terms or preferred-frame curvature combinations must be suppressed by the hierarchy scale, not used to repair a failed ADM bracket. The leading curvature operator must be the EH operator, with α absorbed into the measured G or identified as a deviation/failure. The forbidden local background leaks are:

$$\boxed{U^\mu_{\mathcal{M}}U^\nu_{\mathcal{M}}\nabla_\mu\Psi\nabla_\nu\Psi, \quad v^\mu v^\nu R_{\mu\nu}, \quad \dot{N}^2, \quad \dot{N}_i\dot{N}^i, \quad g^{\mu\nu}_A\partial_\mu\Psi_A\partial_\nu\Psi_A, \quad C^{\mu\nu}k_\mu k_\nu \not\propto g^{\mu\nu}_{\mathrm{rec}}k_\mu k_\nu.}$$

Each such term must be absent, constrained, gauge, boundary-only, integrated-out/internal, high-energy suppressed or large-scale-only. Otherwise the two-derivative fixed point is not GR and Root B fails in the local tested core. Thus the local closure chain is

$$\boxed{\textit{one record metric} + \textit{background-free action} + \textit{two-derivative hierarchy} \Rightarrow S_{\mathrm{EH}} + S_{\mathrm{matter}}.}$$

Remark (Root B closure status under ORC). Conditional on the Operational Relational Covariance postulate of Section "Primitive Postulates", the Root B open theorem closes as a derived Result. The chain is: (i) ORC's "slicing/labels/parametrization are descriptions, not extra physical structures" forces lapse N and shift N^i into the role of Lagrange multipliers, so $\delta S/\delta N = 0$ and $\delta S/\delta N^i = 0$ yield $\mathcal{H} \approx 0$, $\mathcal{H}_i \approx 0$; (ii) ORC's exclusion of unobservable substrate handles forbids the anomaly $A[N, M]$, so the hypersurface-deformation algebra closes; (iii) by Kuchar–Teitelboim's "geometrodynamics regained" theorem (1976), the unique local two-derivative Hamiltonian with first-class HDA closure and Riemannian configuration space is Einstein–Hilbert plus minimally coupled matter, with DeWitt coefficient $\lambda = 1/2$; (iv) at the linearized level, $h_+, h_\times$ are the only propagating gravitational modes; scalar/vector/breathing sectors are constrained.

Open Theorem 26 (Root C: total-energy source projection)**.** *Derive the far-tail/source theorem:*

$$\boxed{Q_A^{(0)} = \eta E_A/c^2, \qquad S_u = \rho_E,}$$

where E_A includes rest, kinetic, binding and internal/confined energy. No unsuppressed species-private source may remain. If Root C closes, the equivalence-principle source gate and compact internal-energy effacement gate close.

Root C inheritance from Roots A and B. Root C uses, but does not re-prove, the previous root outputs:

one record metric g_{rec}	$\to$	$C1$ matter action target,
$N = e^{-u}$ or equivalent bridge	$\to$	$C2$ lapse variation,
ADM matter form with N multiplier	$\to$	$C3$ source theorem,
no scalar/vector vacuum radiation	$\to$	$C5$ anomaly control,
$\Pi_{\mathcal{R}}$ record/internal split	$\to$	internal energy accounting.

The equivalence principle, $q_A \propto E_A/c^2$, and $\sigma_{\mathrm{species}} = 0$ are outputs of Root C, not inputs.

Open Theorem 27 (Root C finite source chain)**.** *Root C reduces to five source conditions:*

$$\boxed{\begin{array}{ll} C1 & S_{\mathrm{m}} = S_{\mathrm{m}}[g_{\mathrm{rec}}, \Psi] \quad \textit{universal matter action}, \\ C2 & N = e^{-u} \quad \textit{or equivalent compression/lapse bridge}, \\ C3 & S_u = (N\sqrt{h})^{-1}\delta S_{\mathrm{m}}/\delta u = \rho_E, \\ C4 & \nabla^2 u = -\kappa \rho_E \quad \textit{far-tail Gauss law}, \\ C5 & \sigma_{\mathrm{species}}^{\mathrm{long,local}} = 0. \end{array}}$$

The bridge in $C2$ must come from the Root B handoff: N is the operational lapse of the record metric, expressed through the compression variable u. An "equivalent" bridge means a redefinition that produces the same shared lapse variation for every stable matter family; it cannot hide a species-dependent lapse. Then an isolated source obeys

$$\boxed{\oint_{\partial B_R} \nabla u \cdot dS = -\kappa \int_{B_R} \rho_E \, d^3x, \qquad u(r) = \frac{Q^{(0)}}{r} + O(r^{-2}),}$$

and therefore

$$\boxed{Q^{(0)} = \eta \, E_{\mathrm{total}}/c^2.}$$

This far-tail regime means $r \gg R_{\text{source}}$, $r \gg \ell_*$, and a convergent multipole expansion with higher multipoles suppressed by the relevant powers of R_{source}/r. A fitted $1/r$ tail may not be used backwards to infer the local source theorem. The substantive content of Root C is not the far $1/r$ integral but the requirement that the compression/lapse source equals the total physical energy for all stable record matter:

$$\boxed{S_u = \rho_E}$$

where the source density in the ADM theorem is measured by the lapse normal,

$$\boxed{\rho_E = T_{\mu\nu} n^\mu n^\nu,}$$

with n^μ the ADM unit normal in the record metric. For radiation and other traceless fields, the source is this energy density, not the trace $T^\mu{}_\mu$. with no unsuppressed species-private remainder:

$$\boxed{S_u \neq \rho_E + \sigma_{\text{species}} \quad \textit{for physical long-range local } \sigma_{\text{species}}.}$$

Species-specific structure is allowed only if it is an internal divergence, gauge, boundary multipole, high-energy suppressed, short-range/confined or already included in ρ_E. Internal, binding, gluon-like and confined energy must contribute to E_{total} while not producing an independent external cone:

$$\boxed{E_{\text{total}} = E_{\text{rest}} + E_{\text{kinetic}} + E_{\text{binding}} + E_{\text{internal}}.}$$

Here binding includes self-gravitational binding in the strong-equivalence audit. Self-gravitating bodies must not fall differently merely because a different fraction of their total mass is stored as gravitational binding energy. In local tests this is the Nordtvedt-type failure mode; in compact binaries it appears as a sensitivity/dipole-radiation audit.

Self-gravitational binding is not assumed to be an ordinary local matter density. Root C treats it through the canonical total energy or quasi-local charge appropriate to the record metric and through compact-object sensitivities. The pass condition is that this binding changes the external monopole only through E_{total}, not through an extra composition-readable source charge.

Open Theorem 28 (C3 lapse-variation source theorem)**.** *Assume the stable record matter sector satisfies*

$$\boxed{S_{\rm m} = S_{\rm m}[g_{\rm rec}, \Psi], \qquad N = e^{-u},}$$

and has no unsuppressed explicit species-private u-coupling outside the record metric. In canonical form,

$$\boxed{S_{\rm m} = \int dt\, d^3x \left(\pi_A \dot{\psi}_A - N\mathcal{H}_{\rm m} - N^i \mathcal{H}_{i,\rm m}\right).}$$

At fixed $h_{ij}, N^i, \psi_A, \pi_A$, treating u as the independent compression variable and N as fixed by the bridge $N = e^{-u}$,

$$\boxed{\delta N = -N\delta u, \qquad \delta_u S_{\rm m} = \int dt\, d^3x\, N\mathcal{H}_{\rm m}\delta u.}$$

Therefore

$$\boxed{S_u = \frac{1}{N\sqrt{h}} \frac{\delta S_{\rm m}}{\delta u} = \frac{\mathcal{H}_{\rm m}}{\sqrt{h}} = \rho_E.}$$

This is the Root C bridge: changing compression/lapse changes the local process rate, and the generator of local process-time translation is the Hamiltonian density. Internal, binding, gluon-like and confined energy are included because

$$\boxed{\mathcal{H}_{\rm m} = \mathcal{H}_{\rm rest} + \mathcal{H}_{\rm kinetic} + \mathcal{H}_{\rm binding} + \mathcal{H}_{\rm internal}.}$$

For quantum matter, $\mathcal{H}_{\rm m}$ means the expectation value of the matter Hamiltonian density in the relevant matter state. Vacuum subtractions must be handled consistently with the cosmological term in the target action; vacuum energy cannot be silently counted in local source tests and then removed in the cosmology window. Root C fails only if a physical long-range local remainder survives:

$$\boxed{S_u = \rho_E + \sigma^{\rm long,local}_{\rm species}, \qquad \sigma^{\rm long,local}_{\rm species} \neq 0.}$$

Thus the C3 gate is: all stable record matter must see u only through the same record metric/lapse, except for gauge, constrained, boundary, internal, short-range or suppressed terms already accounted for in ρ_E.

Open Theorem 29 (Metric-exhaustion and no species-private u-coupling)**.** *If the stable local matter sector is exhausted by one record metric,*

$$\boxed{S_{\rm m} = S_{\rm m}[g_{\rm rec}, \Psi],}$$

then all allowed local u-dependence is metric/lapse dependence. Species labels may appear in masses, charges, spin, internal Hamiltonians, confined sectors, kinetic matrices, multipoles and finite-size corrections, but not as a separate long-range local term

$$\boxed{\Delta S_A = \int d^4x\, N\sqrt{h}\, F_A(u, \Psi_A).}$$

If such a term is physical and unsuppressed, then

$$\boxed{S_u = \rho_E + \sigma_A,}$$

so u becomes a species-readable scalar background rather than just the record lapse/compression variable. This is a fifth-force/source anomaly: two systems with the same total energy but different composition would source or fall differently. Therefore the local Root C gate is

$$\boxed{F_A^{\rm long,local} = 0 \quad \Longleftrightarrow \quad \sigma^{\rm long,local}_{\rm species} = 0,}$$

unless the term is already included in ρ_E, or is gauge, constrained, boundary, internal, short-range/confined or high-energy suppressed. In the common pressure projection language,

$$\boxed{\Pi_{\rm grav}(T_A) = q_A \mathcal{G}, \qquad q_A = \eta E_A/c^2}$$

must hold at monopole order; species-specific tail data may survive only as spin, charge, color, phase, multipole, internal, short-range or suppressed structure.

The allowed classifications are not labels of convenience. A candidate $\sigma_{\rm species}$ term must either be part of the physical Hamiltonian density ρ_E, vanish on reduced observables, be a boundary/multipole term rather than a local monopole source, be internal/confined with no external u-tail, be short-range, or be suppressed below the relevant equivalence-principle, fifth-force, Solar-System or binary-pulsar gate. If no such test is supplied, the term remains a Root C anomaly.

Open Theorem 30 (Root C matter-family and sensitivity audit)**.** *Root C must be checked for pointlike massive matter, photons and fields, bound systems, internal/confined sectors, composite macroscopic bodies, self-gravitating bodies and compact objects. Photons and other traceless fields source u through $T_{\mu\nu}n^\mu n^\nu$, not through a trace coupling. Bound, internal, confined and field energy must enter $\mathcal{H}_\mathrm{m}$ and hence ρ_E, while not creating a private external cone or private u-source.*

For compact binaries define the long-range source charge per unit total energy

$$\boxed{\alpha_A = \frac{q_A}{E_A/c^2}.}$$

The Root C target is $\alpha_A = \alpha_B$ for all stable compact sources, or else $\alpha_A - \alpha_B$ must be gauge, constrained, short-range, gapped, damped or below binary-pulsar and waveform bounds. Otherwise the binary loses energy through a leading scalar dipole or monopole channel rather than through the protected tensor emission channel.

Remark (Root C closure status under ORC). Conditional on the Operational Relational Covariance postulate, the Root C open theorems close as derived Results once Root A and Root B are in place. The argument is: (i) Root A under ORC yields a single record metric $g^{\mu\nu}_{\mathrm{rec}}$; Root B under ORC yields $S = S_{\mathrm{EH}}[g_{\mathrm{rec}}] + S_{\mathrm{matter}}[g_{\mathrm{rec}}, \Psi]$; (ii) the variational stress-energy tensor $T^{\mu\nu} = (2/\sqrt{-g})\,\delta S_{\mathrm{matter}}/\delta g_{\mu\nu}$ is the unique gravitational source compatible with the diffeomorphism invariance of S_{matter} and contains rest, kinetic, binding, internal/confined and field energies of every stable matter species; (iii) ORC's "one operational geometry for every observer" clause forbids species-private cones and species-private u-coupling, so $C5$ ($\sigma^{\mathrm{long,local}}_{\mathrm{species}} = 0$) is automatic; (iv) $\nabla^2 u = -\kappa\rho_E$ and $Q^{(0)} = \eta E_{\mathrm{total}}/c^2$ then follow from the linearized Einstein equation in the weak static limit. Residual quantitative items — the numerical value of G, the smallness of Λ, and Standard-Model species-by-species verification of the universal coupling — remain open as part of the broader G-closure and matter-emergence programs.

Open Theorem 31 (Local gravity core closure)**.** *If Root A, Root B and Root C are derived from $\mathcal{M}$, then the local RMG gravity core is closed:*

$$\boxed{\textit{one operational spacetime} \quad + \quad S_{\mathrm{EH}} + S_{\mathrm{matter}} \quad + \quad \textit{universal total-energy source.}}$$

The remaining sectors – strong-field compact interiors, MOND reentry, cosmology and detailed matter emergence – then become extension windows, not loose foundations of the local gravitational core.

Remark (full gravity-core closure under ORC). Conditional on the Operational Relational Covariance postulate, the three roots close as derived Results (remarks above on Root A, Root B, Root C), and therefore so does this "Local gravity core closure" openthm. The unified statement is: under ORC plus the minimal RMG structural commitments (one base medium $\mathcal{M}$ with stable phase $\mathcal{M}_0$; locality and two-derivative low-energy hierarchy), the low-energy operational sector is exactly

$$\boxed{S_{\mathrm{low}} = \frac{c^4}{16\pi G}\int d^4x\,\sqrt{-g}\,(R - 2\Lambda) + S_{\mathrm{matter}}[g, \Psi] + \text{suppressed corrections,}}$$

with one record cone, two TT graviton modes at $c_{\mathrm{GW}} = c$, no scalar or vector gravity waves, no preferred frame, universal free fall, and gravitational source equal to the total physical energy-momentum. This is the tested local GR core.

Status note. The ORC postulate is the single controlling axiom of this closure; its microscopic derivation from a deeper relational substrate remains the central long-term research program. Candidate routes include emergent gauge redundancy from a UV principle, ORC as an IR fixed point of a nonlinear relational substrate, and consistency arguments showing that ORC-violating theories produce internal pathologies. No such derivation has yet been completed. Until then, the gravitational core is closed conditional on ORC.

Open Theorem 32 (Non-circular local-core proof order)**.** *The local gravity proof must close in the following order:*

$$\boxed{\mathcal{M}_{\text{stable}} \Rightarrow \textit{Root A} \Rightarrow g_{\text{rec}} \Rightarrow \textit{Root B} \Rightarrow S_{\text{EH}} + S_{\text{matter}} \Rightarrow \textit{Root C} \Rightarrow S_u = \rho_E.}$$

Equivalently:

$$\boxed{\begin{array}{lcl} \textit{Root A} & : & \textit{one common external record carrier/metric,} \\ \textit{Root B} & : & \textit{record re-embedding gauge} \;\Rightarrow\; \textit{ADM/EH,} \\ \textit{Root C} & : & \textit{metric/lapse matter coupling} \;\Rightarrow\; \textit{total-energy source.} \end{array}}$$

The proof must obey the no-circularity rules:

$$\boxed{\begin{array}{l} \textit{do not assume ADM to prove Root A,} \\ \textit{do not assume Einstein equations to prove Root B,} \\ \textit{do not assume the equivalence principle to prove Root C,} \\ \textit{do not use MOND/cosmology to compensate for a local-core failure.} \end{array}}$$

The microscopic object still needed from the base medium is an effective record action

$$\boxed{S^{\text{rec}}_{\text{eff}}[\mathcal{M}_0]}$$

with one record metric, no local background leak, a covariant presymplectic form, universal matter metric coupling and no explicit species-private u-coupling. This is the next exact derivation target.

12 Leading-Order Relational Base Derivation

The current internal result is a leading-order derivation of the tested local gravity core from a relational base-medium action candidate. It is not the final microscopic theory: the exact coefficients, matter spectrum, MOND scale and strong-field branch remain to be computed.

Result 7 (Relational base action candidate)**.** *The base-medium is represented without external rods, clocks or background space by relational variables*

$$\boxed{R_{ab}, \qquad \vartheta_{ab}, \qquad K_{ab}.}$$

Here R_{ab} denotes adjacency/separation records, ϑ_{ab} the common rhythm/phase records, and K_{ab} internal coupling/shape records. The candidate base action has Jacobi/constraint form

$$\boxed{S^{\text{rel}}_{\mathcal{M}} = \int d\lambda \, \Big[\Theta_{\mathcal{M}}(R, \vartheta, K; \dot{R}, \dot{\vartheta}, \dot{K}) - \alpha C_0 - \beta^a C_a - \gamma C_\nu\Big].}$$

λ is only a bookkeeping parameter, not physical time. The constraints have the intended meanings

$$\boxed{C_0 = \textit{common process constraint}, \qquad C_a = \textit{relabel/re-embedding constraint}, \qquad C_\nu = \textit{one-rhythm/harmonic sele}}$$

Thus the base does not begin with spacetime. Spacetime variables are allowed only as coarse-grained record variables.

Result 8 (Continuum core route). *Under smooth coarse-graining, local isotropy, one common record carrier and no physical background-label leak, the continuum record variables are*

$$\boxed{h_{ij} = h_{ij}[R, \vartheta, K], \qquad \pi^{ij} = \pi^{ij}[R, \vartheta, K].}$$

Relabel/re-embedding redundancy gives the momentum constraint:

$$\boxed{C_a \longrightarrow H_i.}$$

Small relational loop defects give curvature:

$$\boxed{\Omega_{\text{loop}} - 1 \longrightarrow R^i{}_{jkl}(h), \qquad V_{\text{loop}} \longrightarrow \int d^3x\,\sqrt{h}\,(R(h) - 2\Lambda).}$$

The relational kinetic term decomposes into physical shear plus constrained volume/gauge directions. In the protected continuum limit this gives the DeWitt kinetic form

$$\boxed{T_{\text{rec}} \longrightarrow \frac{1}{\sqrt{h}}\left(\pi_{ij}\pi^{ij} - \frac{1}{2}\pi^2\right).}$$

Therefore the leading-order local record Hamiltonian is

$$\boxed{H = \frac{1}{\sqrt{h}}\left(\pi_{ij}\pi^{ij} - \frac{1}{2}\pi^2\right) - \sqrt{h}\,(R(h) - 2\Lambda) + H_{\text{matter}}.}$$

Together with H_i, this is the ADM/Einstein-Hilbert local fixed point at leading order.

Result 9 (Radiation and source consequence). *The metric perturbation splits operationally as*

$$\boxed{\delta h_{ij} = s_{ij} + \frac{1}{3}h_{ij}b + \textit{gauge/constraint pieces.}}$$

The common breathing variable b is not a separate free radiation field in the tested local sector; it is hidden by common-mode self-calibration and by the Hamiltonian constraint. The observable radiation is the shear sector

$$\boxed{s_{ij} = h_{ij}^{TT},}$$

which carries two tensor modes. Matter couples to the same record metric, so the local source is total energy rather than a species-private tail charge.

Gate 3 (Leading-order status). *RMG currently has a conditional leading-order route*

$$\boxed{\mathcal{M}_{\text{rel}} \Rightarrow \textit{record geometry} \Rightarrow \textit{ADM/EH local core} \Rightarrow \textit{two tensor GW modes} \Rightarrow \textit{total-energy source.}}$$

This is the present leading-order core result. It becomes a full derivation only after the exact coarse-graining map, the constants G, Λ, the stable spectrum, the matter couplings, the MOND reentry scale and the strong-field branch are computed from the base-medium rather than inserted as targets.

Open Theorem 33 ($S_{\mathrm{eff}}^{\mathrm{rec}}$ derivation pipeline)**.** *Start from a base-medium action/evolution functional*

$$\boxed{S_{\mathcal{M}}[\Phi_{\mathcal{M}}], \qquad \Phi_{\mathcal{M}} = \Phi_0 + \delta\Phi,}$$

around a stable phase $\mathcal{M}_0$. Linearization gives

$$\boxed{\delta^2 S_{\mathcal{M}} = \frac{1}{2}\langle \delta\Phi, \mathcal{L}_{\mathcal{M}_0}\delta\Phi\rangle.}$$

The fluctuation space must split into record and internal sectors:

$$\boxed{\delta\Phi = R + I, \qquad R = \Pi_{\mathcal{R}}\delta\Phi, \qquad I = \Pi_{\mathcal{I}}\delta\Phi,}$$

with

$$\boxed{\Pi_{\mathcal{R}} + \Pi_{\mathcal{I}} = 1, \qquad \Pi_{\mathcal{R}}\Pi_{\mathcal{I}} = 0.}$$

The record effective action is then the low-energy functional

$$\boxed{e^{iS_{\mathrm{eff}}^{\mathrm{rec}}[R]} = \int \mathcal{D}I\, e^{iS_{\mathcal{M}}[R,I]}.}$$

Equivalently,

$$\boxed{S_{\mathrm{eff}}^{\mathrm{rec}}[R] = -i\log \int \mathcal{D}I\, e^{iS_{\mathcal{M}}[R,I]}.}$$

The record-sector principal kernel must factorize:

$$\boxed{S_{\mathcal{R}}^{(2)} \sim \int R_A\, Q_{AB}^{\mu\nu}\partial_\mu\partial_\nu R_B, \qquad Q_{AB}^{\mu\nu} = G^{\mu\nu}K_{AB},}$$

so that one record metric is defined by $g_{\mathrm{rec}}^{\mu\nu} \sim G^{\mu\nu}$. The internal integration may renormalize constants, add binding energy, multipoles, finite-size terms, high-energy suppressed operators or large-scale-only corrections. It must not leave local unsuppressed

$$\boxed{U_{\mathcal{M}}^{\mu}, \quad v^\mu v^\nu, \quad \textit{species-private metric}, \quad \textit{explicit species-private u-coupling}, \quad \dot{N}^2, \quad \dot{N}_i\dot{N}^i, \quad \textit{extra record cone}.}$$

If this pipeline succeeds, the same microscopic object supplies the raw material for Root A, Root B and Root C. If it fails, the failure is localized as either non-factorization, background leak, or source anomaly.

Open Theorem 34 (Minimal base-medium input package)**.** *To compute $S_{\mathrm{eff}}^{\mathrm{rec}}$, the base-medium model must provide the input package*

$$\boxed{\mathcal{D}_{\mathcal{M}} = (\Phi_{\mathcal{M}}, S_{\mathcal{M}}, \mathcal{M}_0, \ell_*, \Omega_{\mathcal{M}}, \mathcal{G}_{\mathrm{int}}, \mathcal{O}_{\mathrm{rec}}).}$$

Here $\Phi_{\mathcal{M}}$ denotes the microscopic/collective medium variables, $S_{\mathcal{M}}$ the action or evolution functional, $\mathcal{M}_0$ the stable low-energy phase, ℓ_* the minimal operational node scale, $\Omega_{\mathcal{M}}$ the symplectic or presymplectic structure, $\mathcal{G}_{\mathrm{int}}$ the internal redundancy group, and $\mathcal{O}_{\mathrm{rec}}$ the operational record observables. Around $\mathcal{M}_0$,

$$\boxed{\Phi_{\mathcal{M}} = \Phi_0 + \delta\Phi, \qquad \delta^2 S_{\mathcal{M}} = \frac{1}{2}\langle \delta\Phi, \mathcal{L}_{\mathcal{M}_0}\delta\Phi\rangle.}$$

The spectral target is

$$\boxed{\mathrm{Spec}(\mathcal{L}_{\mathcal{M}_0}) = \mathcal{B}_{\mathrm{rec}} \cup \mathcal{B}_{\mathrm{int}}, \qquad N_{\mathrm{cone}}(\mathcal{B}_{\mathrm{rec}}^{\mathrm{gapless}}) = 1}$$

as one external record carrier/cone, not one scalar field. The projectors must be built from observability and spectrum, not by assuming GR:

$$\boxed{\Pi_{\mathcal{R}} + \Pi_{\mathcal{I}} = 1, \qquad \Pi_{\mathcal{R}}\Pi_{\mathcal{I}} = 0.}$$

At quadratic order, internal integration has the block form

$$\boxed{S^{(2)} = \frac{1}{2}\begin{pmatrix} R & I \end{pmatrix}\begin{pmatrix} L_{RR} & L_{RI} \\ L_{IR} & L_{II} \end{pmatrix}\begin{pmatrix} R \\ I \end{pmatrix}, \qquad L_{\mathrm{eff}} = L_{RR} - L_{RI}L_{II}^{-1}L_{IR}.}$$

The Schur-complement correction may renormalize the record kernel, binding energy, multipoles or suppressed operators. It must not generate an extra cone, local preferred-frame tensor, species-private metric, explicit species-private u-coupling or lapse/shift kinetic terms. Without this input package, RMG-1 has a well-defined proof target; the microscopic derivation remains open.

Status of the package. *At this stage $\mathcal{D}_{\mathcal{M}}$ is a required framework, not yet a completed microscopic specification. Before Root A can be calculated, it must be promoted to at least one explicit ansatz whose quadratic operator, projectors and record/internal split can actually be computed.*

Additional input requirements. *The node scale ℓ_* must be assigned a clear status. In a first toy model it may be treated as a primitive coarse-graining input; in the final theory it should be derived or calibrated from $S_{\mathcal{M}}$ and stable record dynamics. The form $\Omega_{\mathcal{M}}$ must identify physical, gauge, constrained and presymplectic-null directions; the spectrum of $\mathcal{L}_{\mathcal{M}_0}$ alone is not enough, because a Hessian mode may be gauge or constrained rather than physical.*

The record-internal coupling L_{RI} is a central physics input. Its symmetry, sign, scale and derivative order must be specified. The correction $-L_{RI}L_{II}^{-1}L_{IR}$ may renormalize record dynamics, but it must not change the principal cone, create a long-range private source, or flip the sign of a physical kinetic term.

Open Theorem 35 (Candidate base-medium order-parameter ansatz)**.** *A first calculable microscopic target is a single base-medium order parameter, not many independent fundamental substances:*

$$\boxed{\Phi_{\mathcal{M}} = \Psi_{\mathcal{M}} = \sqrt{n}\, e^{i\theta}\chi.}$$

Here n is local node-density/pressure amplitude, θ is phase or medium rhythm, and χ denotes internal shape/orientation data of the same medium. This ansatz is allowed only if χ does not become an independent long-range spacetime carrier. It must be internal, confined, gauge, constrained, suppressed, or part of record matter sharing the same carrier. Before Root A is calculated, χ must be assigned a definite mathematical type: scalar, vector, tensor, group-valued field, constrained composite variable, or topological label. Merely saying "internal shape" is not yet a microscopic specification.

A minimal positive-energy candidate functional is

$$\boxed{E_{\mathcal{M}} = \int d^3X \left[\frac{\kappa_n}{2}(\nabla n)^2 + \frac{\kappa_\theta}{2}n(\nabla\theta)^2 + V(n) + E_{\mathrm{int}}(n,\theta,\chi,\nabla\chi)\right],}$$

with a stable phase

$$\boxed{V'(n_0) = 0, \qquad V''(n_0) > 0.}$$

Schematically,

$$\boxed{S_{\mathcal{M}} = \int d\tau \left[\Theta_{\mathcal{M}}(\Phi_{\mathcal{M}}, \dot{\Phi}_{\mathcal{M}}) - E_{\mathcal{M}}[\Phi_{\mathcal{M}}]\right],}$$

where $\Theta_{\mathcal{M}}$ defines the symplectic structure. The stable phase expansion is

$$\boxed{n = n_0 + \delta n, \qquad \theta = \nu_0\tau + \delta\theta, \qquad \chi = \chi_0 + \delta\chi.}$$

The Root A spectral target is still

$$\boxed{N_{\mathrm{cone}}(\mathcal{B}^{\mathrm{gapless}}_{\mathrm{rec}}) = 1.}$$

This means one observable external carrier/cone, not one scalar excitation. The bookkeeping parameter τ, any preferred flow, and any extra observable carrier branch must not survive as local observables in $S^{\mathrm{rec}}_{\mathrm{eff}}$. The background form $\theta = \nu_0\tau + \delta\theta$ is allowed only if τ is protected from becoming an observable absolute time. The required condition is

$$\boxed{\delta_\tau \mathcal{O}_{\mathrm{rec}} \approx 0}$$

for all physical record observables, up to gauge or constraint terms. The allowed routes are a parametrized action with a reparametrization constraint, a relational clock gauge fixing, or a presymplectic null direction generated by changes of the time label.

The local Lorentz behavior of the record sector must also be explicit. Either it emerges from factorization,

$$\boxed{Q^{\mu\nu}_{AB} = G^{\mu\nu} K_{AB},}$$

with one common principal cone, or a covariant low-energy record action is stated as a target fixed point. It is not enough to assume observable Lorentz invariance while leaving a physical medium time, medium velocity or species-private cone in the local record sector. The long compression variable u should arise from a pressure/density deficit, for example $u \sim -\delta \ln n$, and its weak static equation must be constraint-like:

$$\boxed{\nabla^2 u = -\kappa\rho_E + \textit{allowed corrections},}$$

so that u remains a constraint variable rather than an independent propagating scalar mode. This ansatz is a test model, not a proof: if it leaves a physical medium time, preferred frame or extra cone, it must be constrained or replaced.

Open Theorem 36 (Working toy instantiation of the input package)**.** *The first calculable test bench is allowed to be wrong; its purpose is to make $\mathcal{D}_{\mathcal{M}}$ concrete enough that the Root A spectrum and the spin-2 gate can be tested. Take*

$$\boxed{\Phi^{\mathrm{toy}}_{\mathcal{M}} = (\phi, \theta, Q_{ij}), \qquad \phi = \sqrt{n},}$$

where Q_{ij} is a real symmetric traceless internal strain tensor,

$$\boxed{Q_{ij} = Q_{ji}, \qquad Q^{i}{}_{i} = 0.}$$

The toy stable phase is isotropic:

$$\boxed{\phi = v, \qquad \theta = \nu_0\tau, \qquad Q_{ij} = 0.}$$

Tensor information, if any, must therefore come from fluctuations of Q_{ij} and their record-visible transverse-traceless projection, not from a preferred background direction.

A minimal schematic toy energy is

$$\boxed{E_{\text{toy}} = \int d^3X \left[\frac{k_\phi}{2}(\nabla\phi)^2 + \frac{k_\theta}{2}\phi^2(\nabla\theta)^2 + \frac{\lambda_\phi}{4}(\phi^2 - v^2)^2 + \frac{k_Q}{2}(\nabla Q_{ij})^2 + \frac{\mu_Q^2}{2}Q_{ij}Q_{ij} + \frac{g_Q}{2}\phi^2 Q_{ij}Q_{ij} + E_{\text{mix}} \right]}.$$

The action must be parametrized or presymplectic, for example

$$\boxed{S_{\text{toy}} = \int d\tau \left[\Theta_{\text{toy}} - \alpha C_0 - \beta^a C_a - E_{\text{toy}}\right],}$$

so that the bookkeeping label τ and relabeling variables do not become physical background structures.

Expanding

$$\boxed{\phi = v + \eta, \qquad \theta = \nu_0\tau + \vartheta, \qquad Q_{ij} = q_{ij},}$$

the quadratic target has the schematic form

$$\boxed{\delta^2 E_{\text{toy}} = \frac{1}{2}\eta(-k_\phi\nabla^2 + m_\eta^2)\eta + \frac{1}{2}\vartheta(-k_\theta v^2\nabla^2)\vartheta + \frac{1}{2}q_{ij}(-k_Q\nabla^2 + m_Q^2)q_{ij} + L_{\text{mix}}^{(2)},}$$

with

$$\boxed{m_\eta^2 = 2\lambda_\phi v^2, \qquad m_Q^2 = \mu_Q^2 + g_Q v^2.}$$

The conservative spectrum reading is: η is gapped if $m_\eta^2 > 0$; ϑ is the candidate common phase/cone branch but not the graviton; q_{ij} is gapped if $m_Q^2 > 0$; and $L_{\text{mix}}^{(2)}$ must not change the principal cone. For the conservative Root A toy, $m_Q^2 > 0$ is required unless the Q-sector is explicitly being tested as the shared-cone TT candidate. An unintended massless Q_{ij} branch carries five symmetric-traceless components and can create extra long-range branches, threatening $N_{\text{cone}} = 1$. This version may help Root A while still failing to produce tensor gravity.

Allowed toy mixing terms must be derivative-order and symmetry controlled, for example

$$\boxed{a_1\eta(\nabla\vartheta)^2, \quad a_2\eta Q_{ij}Q_{ij}, \quad a_3 Q_{ij}\partial_i\vartheta\,\partial_j\vartheta, \quad a_4\eta\,\partial_k Q_{ij}\partial_k Q_{ij}.}$$

They are allowed only if, after expansion and internal integration, they do not split the record cone, introduce a preferred tensor or create a long-range species-private source.

To test the tensor gate, decompose

$$\boxed{q_{ij} = q_{ij}^{TT} + \partial_i V_j + \partial_j V_i + \left(\partial_i\partial_j - \frac{1}{3}\delta_{ij}\nabla^2\right)B + \frac{1}{3}\delta_{ij}H.}$$

The tensor candidate passes only if

$$\boxed{\operatorname{rank} K_{Q-\text{strain},TT} = 2, \qquad \textit{positive kinetic sign}, \qquad c_{TT} = c_{\text{rec}},}$$

and all scalar/vector pieces are gauge, constrained, near-zone or suppressed. Rank zero means no tensor gravity; rank greater than two means extra local shear/scalar/vector radiation; a wrong sign, mass term or different cone also fails. Thus the toy model localizes failure instead of hiding it: extra cones, ghosts, preferred frames or species-private sources identify which part of $\mathcal{D}_{\mathcal{M}}$ must be changed.

Result 10 (Stable-record coherent amplitude completion). *The ansatz $\Phi_{\mathcal{M}} = \sqrt{n}e^{i\theta}\chi$ by itself is not yet a microscopic proof of the square law. The square law is derived if the projected long-range stable record sector has the coherent endpoint-counting completion*

$$\boxed{\rho := |\Phi_{\mathcal{M}}| = \sqrt{n}}$$

and the first-order record action

$$\boxed{S_{\text{rec}} = \int d\lambda\, d^3q \left[P_A \dot{R}_A - \alpha\, \mathcal{C}_{\text{rec}}\right],}$$

with

$$\boxed{\mathcal{C}_{\text{rec}} = \frac{1}{2}\left[K^{AB}P_AP_B + K_{AB}\left(c^2\rho^2\nabla_{\text{rec}}R_A\nabla_{\text{rec}}R_B + m_A^2\rho R_AR_B\right)\right].}$$

The multiplier α is the common process multiplier, not an external time. In local common-rhythm proper gauge, $d\tau = \alpha d\lambda$, the momenta eliminate to

$$\boxed{S_{\text{rec}}^{(2)} = \frac{1}{2}\int d\tau\, d^3q\, K_{AB}\left[\partial_\tau R_A\partial_\tau R_B - c^2\rho^2\nabla_{\text{rec}}R_A\nabla_{\text{rec}}R_B - m_A^2\rho R_AR_B\right].}$$

Therefore

$$\boxed{C_t = 1, \qquad C_0 = \rho, \qquad C_x = \rho^2.}$$

The local dispersion relation is

$$\boxed{\omega^2 = c^2\rho^2k^2 + m_A^2\rho.}$$

Consequently, with $s = \rho^{1/2} = n^{1/4}$,

$$\boxed{\omega_0 \sim s, \qquad \ell_{\text{coord}} \sim s, \qquad c_{\text{coord}} = cs^2.}$$

This is the stable-record completion underlying the reciprocal clock/rod response. The order parameter alone does not yield the result; the endpoint-counting record completion is the additional calculable structure.

Result 11 (Endpoint-counting exponent selection). *The powers in the coherent completion have a relational node-link origin. A local gap term has one active endpoint; hence its leading coherent amplitude weight is*

$$\boxed{L_a \propto \rho_a.}$$

A spatial link has two active endpoints and must vanish if either endpoint is inactive. Endpoint symmetry and minimal coherent order then give

$$\boxed{J_{ab} \propto \rho_a\rho_b.}$$

In a smooth patch, $\rho_a, \rho_b \to \rho$, so the stable record sector has

$$\boxed{\textit{gap} \sim \rho, \qquad \textit{link stiffness} \sim \rho^2.}$$

This selects $C_t : C_0 : C_x = 1 : \rho : \rho^2$, and therefore the square law $c_{\text{coord}}/c = s^2$ with $s = \rho^{1/2}$.

Open Theorem 37 (Candidate stable-phase spectrum audit)**.** *For the candidate*

$$\boxed{\Phi_{\mathcal{M}} = \sqrt{n}\, e^{i\theta}\chi,}$$

write

$$\boxed{n = n_0 + \eta, \qquad \theta = \nu_0 \tau + \vartheta, \qquad \chi = \chi_0 + \xi.}$$

The quadratic spectrum should split schematically as

$$\boxed{\delta^2 S_{\mathcal{M}} = S^{(2)}_{\eta\eta} + S^{(2)}_{\vartheta\vartheta} + S^{(2)}_{\xi\xi} + S^{(2)}_{\text{mix}}.}$$

The desired Root A pattern is:

sector	*required low-energy status*
η	*gapped, constrained, near-zone or suppressed*
ϑ	*common rhythm/cone variable, not by itself the graviton*
ξ	*internal/confined/gauge/gapped/suppressed or shared-cone shear/matter*
S_{mix}	*renormalization only; no extra cone*

The phase branch may have

$$\boxed{\omega^2 = c^2_{\text{rec}} k^2 + O(k^4), \qquad G^{\mu\nu} k_\mu k_\nu = 0,}$$

and can serve as the candidate common record cone. But this is only Root A input. It does not by itself prove gravitational dynamics:

$$\boxed{\textit{one cone} \neq \textit{two tensor gravitons.}}$$

Identifying the phase/acoustic branch with the graviton would reduce the framework to scalar acoustic gravity, which is excluded by the two-tensor-mode requirement. The local-GR route is

$$\boxed{\textit{one common record cone} \xrightarrow{\textit{Root B}} \textit{ADM constraints} \xrightarrow{\textit{two-derivative fixed point}} S_{\text{EH}} \Rightarrow \textit{two tensor gravitational modes.}}$$

This candidate passes the first spectrum check only if the amplitude mode is not a long-range scalar gravity mode, internal modes do not generate extra cones, the acoustic cone is universal for record matter, and the tensor sector is derived by Root B rather than assumed.

Open Theorem 38 (Composite record metric and spin-2 emergence gate)**.** *The record metric must be a composite response of the base medium:*

$$\boxed{g^{\mu\nu}_{\text{rec}} = \mathcal{G}^{\mu\nu}[\Phi_{\mathcal{M}}], \qquad g^{\mu\nu}_{\text{rec}} = \bar{g}^{\mu\nu} + \delta g^{\mu\nu}_{\text{rec}}.}$$

The phase/acoustic branch may define the common cone, but the gravitational field is the low-energy dynamics of the composite record geometry, not simply the scalar phase ϑ:

$$\boxed{\delta g^{\mu\nu}_{\text{rec}} = \frac{\delta \mathcal{G}^{\mu\nu}}{\delta \Phi_{\mathcal{M}}} \delta \Phi_{\mathcal{M}} + \cdots .}$$

In spatial language,

$$\boxed{\delta h_{ij} = h^{TT}_{ij} + \partial_i V_j + \partial_j V_i + \left(\partial_i \partial_j - \frac{1}{3}\delta_{ij}\nabla^2\right) B + \frac{1}{3}\delta_{ij} H.}$$

Root B must make the scalar and vector pieces constraint, gauge or near-zone, leaving only

$$\boxed{h_{ij}^{TT}}$$

as propagating vacuum gravitational data:

$$\boxed{\#(\text{propagating vacuum gravitational DOF}) = 2.}$$

Possible microscopic sources of this tensor/shear sector are anisotropic stress/strain of χ, collective deformation of the record propagation kernel $G^{\mu\nu}$, topological/oscillon-network elasticity, or emergent embedding geometry after quotienting record histories. None may create a second long-range matter cone or private spacetime. Thus:

$$\boxed{\textit{acoustic cone} = \textit{kinematics}, \qquad \textit{ADM closure} = \textit{gravitational dynamics.}}$$

If the candidate supplies only a scalar acoustic mode, it fails the spin-2 gate.

Gate 4 (Tensor-closure reduction)**.** *The spin-2 issue is not closed in RMG-1. It is reduced to a finite calculation on the candidate record action. The candidate may pass only if the record-visible χ-strain has a transverse-traceless projection and its quadratic kinetic matrix has exactly two positive, luminal, common-cone radiative modes. Equivalently:*

$$\boxed{\textit{candidate tensor closure} \Longleftrightarrow \operatorname{rank} K_{\chi-\mathrm{strain}} = 2 \ \textit{on the physical record sector.}}$$

Rank zero means no tensor gravity; rank greater than two means extra local shear/scalar/vector radiation; a wrong sign, mass term, or different cone also fails. The detailed proof attempt and projector algebra belong in the work dossier, not in the core file, until the kinetic-rank test is actually derived from $S_{\mathcal{M}}$ or $S_{\mathrm{eff}}^{\mathrm{rec}}$.

This theorem remains open in RMG-1.

13 Allowed Deviation Windows

Additional sectors may be introduced only if they do not modify the local GR core.

Allowed windows:

MOND-like low-acceleration response,

cosmological background state of the medium,

matter-sector emergence,

high-energy corrections.

Rule:

$$\boxed{\text{deviations must not violate the local universal metric, } \gamma = \beta = 1, \text{ or } c_{\mathrm{GW}} = c.}$$

They also must not introduce an independent local $u^\mu_{\mathcal{M}}$ into the tested core.

Gate 5 (Flow/vorticity split)**.** *Medium flow and vorticity are allowed only in two protected roles. In the tested local regime they must be hidden inside ADM constraint variables or near-zone fields:*

$$\boxed{\mathcal{A}_i,\ u \quad \textit{constraint/near-zone only, not independent vacuum radiation.}}$$

In large-scale or low-acceleration regimes they may re-enter as an effective medium correction:

$$\boxed{\mathbf{a} = \mathbf{a}_{\mathrm{GR}} + \mathbf{a}_{\mathrm{med}}.}$$

The correction must satisfy:

$$\boxed{\mathbf{a}_{\mathrm{med}} \to 0 \quad \textit{in local tested high-acceleration regimes,}}$$

while it may become nonzero in galactic low-acceleration or cosmological background regimes.

Gate 6 (Local flow decoupling)**.** *In the tested local core, medium flow and vorticity may enter only as constraint, shift, near-zone or lower-order transport data. They must not modify the physical record-sector principal tensor:*

$$\boxed{\Delta P_A^{\mathrm{local}}(k) \not\supset \alpha_A v^\mu v^\nu k_\mu k_\nu \quad \textit{for external record-forming } A.}$$

Equivalently, the preferred-frame PPN target is:

$$\boxed{\alpha_1 = \alpha_2 = \alpha_3 = 0 \quad \textit{at the GR/local fixed point.}}$$

Large-scale medium corrections are allowed only through a reentry function $\mathcal{F}$ satisfying:

$$\boxed{\mathcal{F} \to 0 \quad \textit{in lab, Solar-System, binary-pulsar and local GW regimes.}}$$

This gate is open from $\mathcal{M}$; if it fails, flow/vorticity becomes an observable preferred frame.

Open Theorem 39 (MOND-safe reentry)**.** *Derive a scale or acceleration gate:*

$$\boxed{\mathbf{a}_{\mathrm{med}} = \mathcal{F}\left(\frac{|\nabla\Phi|}{a_0}, \frac{L}{L_{\mathrm{bg}}}, \omega_{\mathrm{flow}}\right)\mathbf{a}_{\mathrm{flow}},}$$

with:

$$\boxed{\mathcal{F} \to 0 \quad \textit{in local tested regimes,} \qquad \mathcal{F} \neq 0 \quad \textit{in galactic low-acceleration regimes.}}$$

The exact function $\mathcal{F}$ is not fixed in RMG-1.

14 Strong-Field Compact-Object Contract

RMG-1 does not yet select the strong-field compact-object branch. The branch must be derived from the base-medium equilibrium law and the record/internal projection.

The currently allowed branches are:

$$\boxed{\text{GR/Kerr-like exterior,} \qquad \text{horizonless exponential branch,} \qquad \text{hybrid branch,} \qquad \text{alternative branch.}}$$

Candidate strong-field signatures — horizonless static exteriors, exponential freezing, curvature-soft deep boundaries, shifted ISCOs and breathing-mode tests — are listed as targets, not predictions. They become RMG-1 predictions only if they are derived from $\mathcal{M}$.

Gate 7 (Strong-field compact projection)**.** *For compact objects, the internal sector may dominate bound medium structure and contribute to total energy:*

$$\boxed{E_{\text{total}} = E_{\mathcal{R}} + E_{\mathcal{I}} + E_{\text{bind}}.}$$

However, the external observable cone must be defined by the record projection:

$$\boxed{\Pi_{\mathcal{R}} P_{\text{full}}(k) \Pi_{\mathcal{R}} = (G^{\mu\nu} k_\mu k_\nu)\, K_{\mathcal{R}},}$$

while internal compact modes may enter the gravitational source only through conserved total charges such as energy, angular momentum and allowed gauge charges. They must not define an independent external spacetime cone.

Open Theorem 40 (Hybrid compact-object program)**.** *Derive, or rule out, the hybrid strong-field branch:*

$$\boxed{\textit{external record geometry is GR/Kerr-like to current precision}}$$

while

$$\boxed{\textit{deep internal base-medium structure may freeze, soften or store energy.}}$$

The required proof is:

$$\boxed{\Pi_{\mathcal{I}} \textit{ contributes to } H_{\text{phys}} \quad \textit{but} \quad \Pi_{\mathcal{I}} \textit{ does not create a separate external cone.}}$$

If this fails, RMG must either reduce to a fully GR-like strong-field branch or derive and observationally confront an alternative strong-field branch.

15 Statement Status Map

This section is the integrated WP0 audit map. It is not a proof. It records which statements are postulates, which are conditional bridges, which are open theorems, and which gates block the first local-gravity MVP.

The sortable audit status vocabulary is restricted to

$$\boxed{\text{postulate, conditional result, derived result, target, open theorem, fail-risk.}}$$

Free-form context belongs in notes, not in the status label. Priority is used only for proof management:

$$\boxed{P0 = \text{MVP blocker}, \quad P1 = \text{first local package}, \quad P2 = \text{important later item}, \quad P3 = \text{extension window.}}$$

Foundation and local bridge.

- **RMG-001** one base-medium $\mathcal{M}$: postulate.
- **RMG-002** stable low-energy phase: P0 target. It is introduced as a primitive requirement, but the audit treats its physical realization as a condition to verify before Root A can be calculated.
- **RMG-004** Operational Relational Covariance: P0 target. It must be derived from microscopic record equivalence or replaced by a more explicit protection theorem.

- **RMG-005** positive physical energy: P0 target. It is introduced as a primitive requirement, but physical propagating modes still require a ghost/gradient-stability audit.
- **RMG-007–RMG-014** operational and weak-field bridge: conditional or derived local results. The weak bridge gives $u(r) = G_{\mathrm{RMG}}M/(rc^2)$ and $\gamma = \beta = 1$, but their final status remains conditional on Roots A–C.

Root blockers.

- **RMG-015** Root A: derive one external record carrier/cone from the stable phase. Blocks factorization, no-birefringence, the composite metric, and the one-cone part of Root B and Root C.
- **RMG-016** Root B: derive record-embedding ADM closure, including $\Gamma_{\mathcal{R}}$, $\Omega_{\mathcal{R}}$, embedding generators, ADM brackets, hyperbolicity and no ghost/extra-mode leakage. Blocks the ADM/EH fixed point and the two-TT-mode statement.
- **RMG-017** Root C: derive the total-energy source projection for all stable record matter. Blocks equivalence-principle source universality, compact-binary source tests and the no-fifth-force gate.
- **RMG-018** non-circular proof order: proof discipline target. Do not assume ADM to prove Root A, do not assume Einstein equations to prove Root B, and do not assume the equivalence principle to prove Root C.

Local-core gate clusters.

- **RMG-A-Factorization: RMG-019–RMG-023.** Prove $Q^{\mu\nu}_{AB} = G^{\mu\nu}K_{AB}$, eliminate the failure matrix $B^{\mu\nu}_{AB}$, forbid low-energy birefringence, remove extra observable rank-two tensors, and decouple local flow/vorticity from the tested PPN core.
- **RMG-B-ADM: RMG-024–RMG-027.** Prove $\Delta_{\mathcal{M}} = 0$ or suppression, close the hypersurface-deformation algebra, derive $\lambda = 1/2$, and obtain the ADM/EH fixed point.
- **RMG-GW-Tensor: RMG-028–RMG-030.** Prove exactly two positive, luminal transverse-traceless tensor modes and a detector-visible TT/shear response; forbid scalar/vector, massive spin-2, negative-energy or extra shear radiation in the tested local sector.
- **RMG-C-Source: RMG-031–RMG-037.** Prove $q_A \propto E_A/c^2$, the far-tail energy monopole, local $\sigma_{\mathrm{grav}} \propto \rho_E/c^2$, no species-private source, internal and binding-energy effacement, no species-private u-coupling, and matter-coupled ADM closure.
- **RMG-Micro: RMG-038–RMG-041.** Specify the microscopic input package, test the candidate order parameter $\sqrt{n}e^{i\theta}\chi$, derive $S^{\mathrm{rec}}_{\mathrm{eff}}[\mathcal{M}_0]$, and construct the composite record metric $g^{\mu\nu}_{\mathrm{rec}} = \mathcal{G}^{\mu\nu}[\Phi_{\mathcal{M}}]$.

Coverage rows added by the audit. The following rows are now tracked explicitly, even where the body of RMG-1 has not yet promoted them to standalone theorems:

- **RMG-045** GW emission and radiation reaction: derive tensor quadrupole energy loss, radiation reaction and compact-binary waveform phase from the same record metric and total-energy source. The audit must show $G_{\mathrm{Newton}} = G_{\mathrm{radiation}} = G_{\mathrm{record}}$, $dE_{\mathrm{orbit}}/dt = -P_{TT}$, the angular-momentum flux for eccentric timing systems, the source-mass identification $m_A = E_A/c^2$, and no scalar/vector dipole or monopole channel. Propagation alone is not enough.

- **RMG-046** cosmological background reduction: specify the large-scale background state, expansion law and any dark-energy-like sector without using cosmology to cover a local-core failure.
- **RMG-047** finite strong-field tests: identify the ISCO, shadow, ringdown, strong-lensing and late-inspiral observables that must remain GR/Kerr-like or become explicit falsification targets.
- **RMG-048** quantum-extension boundary: state whether quantum claims are outside the MVP, imported as targets or derived later from the base-medium record action.
- **RMG-049** concrete Solar-System observation map: connect Newtonian limit, redshift, light bending, Shapiro delay, perihelion/ranging and frame-dragging tests to explicit RMG gates.

Observational sub-items.

- **OBS-001** Newtonian/orbital limit: checks RMG-012 and RMG-031–RMG-033.
- **OBS-002** gravitational redshift: checks RMG-009, RMG-010 and RMG-013. Failure means either species-dependent clock response or a physical split between the lapse N and compression e^{-u}.
- **OBS-003** light bending and Shapiro delay: checks RMG-008, RMG-013 and RMG-049.
- **OBS-004** perihelion, ranging and PPN β: checks RMG-013, RMG-027 and RMG-049.
- **OBS-005** frame dragging and preferred-frame null tests: check RMG-016, RMG-023, RMG-027 and RMG-049. This is a conditional-result row: it is consistent with frame-dragging data if Root B closes with $\Delta_{\mathcal{M}} = 0$ or a suppressed/gauge/constrained remnant.
- **OBS-006** binary-pulsar orbital decay: checks RMG-017, RMG-031, RMG-035 and RMG-045; no leading scalar dipole loss is allowed. The first quantitative emission MVP is a PSR B1913+16-style $\dot{P}_b$ comparison, with tensor damping and no private scalar/vector flux.
- **OBS-007** GW propagation: checks RMG-028 and RMG-030; requires $c_{\rm GW} = c$ and exactly two tensor polarizations; the multimessenger benchmark is $|c_{\rm GW}/c - 1| \lesssim 10^{-15}$.
- **OBS-008** compact-binary waveform phase: checks RMG-028–RMG-030 and RMG-045; propagation and emission must be consistent, with chirp mass built from $m_A = E_A/c^2$ and phase evolution from tensor energy balance.
- **OBS-009** strong-field finite tests: checks RMG-043, RMG-044 and RMG-047.
- **OBS-010** MOND transition: checks RMG-023 and RMG-042; any large-scale correction must not leak into the local core.
- **OBS-011** cosmological background: checks RMG-046.
- **OBS-012** high-energy corrections: any higher-derivative or high-energy operator must be below the relevant local bound and may not modify local PPN, GW speed or GW emission.
- **OBS-013** compact-object branch: finite-size, spin, tidal and strong-field internal structure may be tested only after the local core is protected; the branch may not create a private cone or scalar/vector loss channel.

- **OBS-014** quantum-extension boundary: the local-gravity MVP is classical. Quantum extension is future work and may not be used as an unsupported premise inside the classical proof.

For OBS-004 the chain must be explicit:

$$\boxed{\lambda = \frac{1}{2} \Rightarrow \text{first-class ADM closure} \Rightarrow \text{EH two-derivative local fixed point} \Rightarrow \text{nonlinear weak-field expansion} \Rightarrow \beta =}$$

If any arrow is imported rather than derived or audited, the corresponding observation row remains conditional.

Observation matching discipline. Every observation row must name its Status Map IDs, proof gate and failure condition. The word "partial" in the dependency map means background compatibility, not a completed proof: for example Root A is partial in the Newtonian row because one operational metric is needed before a far-tail source can be read geometrically, while Root C is only partial in the Shapiro/light-bending row because source normalization enters the weak-field potential but the bending coefficient is mainly a Root A/B gate. Cosmology partial rows mean Root A/B/C must be compatible with an FLRW-like background, but cosmological dynamics remain an extension window.

Working observational anchors. The following numerical values are working anchors for future calculation memos, not final data-source citations; each execution artifact must re-check the exact convention and source before using them:

- Cassini-style Shapiro: $\gamma - 1 = (2.1 \pm 2.3) \times 10^{-5}$.
- LLR/perihelion-style PPN: $\beta - 1 \sim (1.2 \pm 1.1) \times 10^{-4}$ as a working anchor.
- GP-B frame dragging: observed about 37.2 ± 7.2 mas/yr, with the GR target about 39.2 mas/yr.
- GW170817/GRB timing: $-3 \times 10^{-15} \lesssim (c_{\rm GW} - c)/c \lesssim 7 \times 10^{-16}$ as a working benchmark.
- PSR B1913+16-style timing: $\dot{P}_b \sim -2.4025 \times 10^{-12}\,$s/s, with tensor damping agreement at the sub-percent scale after standard corrections.
- MICROSCOPE-style weak-equivalence anchor: $\eta \sim (-1.5 \pm 2.3 \pm 1.5) \times 10^{-15}$.
- LLR Nordtvedt anchor: $\eta_N \sim (2 \pm 5) \times 10^{-4}$.
- PSR J0337+1715-style strong-field free-fall anchor: $|\Delta a/a| < 2.6 \times 10^{-6}$ as a conservative working bound.

MVP observation split. The first observation MVP is local and narrow:

- Newtonian $1/r$ source with $M = E_{\rm total}/c^2$;
- redshift/lapse universality;
- light bending and Shapiro with one cone and $\gamma = 1$;
- perihelion/ranging with $\beta = 1$ through EH local closure;

- no preferred-frame local flow;
- GW propagation with $c_{\rm GW} = c$ and two TT modes;
- binary-pulsar scalar-dipole exclusion from Root C;
- binary-pulsar tensor $\dot{P}_b$ calculation from the GW-emission gate.

Compact-binary high-PN waveform templates, merger/ringdown, strong-field branch selection, MOND and cosmology are not MVP blockers unless they leak into these local tests.

Falsification priority. After the local core is protected, RMG must either derive the GR-like local core from the base medium or produce controlled deviation targets. Priority rows are:

- **P1** PSR B1913+16-style timing: tensor $\dot{P}_b$ from Root B/C/GW emission without hidden scalar/vector loss.
- **P2** high-precision 2PN Shapiro/delay: any calculable suppressed post-core correction after Roots A/B/C close.
- **P3** composition tests: $\eta = 0$ from Root C source universality at the 10^{-15} scale.
- **P4** GW170817 speed: one record cone gives $c_{\rm GW} = c$; this is not distinctive from GR if the local core passes, but it is a hard failure gate.
- **P5** neutron-star tidal/compact branch: allowed finite-size response must be predicted explicitly and must not hide a local-core failure.

Medium-to-target verification interface. The microscopic input package must be tested against the low-energy target by explicit checks: compute $\mathrm{Spec}(\mathcal{L}_{\mathcal{M}_0})$ and count the record cones; derive $\Pi_{\mathcal{R}}$ and $\Pi_{\mathcal{I}}$ and show that internal branches do not become private spacetimes; identify presymplectic null directions and first-class generators; compute the Schur-complement correction from internal integration and verify that it does not alter the local principal tensor or ADM bracket; compute the tensor kinetic rank, sign, mass and cone; and vary the lapse/metric to check that rest, kinetic, binding, internal/confined and field energy enter the same source. In particular, the chain

$$\boxed{\mathcal{D}_{\mathcal{M}} \to S_{\rm eff}^{\rm rec} \to S_{\mathcal{R}}[g_{\rm rec}, \Psi_{\rm rec}]}$$

must be explicit enough to run the Root B presymplectic, bracket and hyperbolicity audits.

MVP linearization scope. The first local-gravity MVP starts with a Minkowski local core for the principal-symbol, mode-count and hyperbolicity tests; a weak static local background is then used for the PPN/nonlinear-coefficient check; FLRW belongs to the cosmology window. The first pass requires the quadratic action in metric perturbations, the linearized constraints and their relevant quadratic-order closure, leading weak-field matter coupling, exactly two TT propagating modes, and a leading tensor-emission audit. Gauge fixing, such as harmonic gauge, may be chosen only after constraints are derived and may not hide kinetic lapse/shift or scalar/vector modes. Higher post-Newtonian terms and full nonlinear closure remain later targets.

RMG-distinctive observables. The protected local core is deliberately GR-like. RMG becomes observationally distinctive only if controlled windows produce falsifiable deviations after the local core is protected:

- high-precision delay tests may search for suppressed post-GR corrections only after Root A/B/C close;
- compact-binary waveform residuals, ISCO shifts, shadows and ringdown features belong to the compact-object branch;
- galaxy rotation belongs to MOND-like large-scale reentry and must not leak into Solar-System, binary-pulsar or local GW tests;
- cosmological acceleration belongs to the background-state closure, where Λ must be derived, regulated or replaced by a medium background law;
- preferred-frame null tests remain local-core failure tests, not deviation-window successes.

The immediate P0 blockers are therefore the stable phase, relational covariance/protection, positive-energy physical modes, the local effective hierarchy, Roots A–C, factorization, $\Delta_{\mathcal{M}} = 0$, ADM/EH closure, total-energy source universality, and the microscopic input package.

16 MVP Local-Core Proof Tasks

This section converts the specification-stage memos into an execution plan. It adds no new physical claim. Its purpose is to say which calculations must be performed, in which order, what each calculation must output, and what counts as failure. The central execution question is:

Which calculations turn the local gravity mechanism from a specification into a proof attempt?

The first MVP is deliberately narrower than the full RMG program. It targets only the local gravity core: one external record cone; linearized ADM/Einstein–Hilbert closure; universal total-energy sourcing; no leading scalar/vector dipole radiation; leading tensor radiation-reaction consistency; and observation matching only at the level supported by the completed gates. MOND, cosmology, strong-field branches and quantum extensions may not repair a failure of this local core.

Non-circular execution rules. The following rules are stronger than style requirements; if a task violates one of them, it is not counted as a proof.

- Root A may not assume ADM closure. One record cone must be obtained from the medium spectrum and operational record observables.
- Root B may not assume Einstein equations. ADM closure and the EH fixed point must come from the projected record action.
- Root C may not assume the equivalence principle. Universal free fall and universal source strength are output gates.
- GW emission may not assume the quadrupole formula. Tensor flux and radiation reaction must be derived from the radiative channel.
- Observations may not outrun proof gates. An observation row is not marked passed unless every linked local gate is derived or explicitly kept as a conditional target.

Dependency chain. The execution chain is:

$$\boxed{\mathcal{D}_{\mathcal{M}} \to \text{Root A} \to \text{Root B} \to \text{Root C} \to \text{GW emission} \to \text{observation matching.}}$$

If Root A fails, the local core has no unique record spacetime. If Root B fails, one cone is not enough to obtain the ADM/EH local dynamics. If Root C fails, the metric dynamics may be right but matter sourcing and free fall are wrong. If GW emission fails, propagation may be correct while binary-pulsar and inspiral energy loss fail. If observation matching fails, the completed proof gates are insufficient or a local anchor has been missed.

MVP task blocks. The execution program is divided into six artifacts. The effort estimates are planning estimates, not proof claims.

- **MVP-0:** **`MVP_Input_Freeze.md`.** Freeze the exact symbols, assumptions, conditional inputs, normalizations and anchors allowed in the MVP. Planning scale: about one week.
- **MVP-A:** **`Root_A_MVP_Spectrum_Calculation.md`.** Execute the low-energy spectrum audit on the locked toy or candidate $\mathcal{D}_{\mathcal{M}}$. Expected output: one gapless external record cone. Planning scale: about four to eight weeks.
- **MVP-B:** **`Root_B_Linearized_ADM_MVP.md`.** Execute the linearized ADM closure audit on the projected record action $S_{\mathcal{R}}$. Expected output: first-class linearized constraints, two TT modes and record-cone principal symbol. Planning scale: about eight to twelve weeks.
- **MVP-C:** **`Root_C_Source_MVP_Calculation.md`.** Execute the lapse/source and matter-family audit. Expected output: the source equals total physical energy density and has no species-private tail. Planning scale: about six to ten weeks.
- **MVP-E:** **`PSR_B1913_Emission_MVP.md`.** Derive leading tensor flux and radiation reaction. The strict input is the completed Root B tensor channel plus the completed Root C source; a conditional draft is allowed only under the fast-track rule below. Planning scale: about six to ten weeks.
- **MVP-O:** **`MVP_Observation_Checklist.md`.** Compare only supported gates against local anchors and keep every row no stronger than its proof-gate status. Planning scale: about two weeks.

MVP-0 input freeze. The first artifact prevents later calculations from importing hidden assumptions. It must lock: selected medium variables and background; the coarse-graining scale; dimensional normalizations for $G_{\rm RMG}$, c, ℓ_* and Λ, or their defining relations; the record/internal split convention; the allowed internal gauge group and observables; the target low-energy action form; the primary local observation anchor; and the exact status vocabulary.

The freeze tasks are:

- list every primitive input used by the MVP;
- list every conditional bridge and forbid presenting it later as derived;
- list every deferred branch and forbid using it to repair a local-core failure;
- lock notation for $\mathcal{D}_{\mathcal{M}}$, $S_{\mathcal{R}}$, $g_{\rm rec}$, N, u, ρ_E and TT fields;

- lock the toy or candidate $\mathcal{D}_{\mathcal{M}}$: fields, tensor types, potential terms, interaction terms, background $\mathcal{M}_0$ and scale hierarchy;
- lock the dimensional normalization of G_{RMG}, c, ℓ_* and Λ;
- lock PSR B1913+16 as the primary binary-pulsar emission anchor; other pulsars are extension targets, not MVP replacements.

Frozen MVP toy input. The first Root A execution pass uses a conservative toy input, not a final microscopic ontology:

$$\boxed{\Phi_{\mathcal{M}}^{\mathrm{MVP}} = (\phi, \theta, Q_{ij}), \qquad \phi = \sqrt{n}, \qquad Q_{ij} = Q_{ji}, \qquad Q^i{}_i = 0.}$$

The selected phase is

$$\boxed{\mathcal{M}_0: \quad \phi = v, \qquad \theta = \nu_0 \tau, \qquad Q_{ij} = 0.}$$

Background isotropy means $\phi = v$ is spatially constant, $\partial_i \theta = 0$, and $Q_{ij} = 0$; hence spatial rotations act trivially on $\mathcal{M}_0$. The coarse-graining scale ℓ_* is a primitive input for the MVP, not a fit parameter to repair local failures.

The first toy energy is frozen as

$$E_{\mathrm{toy}} = \int d^3X \left[\frac{k_\phi}{2} (\nabla \phi)^2 + \frac{k_\theta}{2} \phi^2 (\nabla \theta)^2 + \frac{\lambda_\phi}{4} (\phi^2 - v^2)^2 \right.$$
$$\left. + \frac{k_Q}{2} (\nabla Q_{ij})^2 + \frac{\mu_Q^2}{2} Q_{ij} Q_{ij} + \frac{g_Q}{2} \phi^2 Q_{ij} Q_{ij} \right],$$

with $E_{\mathrm{mix}} = 0$ in the first spectrum calculation. Cubic and quartic Q-invariants, such as $\mathrm{Tr}(Q^3)$ and $(\mathrm{Tr}\, Q^2)^2$, are deferred. Any later nonzero mixing term, for example $\eta (\nabla \theta)^2$, $\eta Q_{ij} Q_{ij}$, or $Q_{ij} \partial_i \theta \, \partial_j \theta$, requires a revised cone-splitting and suppression audit.

Numerical first-pass regime. The first Root A calculation uses dimensionless units

$$\boxed{\ell_* = 1}$$

and the locked parameter regime

$$\boxed{v \ell_* = 1, \quad \lambda_\phi = \frac{1}{2}, \quad k_\phi = k_\theta = k_Q = 1, \quad \mu_Q^2 \ell_*^2 = 4, \quad g_Q = 0.}$$

Therefore

$$\boxed{m_\eta^2 = 2 \lambda_\phi v^2 = 1\, \ell_*^{-2}, \qquad m_Q^2 = \mu_Q^2 + g_Q v^2 = 4\, \ell_*^{-2}.}$$

The amplitude branch η and the Q_{ij} branch are gapped in this conservative first pass, while the phase fluctuation ϑ is the candidate gapless record branch. These values make the first Hessian calculation unambiguous; they are not asserted as measured constants.

Presymplectic working form. The toy action is frozen as

$$\boxed{S_{\mathrm{toy}} = \int d\tau \, d^3X \, \left[\pi_\phi \dot{\phi} + \pi_\theta \dot{\theta} + \pi_Q^{ij} \dot{Q}_{ij} - \alpha C_0 - \beta^a C_a - E_{\mathrm{toy}} \right].}$$

Both Q_{ij} and π_Q^{ij} are symmetric traceless 3×3 tensors with five independent components. Before imposing constraints, the toy phase space has $2+2+10=14$ components. The working constraint forms are

$$\boxed{C_a = \pi_\phi \partial_a \phi + \pi_\theta \partial_a \theta + \pi_Q^{ij} \partial_a Q_{ij},}$$

and

$$\boxed{C_0 = \pi_\phi \dot{\phi} + \pi_\theta \dot{\theta} + \pi_Q^{ij} \dot{Q}_{ij} - \mathcal{L}_{\rm toy}.}$$

These are reparametrization/relabeling constraint candidates, not a completed ADM derivation. Root B must still test whether the resulting constraint algebra is first-class.

Conservative Q-route firewall. The first MVP chooses the conservative route:

$$\boxed{Q_{ij} \text{ is gapped/internal in Root A.}}$$

This can support a one-cone Root A pass, but it cannot by itself prove two tensor gravitons. The alternative TT-candidate route would allow a controlled gapless q_{ij}^{TT} projection only after revising the input freeze and proving

$$\boxed{\operatorname{rank} K_{Q-\text{strain},TT} = 2, \quad \text{positive kinetic sign}, \quad c_{TT} = c_{\rm record},}$$

with no scalar/vector leakage and no second metric tensor mode. The two routes are mutually exclusive within a single MVP input freeze.

Root A start gate. Root A may start only from this package:

$$\boxed{\begin{array}{c} (\phi, \theta, Q_{ij}), \quad \mathcal{M}_0 = (v, \nu_0 \tau, 0), \quad E_{\rm toy}, \quad E_{\rm mix} = 0, \\ \ell_* = 1, \quad v = 1, \quad \lambda_\phi = \frac{1}{2}, \quad k_\phi = k_\theta = k_Q = 1, \quad \mu_Q^2 = 4, \quad g_Q = 0, \\ m_\eta^2 = 1\, \ell_*^{-2}, \quad m_Q^2 = 4\, \ell_*^{-2}. \end{array}}$$

The next calculation must output:

- the explicit quadratic Hessian;
- the spectrum classification;
- the record/internal projector candidate;
- the suppression catalog;
- the decision whether Root B may proceed, proceed conditionally or stop.

A clean failure is acceptable if it identifies which part of $\mathcal{D}_\mathcal{M}$ must be changed.

First Root A MVP spectrum result. With

$$\phi = 1 + \eta, \qquad \theta = \nu_0 \tau + \vartheta, \qquad Q_{ij} = q_{ij},$$

the frozen conservative toy gives the spatial quadratic Hessian

$$\boxed{\delta^2 E_{\rm toy} = \frac{1}{2}\eta[-\nabla^2 + 1]\eta + \frac{1}{2}\vartheta[-\nabla^2]\vartheta + \frac{1}{2} q_{ij}[-\nabla^2 + 4] q_{ij}.}$$

The amplitude expansion is
$$\frac{1}{8}(\phi^2-1)^2=\frac{1}{2}\eta^2+\frac{1}{2}\eta^3+\frac{1}{8}\eta^4,$$
so $m_\eta^2=1$. The Q-sector has $m_Q^2=\mu_Q^2+g_Qv^2=4$. Since $g_Q=0$, there is no ϕ-Q quadratic mixing; any $\eta q_{ij}q_{ij}$ term would be cubic and is outside the first Hessian audit.

In Fourier space,

$$\boxed{\lambda_\eta(k)=k^2+1,\qquad \lambda_\vartheta(k)=k^2,\qquad \lambda_Q(k)=k^2+4}$$

with five Q-components. The first projector candidate is

$$\boxed{\Pi_{\mathcal{R}}(\eta,\vartheta,q_{ij})=(0,\vartheta,0),\qquad \Pi_{\mathcal{I}}(\eta,\vartheta,q_{ij})=(\eta,0,q_{ij}),}$$

which satisfies
$$\Pi_{\mathcal{R}}+\Pi_{\mathcal{I}}=1,\qquad \Pi_{\mathcal{R}}\Pi_{\mathcal{I}}=0,\qquad \Pi_{\mathcal{R}}^2=\Pi_{\mathcal{R}},\qquad \Pi_{\mathcal{I}}^2=\Pi_{\mathcal{I}}.$$
Thus the conservative toy has

$$\boxed{B_{\mathrm{rec}}^{\mathrm{gapless}}=\{\vartheta\},\qquad N_{\mathrm{gapless}}^{\mathrm{spatial}}(B_{\mathrm{rec}})=1.}$$

This is an A4 pass: one gapless branch in the spatial Hessian. It is not yet an A5 pass, because a full cone count requires the temporal kinetic/principal symbol.

Temporal completion target. The recommended next completion is the second-order kinetic test
$$\delta^2L_{\mathrm{kin}}=\frac{1}{2}Z_\eta\dot{\eta}^2+\frac{1}{2}Z_\theta\dot{\vartheta}^2+\frac{1}{2}Z_Q\dot{q}_{ij}\dot{q}_{ij}.$$
The canonical first test is
$$\boxed{Z_\eta=Z_\theta=Z_Q=1,}$$
which would give

$$\boxed{\omega_\eta^2=k^2+1,\qquad \omega_\vartheta^2=k^2,\qquad \omega_Q^2=k^2+4,\qquad c_\eta^2=c_{\mathrm{record}}^2=c_Q^2=1.}$$

If Z_η, Z_θ and Z_Q differ in the leading local regime, the sectors have different high-frequency cones; that is a Root A one-cone failure unless the non-record branch is demonstrably absent from $\mathcal{O}_{\mathrm{rec}}$ and cannot form external clocks, signals or detector records. At this stage the temporal completion is a conditional bridge, not a derived result.

K1 status is therefore:

$$\boxed{\text{specified for the MVP: yes,}\qquad \text{derived from } \mathcal{D}_{\mathcal{M}}:\ \text{no.}}$$

To remove this conditionality, a later derivation must construct the medium Lagrangian density from $\mathcal{D}_{\mathcal{M}}$ and show that the leading kinetic block is sector-universal:

$$\boxed{Z_\eta=Z_\theta=Z_Q.}$$

If the medium gives sector-dependent Z-coefficients in the leading local regime, the one-cone K1 result fails.

With K1 the branch operators are

$$\boxed{P_\eta=-\omega^2+k^2+1,\qquad P_\vartheta=-\omega^2+k^2,\qquad P_Q=-\omega^2+k^2+4.}$$

The principal symbol is meant in the PDE sense: only the highest-derivative terms define the principal cone. The mass terms 1 and 4 are lower-order terms; they change low-energy dispersion but not the causal principal cone. Thus every branch has

$$\boxed{P_{\text{principal}}(\omega, k) = -\omega^2 + k^2.}$$

For gapped branches the phase velocity may exceed one at low momentum,

$$v_{\text{phase}} = \frac{\omega}{k} = \sqrt{1 + \frac{m^2}{k^2}} > 1,$$

but the information-carrying group velocity is

$$v_{\text{group}} = \frac{d\omega}{dk} = \frac{k}{\sqrt{k^2 + m^2}} \le 1.$$

So the massive-mode phase velocity is not a superluminal signal. The normalization $c_{\text{record}} = 1$ is the canonical K1 unit choice; matching it to the measured speed of light belongs to the later static/radiative normalization bridge.

The K1 quadratic Hamiltonian is positive semidefinite:

$$\begin{aligned}
H_\eta &= \frac{1}{2}\pi_\eta^2 + \frac{1}{2}(\nabla\eta)^2 + \frac{1}{2}\eta^2,\\
H_\vartheta &= \frac{1}{2}\pi_\vartheta^2 + \frac{1}{2}(\nabla\vartheta)^2,\\
H_Q &= \frac{1}{2}\pi_Q^{ij}\pi_Q^{ij} + \frac{1}{2}(\nabla q_{ij})(\nabla q_{ij}) + 2q_{ij}q_{ij}.
\end{aligned}$$

The only flat direction is the constant phase mode of ϑ, the expected shift/phase zero mode.

Root A MVP verdict. The conservative toy passes the static/spatial Hessian and gap audit: η is gapped with $m_\eta = 1\,\ell_*^{-1}$, q_{ij} is gapped with $m_Q = 2\,\ell_*^{-1}$, and ϑ is the only spatial gapless record candidate. If ℓ_* is microscopic, these gapped branches decouple from lab, Solar-System and binary propagation; the physical interpretation of that decoupling remains conditional on the final scale assignment.

The single-branch spatial factorization is trivial:

$$\Pi_{\mathcal{R}} L_{\text{spatial}}(k) \Pi_{\mathcal{R}} = k^2, \qquad K_{\mathcal{R}} = 1.$$

With K1 the full projected operator is

$$\boxed{\Pi_{\mathcal{R}} P(\omega, k) \Pi_{\mathcal{R}} = -\omega^2 + k^2,}$$

equivalently

$$P(\omega, k) = \mathrm{diag}(-\omega^2 + k^2 + 1,\ -\omega^2 + k^2,\ -\omega^2 + k^2 + 4),$$

and

$$\Pi_{\mathcal{R}} P \Pi_{\mathcal{R}} = \mathrm{diag}(0,\ -\omega^2 + k^2,\ 0).$$

This does not prove all-species factorization, because photons, atoms, rods, clocks and detector matter are not included as separate matter families in the toy. That part must be audited in Root C when the matter Hamiltonian and record observables are included, and in Root B when the record action is constructed.

Operational consequences of the K1 one-cone result are limited but concrete:

- record signals built from the ϑ branch propagate with $c = 1$ in canonical units;
- η and q_{ij} are gapped, with ranges of order $1/m_\eta$ and $1/m_Q$;
- at distances much larger than those ranges, only the gapless ϑ branch remains in the conservative toy;
- the one-branch record sector has no low-energy birefringence;
- the isotropic background $Q_{ij} = 0$ introduces no local preferred spatial direction.

Open operational tests remain: matter species must be shown to couple through the same record structure; light bending and clock comparison require matter and clock probes; nonlinear weak-field coefficients require Root B; and GW speed/tensor modes require the tensor route, which the conservative Q branch does not provide.

Operationally, $\vartheta = \theta - \nu_0 \tau$ is a Goldstone-like clock/phase fluctuation of the record medium, not yet a metric perturbation by itself. A gapless phase mode would be a fifth-force risk if it coupled directly to matter as a private scalar. The MVP must later show that ϑ enters matter only through the common record metric/clock structure, or that a shift/record symmetry forbids species-private matter coupling. The most likely conservative-MVP route is the shift/record symmetry

$$\boxed{\theta \to \theta + \text{constant.}}$$

Root C must test that the matter Hamiltonian may depend on derivatives or record-metric combinations of θ, but not on a species-private explicit θ-charge. Such a private explicit phase coupling would fail the fifth-force gate.

The C_0, C_a constraint caveat remains open. Root B must compute

$$\boxed{\{C_0, C_0\}, \qquad \{C_0, C_a\}, \qquad \{C_a, C_b\},}$$

verify closure on the constraint surface, verify that ϑ is not removed as the record branch, verify that no extra scalar/vector propagating mode is introduced, and verify that the record principal symbol is preserved.

Root B may proceed only conditionally. It may import that ϑ is the record candidate and that η, q_{ij} are gapped in the conservative pass, but it may not claim two tensor modes from the conservative Q-sector, full ADM/EH closure, all-species one-cone universality, or a derived g_{rec} with both temporal and spatial components. The recommended next artifact is the conditional Root B record-action bridge; deriving K1 from $\mathcal{D}_\mathcal{M}$ first is stricter, but may not rescue the local core if the Root B bridge fails even under K1.

Root B conditional record-action bridge. The Root B bridge executes the MVP-B B0 task: construct a conditional record-action object precise enough for the Root B ADM audit without treating ADM closure, the Einstein–Hilbert action, two TT modes or all-species universality as already proven. The next artifact after this bridge is the linearized Root B audit, which executes B1–B7.

Root B may import from Root A/K1 only the following facts, with the following status:

$$\boxed{\begin{gathered}\vartheta:\ \text{conditional record branch,} \qquad P_\mathcal{R}(\omega, k) = -\omega^2 + k^2, \qquad c_{\text{record}} = 1,\\ \eta,\ q_{ij}:\ \text{gapped/internal in the conservative route,}\\ K1:\ \text{specified bridge, not yet derived from } \mathcal{D}_\mathcal{M},\\ \text{all-species universality and } C_0, C_a \text{ closure: open.}\end{gathered}}$$

It may not import ADM closure, an EH action, two TT gravitons or a universal matter metric.

The bridge object is

$$\boxed{\begin{aligned} S_{\mathcal{R}}^{\text{bridge}} &= S_{K1}[\vartheta;\bar{g}] + S_{\text{gap}}[\eta, q_{ij};\bar{g}] + S_{\text{geom}}^{\text{cond}}[g_{\text{rec}}] \\ &\quad + S_{\text{coupling}}^{\text{cond}}[g_{\text{rec}}, \Psi_{\text{rec}}] + S_{\text{rem}}. \end{aligned}}$$

Here S_{K1} carries the one-cone phase result; S_{gap} carries the gapped internal branches; $S_{\text{geom}}^{\text{cond}}$ is the conditional metric-dynamics sector to be audited by Root B; $S_{\text{coupling}}^{\text{cond}}$ is only a placeholder for matter/record probes and must not assume Root C; and S_{rem} must be classified as gauge, constrained, boundary, gapped, high-energy suppressed, large-scale-only or fail-risk.

The K1 cone reconstructs only the local background cone metric:

$$\boxed{-\omega^2 + k^2 = 0 \quad \Longleftrightarrow \quad \bar{g}^{\mu\nu} = \eta^{\mu\nu} = \operatorname{diag}(-1,1,1,1)}$$

in canonical K1 units. Thus the bridge takes $\bar{g} = \eta$ as the natural local background choice. A constant conformal rescaling is harmless only if the measured normalization of c_{record} is kept fixed. A nontrivial conformal factor, anisotropic background or preferred-frame background must be justified separately and is a possible Root A/Root B failure mode.

To run the Root B audit, the bridge introduces conditional record-geometry variables

$$\boxed{N = 1 + n, \qquad N^i = n^i, \qquad h_{ij} = \delta_{ij} + \gamma_{ij}, \qquad \pi^{ij} \text{ conjugate to } h_{ij}.}$$

Their tensor types are the standard ADM types: N is a scalar lapse, N^i a spatial vector shift, h_{ij} a symmetric spatial two-tensor and π^{ij} the conjugate symmetric tensor density. The unresolved issue is not the ADM tensor type. The unresolved issue is the medium map: whether ϑ maps to N, to h_{ij}, to both through g_{rec}, or to neither as a directly geometric variable.

Metric reconstruction therefore has three possible routes:

1. build $g_{\text{rec}}^{\mu\nu}$ from Goldstone-gradient data such as $\partial_\mu\theta\,\partial_\nu\theta$;
2. build g_{rec} from the K1 dispersion/principal-symbol structure;
3. introduce g_{rec} as a conditional bridge variable and derive it later.

The first MVP uses the third route. The first two routes are deferred to a later metric-from-medium derivation.

Root B may use the two-derivative Hamiltonian form only as an audit filter, not as an assumed result:

$$\boxed{\mathcal{H}_B = \frac{1}{\sqrt{h}}\left(\pi^{ij}\pi_{ij} - \lambda\pi^2\right) - \sqrt{h}(\alpha R - 2\Lambda) + \Delta_{\mathcal{M}}.}$$

The tested target is

$$\boxed{\lambda = \frac{1}{2}, \qquad \alpha = 1, \qquad \Delta_{\mathcal{M}} = 0}$$

or $\Delta_{\mathcal{M}}$ must be a constraint, gauge, boundary, internal, gapped, high-energy suppressed or large-scale-only term with a stated control. For this MVP, the working controls are:

- high-energy suppression requires a stated Λ_{UV}, with $E_{\text{test}}/\Lambda_{\text{UV}} < 10^{-5}$ as the Solar-System PPN-scale working threshold unless a sharper bound applies;

- a gapped branch must satisfy $m > 1/r_{\rm test}$ for the relevant experiment;
- a large-scale-only term is allowed only for $k < H_0/c$ and may not repair a local Solar-System, laboratory, binary-pulsar or GW failure;
- a boundary term must live at a declared boundary and not change the local bracket;
- a gauge or constraint term must be removed by a stated generator or first-class constraint check.

The re-embedding audit acts on the record fields by

$$\boxed{\delta_\xi \Phi_{\mathcal{R}} = \mathcal{L}_\xi \Phi_{\mathcal{R}}}, \qquad \boxed{\Omega_{\mathcal{R}}(\delta\Phi, \delta_\xi\Phi) \approx 0}$$

or equivalently by the generators $H[N] + D[N^i]$. The field types must be declared before this calculation: ϑ is a scalar with $\mathcal{L}_\xi\vartheta = \xi^\mu\partial_\mu\vartheta$; $g_{\mathrm{rec}\,\mu\nu}$ is a symmetric tensor with $\mathcal{L}_\xi g_{\mu\nu} = \nabla_\mu\xi_\nu + \nabla_\nu\xi_\mu$; N, N^i, h_{ij} inherit their transformations from the spacetime metric split; and retained η, q_{ij} branches must be treated as internal/gapped unless a later input freeze revises their role.

The tensor-sector risk is the main conservative-route risk. Root A/K1 gives one scalar-like phase record branch, a gapped Q_{ij} branch and no derived TT graviton from conservative Q. Root B nevertheless requires

$$\boxed{\text{physical DOF} = 2, \qquad \text{rank}\, K_{TT} = 2, \qquad \text{positive tensor kinetic sign}, \qquad P_{TT} = P_{\mathcal{R}},}$$

with no scalar/vector radiation and no second metric. The allowed sequence is:

1. first try a conditional TT construction from γ_{ij}^{TT} as a record-geometry variable, carrying the K1 cone;
2. if the hyperbolicity or rank audit fails on γ_{ij}^{TT}, revise the input freeze toward a gapless TT-candidate Q-route;
3. if only the scalar one-cone sector remains, count this as a blocking failure of the local tensor-gravity MVP.

The forbidden move is to rename the gapped conservative Q-sector as two TT gravitons.

The fifth-force firewall is also only conditional at this bridge stage. The toy action contains derivatives of θ but no explicit θ, so

$$\boxed{\theta \to \theta + \text{constant}}$$

is a shift symmetry of the toy action. The background $\theta = \nu_0\tau$ breaks this symmetry in the selected phase, and ϑ is the Goldstone-like fluctuation. This may protect against a private scalar force, but only if the later matter Hamiltonian depends on derivatives or common record-metric combinations of θ, not on species-private explicit θ-charge. That test belongs to Root C and the record-action bridge; it is not assumed here.

Root B may proceed to the linearized ADM audit only with the following labels: K1 is specified rather than medium-derived; the metric/tensor sector is conditional; two TT modes are an audit target rather than an input; matter universality is not assumed; and every anomaly or $\Delta_{\mathcal{M}}$ must carry a scale or structural reason. Root B must stop if the bridge requires kinetic lapse, a physical shift, a private preferred-frame tensor, scalar/vector radiation in the local vacuum sector, a second metric, or the identification of gapped Q with TT gravity. If only some Root B conditions fail, each failed condition must be documented separately and the revision must be assigned either to the bridge, to the input freeze, or to the program status if only a scalar one-cone conditional theory remains.

The accumulated conditional debt is:

Bridge	**Source**	**Risk**	**Resolution path**
K1 specification	Root A temporal completion	$Z_\eta = Z_\theta = Z_Q = 1$ not derived	derive K1 from the medium
$\bar{g} = \eta$	this bridge	Lorentz invariance not derived from $\mathcal{D}_\mathcal{M}$	medium symmetry analysis
N, N^i, h_{ij}	this bridge	metric origin from medium not derived	metric-from-medium derivation
γ_{ij}^{TT} existence	this bridge	conservative Q is gapped, so no TT source is derived	CB-5 revision or record-geometry route
C_0, C_a closure	Root A temporal completion	constraint algebra not audited	Root B bracket calculation
matter universality	this bridge	matter species not modeled	Root C matter-family audit

No row in this ledger may be silently counted as derived. Each row must be resolved, or explicitly accepted as a final conditional bridge, before any Root B pass claim is allowed.

Root A execution tasks. Root A begins only after the medium choice is locked by MVP-0. The MVP-A calculation must verify that the selected $\mathcal{D}_\mathcal{M}$ provides all needed components; check stability of the selected phase $\mathcal{M}_0$; compute $\mathcal{L}_{\mathcal{M}_0}$ and its branches; verify $\Pi_\mathcal{R} + \Pi_\mathcal{I} = 1$ and $\Pi_\mathcal{R}\Pi_\mathcal{I} = 0$; extract the low-energy dispersion of record branches for the selected stable phase; prove

$$\boxed{N_{\rm cone}(B_{\rm rec}^{\rm gapless}) = 1}$$

for that phase; test $Q_{AB}^{\mu\nu} = G^{\mu\nu} K_{AB}$; build a suppression catalog for every non-record branch; and translate the cone into operational clocks, rods and signals. Alternative stable phases are deferred unless they are selected in MVP-0.

Root B execution tasks. Root B must construct or explicitly condition the bridge

$$\boxed{\mathcal{D}_\mathcal{M} \to S_{\rm eff}^{\rm rec} \to S_\mathcal{R}.}$$

If any part of this bridge is conditional rather than derived, it must already appear in the MVP-0 conditional-bridge table. Root B then extracts $\Theta_\mathcal{R}$ and $\Omega_\mathcal{R}$, identifies embedding null directions, derives $H[N]$ and $D[N^i]$, and computes $\{D, D\}$, $\{H, D\}$ and $\{H, H\}$ from the canonical Poisson or Dirac structure derived from $\Omega_\mathcal{R}$, not from target ADM brackets. The pass target is $A[N, M] = 0$ or a controlled boundary/gauge term, $\lambda = 1/2$, $\alpha = 1$, $\Delta_\mathcal{M} = 0$ or suppressed, two positive TT modes, record-cone propagation and matter-coupled constraint preservation.

Root B linearized ADM MVP scaffold. The linearized Root B MVP is an execution scaffold, not a completed proof. It starts from the conditional $S_\mathcal{R}^{\rm bridge}$ and asks whether the local record sector can support a linearized ADM phase space with non-kinetic lapse/shift, first-class constraints, exactly two positive TT modes, the K1 record cone and no local scalar/vector/preferred-frame leakage. The current status is:

$$\boxed{\begin{array}{c} B0 \text{ bridge imported conditionally}, \qquad \Theta_\mathcal{R}, \Omega_\mathcal{R} \text{ not yet derived}, \\ \text{no kinetic lapse/shift: first hard check}, \qquad \text{two TT modes: conditional/fail-risk}, \\ \text{ADM bracket closure and EH fixed point: open.} \end{array}}$$

The local background is the K1 background

$$\boxed{\bar{g} = \eta, \qquad \eta^{\mu\nu} = \mathrm{diag}(-1, 1, 1, 1).}$$

This is more than notation: the linearized principal operator must be Lorentz-covariant under $x^\mu \mapsto \Lambda^\mu{}_\nu x^\nu$. The K1 choice $Z_\eta = Z_\theta = Z_Q = 1$ gives the common principal symbol $-\omega^2 + k^2$, and the branch dispersions $\omega^2 = k^2 + m^2$ are Lorentz-covariant at the principal-symbol level. A sector-dependent leading Z would create a preferred cone and would prevent $\bar{g} = \eta$ from serving as the common local background.

The conditional ADM perturbations are

$$\boxed{N = 1 + n, \qquad N^i = n^i, \qquad h_{ij} = \delta_{ij} + \gamma_{ij}, \qquad \pi^{ij} \text{ conjugate to } \gamma_{ij}.}$$

The spatial perturbation is decomposed as

$$\boxed{\gamma_{ij} = \gamma_{ij}^{TT} + \partial_i V_j^T + \partial_j V_i^T + \left(\partial_i \partial_j - \frac{1}{3}\delta_{ij}\partial^2\right)\sigma + \frac{1}{3}\delta_{ij}\gamma,}$$

with

$$\boxed{\partial^i \gamma_{ij}^{TT} = 0, \qquad \delta^{ij}\gamma_{ij}^{TT} = 0, \qquad \partial^i V_i^T = 0.}$$

The target mode assignment is: γ_{ij}^{TT} supplies two propagating modes; V_i^T, σ, γ are constrained, gauge or near-zone; and n, n^i are multipliers with no physical kinetic energy. Any unsuppressed scalar, vector, lapse, shift, massive spin-2, second metric or preferred-frame branch is a local Root B failure.

The conditional quadratic action to audit is

$$\boxed{S_{\mathcal{R}}^{(2)} = \int dt\, d^3x \left[\pi^{ij}\dot{\gamma}_{ij} + \pi_\vartheta \dot{\vartheta} - n\mathcal{H}^{(1)} - n^i \mathcal{H}_i^{(1)} - \mathcal{H}_{\mathrm{phys}}^{(2)} - \Delta_{\mathcal{M}}^{(2)}\right].}$$

This is an audit target, not an assumed action. The primary constraints

$$\boxed{\pi_n \approx 0, \qquad \pi_i \approx 0}$$

must arise from the record action itself. The target linear constraints are

$$\boxed{\mathcal{H}_i^{(1)} = -2\partial_j \pi^j{}_i, \qquad \mathcal{H}^{(1)} = -\partial^i \partial^j \gamma_{ij} + \partial^2 \gamma.}$$

The sign convention is

$$\boxed{R^{(1)} = \partial^i \partial^j \gamma_{ij} - \partial^2 \gamma, \qquad \mathcal{H}^{(1)} = -R^{(1)}}$$

up to the overall sign convention for $H[N]$, normalization and boundary terms. Momentum-squared ADM terms are quadratic around the zero-momentum background and enter the quadratic consistency and bracket audit, not the first-order scalar constraint.

Possible sources of $\Delta_{\mathcal{M}}^{(2)}$ are: η-geometry mixing after integrating out the gapped amplitude branch; q_{ij}-geometry mixing after integrating out the gapped strain branch; higher-derivative operators suppressed by ℓ_*; or residual preferred-frame/medium-flow terms left by an incomplete projection. The conservative local target is

$$\boxed{\Delta_{\mathcal{M}}^{(2)} = 0}$$

after gapped/internal fields are integrated out, or else every surviving piece must carry a suppression scale and an observation range.

The B1 presymplectic extraction must compute

$$\boxed{\delta L_{\mathcal{R}} = E_{\mathcal{R}}\delta\Phi_{\mathcal{R}} + d\Theta_{\mathcal{R}}(\delta\Phi_{\mathcal{R}}), \qquad \Omega_{\mathcal{R}} = \int_{\Sigma}\delta\Theta_{\mathcal{R}}.}$$

The comparison target is the canonical form

$$\boxed{\Theta_{\mathcal{R}} = \int d^3x\,\left(\pi^{ij}\delta\gamma_{ij} + \pi_{\vartheta}\delta\vartheta\right) + \text{boundary/constraint/internal terms.}}$$

This form must be derived from

$$\boxed{\mathcal{D}_{\mathcal{M}} \to S^{\text{bridge}}_{\mathcal{R}} \to \Theta_{\mathcal{R}} \to \Omega_{\mathcal{R}}.}$$

If $\Theta_{\mathcal{R}}$ contains an unsuppressed $\pi_n\delta n$ or $\pi_i\delta n^i$ physical kinetic sector, Root B fails before the bracket audit.

The B2–B4 constraint audit follows the Dirac algorithm:

1. derive primary constraints from velocity-momentum relations;
2. preserve π_n and π_i in time, producing $\mathcal{H} \approx 0$ and $\mathcal{H}_i \approx 0$;
3. iterate preservation of $\mathcal{H}, \mathcal{H}_i$;
4. classify all constraints as first class or second class;
5. pass to Dirac brackets if second-class constraints appear;
6. verify that no step creates a propagating lapse, shift, scalar, vector or preferred-frame medium mode.

The abelianized linear check is

$$\boxed{\{D[L], D[M]\} \text{ closes on } D, \qquad \{H[N], D[M]\} \text{ closes on } H, \qquad \{H[N], H[M]\} = 0}$$

at flat linear order. This is not a full Root B pass. The quadratic consistency check must include the first nonlinear corrections in $h_{ij} = \delta_{ij} + \gamma_{ij}$ and verify

$$\boxed{\{H[N], H[M]\} - D[h^{ij}(N\partial_j M - M\partial_j N)] = A[N, M]}$$

with $A[N, M] = 0$ or a controlled boundary/gauge/constraint term.

The B5 filter tests

$$\boxed{\lambda = \frac{1}{2}, \qquad \alpha = 1, \qquad \Delta_{\mathcal{M}} = 0}$$

or a controlled $\Delta_{\mathcal{M}}$. $\lambda \neq 1/2$ leaves a scalar gravitational mode; $\alpha \neq 1$ after measured-G normalization gives the wrong curvature response; and an unsuppressed preferred-frame, scalar/vector, massive-tensor, second-metric or non-ADM operator fails the local core.

The B6 hyperbolicity certificate must prove

$$\boxed{\operatorname{rank} K_{TT} = 2, \qquad \text{positive kinetic sign}, \qquad \text{positive gradient sign}, \qquad P_{TT} = -\omega^2 + k^2,}$$

with no scalar/vector leakage, no massive spin-2 leakage and no second metric. The comparison action is

$$\boxed{S_{TT}^{(2)} = \frac{1}{2}\int dt\, d^3x\, \left[\dot{\gamma}_{ij}^{TT}\dot{\gamma}_{ij}^{TT} - (\partial_k\gamma_{ij}^{TT})(\partial_k\gamma_{ij}^{TT})\right].}$$

This is not a proof unless the TT projector and its positive conjugate momentum are derived from $S_{\mathcal{R}}^{\mathrm{bridge}}$. The possible TT routes are: identify γ_{ij}^{TT} with a TT projection of q_{ij} (blocked in the conservative route because q_{ij} is gapped); build it as a composite from ϕ,θ gradients or other record data; keep it as a new conditional record-geometry variable; or revise the input freeze toward a gapless TT-candidate Q-route.

The B7 matter-preservation check does not prove Root C, but it must not make Root C impossible. It must test

$$\boxed{\dot{\mathcal{H}} \approx 0, \qquad \dot{\mathcal{H}}_i \approx 0}$$

for the matter sectors used in the MVP: massive test bodies, photons/EM field, bound systems, composite macroscopic bodies and, only after the local audit is safe, compact-body effective variables. Matter must not require a private lapse, private shift, species-private metric, nonzero local $A[N,M]$, or explicit species-private θ-charge.

The canonical degree-of-freedom count must include lapse/shift momenta:

$$\boxed{6(\gamma_{ij}) + 6(\pi^{ij}) + 1(n) + 1(\pi_n) + 3(n^i) + 3(\pi_i) = 20}$$

phase-space variables. The standard ADM pattern has four primary constraints π_n,π_i and four secondary constraints $\mathcal{H},\mathcal{H}_i$. If all eight are first class, they remove sixteen phase-space degrees of freedom:

$$\boxed{20 - 8\times 2 = 4}$$

leaving two propagating configuration modes, γ_{ij}^{TT} and its conjugate momentum. This count is valid only if B1–B4 actually produce the required first-class constraints.

The next calculation is triggered only when: $S_{\mathcal{R}}^{\mathrm{bridge}}$ is documented; $\Phi_{\mathcal{R}}$ is fixed; $\bar{g}=\eta$ is locked; the K1 block $Z_\eta = Z_\theta = Z_Q = 1$ is specified; conditional bridges are listed; $\Delta_{\mathcal{M}}^{(2)}$ sources are enumerated; and the B1 output format is fixed. B1 must output an explicit $\Theta_{\mathcal{R}}$, an explicit $\Omega_{\mathcal{R}}$, a yes/no decision on $\pi_n\delta n$ and $\pi_i\delta n^i$, a phase-space dimension count and a stop/continue decision. If B1 is unclear, the scaffold must be revised before starting the bracket calculation.

Root B B1 presymplectic extraction. B1 is the first Root B calculation checkpoint. Its central question is:

$$\boxed{\text{Does } S_{\mathcal{R}}^{(2)} \text{ produce } \Theta_{\mathcal{R}} \text{ and } \Omega_{\mathcal{R}} \text{ with no } \pi_n\delta n \text{ or } \pi_i\delta n^i \text{ sector?}}$$

The answer is a conditional yes under the no-dot rule for $\Delta_{\mathcal{M}}^{(2)}$:

$$\boxed{\begin{gathered}\Theta_{\mathcal{R}} \text{ derived from } S_{\mathcal{R}}^{(2)}: \text{ yes, conditional,}\\ \Omega_{\mathcal{R}} \text{ derived}: \text{ yes, conditional,}\\ \pi_n\delta n,\ \pi_i\delta n^i: \text{ absent in the conditional action,}\\ \pi_n\approx 0,\quad \pi_i\approx 0: \text{ primary constraints,}\\ B1: \text{ conditional pass; Root B proof not passed.}\end{gathered}}$$

The conditional nature of the action is essential:

$$\boxed{\text{IF } S_{\mathcal{R}}^{\mathrm{bridge}} \text{ produces the no-dot quadratic form, THEN } \pi_n = 0,\ \pi_i = 0 \text{ at B1 order.}}$$

B1 does not prove

$$\boxed{\mathcal{D}_{\mathcal{M}} \to S_{\mathcal{R}}^{\text{bridge}} \to \text{ no-dot lapse/shift action.}}$$

If the eventual medium-derived action contains physical $\dot{n}$ or $\dot{n}^i$ terms, the B1 conditional pass is revoked.

For B1 the remainder is split as

$$\boxed{\Delta_{\mathcal{M}}^{(2)} = \Delta_{\text{no-dot}}^{(2)} + \Delta_{\text{lapse-dot}}^{(2)} + \Delta_{\text{shift-dot}}^{(2)} + \Delta_{\text{mixed-dot}}^{(2)}.}$$

B1 requires

$$\boxed{\Delta_{\text{lapse-dot}}^{(2)} = \Delta_{\text{shift-dot}}^{(2)} = \Delta_{\text{mixed-dot}}^{(2)} = 0}$$

or these pieces must be pure boundary/gauge degeneracies with an explicit generator. The conservative conditional branch keeps only $\Delta_{\text{no-dot}}^{(2)}$ at this stage. Candidate no-dot sources include η-geometry mixing, suppressed by $m_\eta = \ell_*^{-1}$; q_{ij}-geometry mixing, suppressed by $m_Q = 2\ell_*^{-1}$; higher-derivative spatial operators suppressed by ℓ_*; and possible ϑ-geometry or preferred-flow terms, which remain open until the B2–B7 and Root C audits.

The variation uses δ as the field-space exterior derivative and d_t as the time derivative. For the metric canonical pair,

$$\boxed{\delta(\pi^{ij}\dot{\gamma}_{ij}) = \dot{\gamma}_{ij}\delta\pi^{ij} - \dot{\pi}^{ij}\delta\gamma_{ij} + d_t(\pi^{ij}\delta\gamma_{ij}).}$$

For the record phase pair,

$$\boxed{\delta(\pi_\vartheta\dot{\vartheta}) = \dot{\vartheta}\,\delta\pi_\vartheta - \dot{\pi}_\vartheta\,\delta\vartheta + d_t(\pi_\vartheta\delta\vartheta).}$$

For lapse and shift,

$$\boxed{\delta(-n\mathcal{H}^{(1)}) = -\mathcal{H}^{(1)}\delta n - n\delta\mathcal{H}^{(1)}, \qquad \delta(-n^i\mathcal{H}_i^{(1)}) = -\mathcal{H}_i^{(1)}\delta n^i - n^i\delta\mathcal{H}_i^{(1)}.}$$

There is no term $d_t(\pi_n\delta n)$ or $d_t(\pi_i\delta n^i)$, because n and n^i have no time derivatives in the conditional B1 action.

Collecting time-boundary terms gives

$$\boxed{\Theta_{\mathcal{R}} = \int d^3x\,\left(\pi^{ij}\delta\gamma_{ij} + \pi_\vartheta\delta\vartheta\right) + \Theta_{\text{boundary}} + \Theta_{\text{internal}} + \Theta_{\Delta\text{no-dot}}.}$$

In the no-dot branch,

$$\boxed{\operatorname{coeff}_{\delta n}\Theta_{\mathcal{R}} = 0, \qquad \operatorname{coeff}_{\delta n^i}\Theta_{\mathcal{R}} = 0,}$$

so

$$\boxed{\pi_n = 0, \qquad \pi_i = 0}$$

as primary constraints. Boundary terms may contribute boundary charges, internal terms may contribute gapped/internal symplectic blocks, and $\Theta_{\Delta\text{no-dot}}$ may contribute classified no-dot corrections; none may generate a local physical δn or δn^i kinetic coefficient.

The field-space derivative yields

$$\boxed{\Omega_{\mathcal{R}} = \int_\Sigma d^3x\,\left(\delta\pi^{ij}\wedge\delta\gamma_{ij} + \delta\pi_\vartheta\wedge\delta\vartheta\right) + \Omega_{\text{boundary}} + \Omega_{\text{internal}} + \Omega_{\Delta\text{no-dot}}.}$$

No physical lapse/shift symplectic block appears:

$$\boxed{\delta\pi_n \wedge \delta n \text{ absent}, \qquad \delta\pi_i \wedge \delta n^i \text{ absent.}}$$

Thus n and n^i are candidate multiplier variables at B1 order. This is not yet a proof that they generate the ADM constraints; B2/B3 must still show embedding gauge and the secondary constraints $\mathcal{H} \approx 0, \mathcal{H}_i \approx 0$.

The B1 bookkeeping has

$$\boxed{20 \text{ metric/lapse/shift phase-space variables} + 2 \text{ record-phase variables } (\vartheta, \pi_\vartheta) = 22.}$$

B1 does not decide whether ϑ is a clock variable, gauge direction, constrained variable, decoupled phase, or a dangerous scalar gravitational mode. If B2/B3 do not reduce or decouple it, ϑ remains a gapless scalar-like local mode, and B6/Root C must show that it does not mediate a species-dependent fifth force.

B1 must be revoked if a later derivation of $S_{\mathcal{R}}^{\text{bridge}}$ produces an unsuppressed term of the form

$$\boxed{\dot{n}^2, \quad \dot{n}_i\dot{n}^i, \quad \dot{n}\,\dot{\gamma}, \quad \dot{n}^i\dot{\gamma}_{ij}, \quad \dot{n}\,\dot{\vartheta}, \quad \dot{n}^i\partial_i\vartheta, \quad \dot{n}^i n_i}$$

unless the term is proven to be pure boundary or gauge-degenerate and absent from $\Omega_{\mathcal{R}}$ on physical observables.

The critical forward tests inherited from B1 are:

- B2/B4 must verify that $\Delta^{(2)}_{\text{no-dot}}$ hides no lapse/shift kinetic sector;
- B2/B3 and Root C must decide whether ϑ is gauge, constrained or decoupled from fifth-force sourcing;
- B7 and Root C must show that no-dot remainders stay safe under matter coupling;
- B3 must derive $\mathcal{H}, \mathcal{H}_i$ rather than merely use their target forms;
- B4 must close $\{H, H\}$ beyond the abelian linear check;
- B6 must make the TT origin compatible with the same presymplectic form.

If any of these tests fails, the corresponding part of the B1 conditional pass fails retroactively.

Root B B2 embedding null test. B2 uses the B1 presymplectic form to test whether linearized re-embedding deformations are gauge directions rather than physical preferred slicing or medium-flow degrees of freedom. The correct canonical test is not off-shell zero:

$$\boxed{\Omega_{\mathcal{R}}(\delta\Phi, \delta_\xi\Phi) \neq 0 \quad \text{in general off shell.}}$$

The correct test is

$$\boxed{\Omega_{\mathcal{R}}(\delta\Phi, \delta_\xi\Phi) = \delta G[\xi], \qquad G[\xi] = H[N] + D[M^i] + \text{boundary},}$$

with $G[\xi] \approx 0$ on the constraint surface. Thus re-embedding becomes a presymplectic null direction only after B3 derives the constraints $H \approx 0$ and $D \approx 0$ from the Dirac consistency step. B2 can identify the candidate generator; it does not yet prove the constraints.

B2 imports the B1 form

$$\boxed{\Omega_{\mathcal{R}} = \int d^3x \, \left(\delta\pi^{ij} \wedge \delta\gamma_{ij} + \delta\pi_\vartheta \wedge \delta\vartheta\right) + \Omega_{\text{boundary}} + \Omega_{\text{internal}} + \Omega_{\Delta\text{no}-\text{dot}},}$$

with no local $\delta\pi_n \wedge \delta n$ or $\delta\pi_i \wedge \delta n^i$ block. If such a block is later derived, B2 is revoked together with B1.

For a spatial embedding shift M^i, the linearized metric perturbation transforms as

$$\boxed{\delta_M \gamma_{ij} = \partial_i M_j + \partial_j M_i.}$$

The momentum transforms by the Lie action, $\delta_M \pi^{ij} = \mathcal{L}_M \pi^{ij}$, but the selected background has $\gamma_{ij} = 0$, $\dot\gamma_{ij} = 0$ and hence $\pi^{ij} = 0$. Therefore $\mathcal{L}_M \pi^{ij}$ is higher order in the first linearized B2 check and is restored only in later nonlinear audits.

Using the metric part of $\Omega_{\mathcal{R}}$,

$$\Omega_{\text{metric}} = \int d^3x \, \delta\pi^{ij} \wedge \delta\gamma_{ij},$$

one obtains

$$\begin{aligned}
\Omega_{\text{metric}}(\delta\Phi, \delta_M\Phi) &= \int d^3x \, \delta\pi^{ij}(\partial_i M_j + \partial_j M_i) \\
&= 2\int d^3x \, \delta\pi^{ij}\partial_i M_j \\
&= \delta \int d^3x \, M^i(-2\partial_j \pi^j{}_i) + \text{boundary}.
\end{aligned}$$

For compactly supported variations the boundary term vanishes; for non-compact variations it is deferred to ADM momentum-charge bookkeeping. Thus

$$\boxed{D[M^i] = \int d^3x \, M^i \mathcal{H}_i^{(1)}, \qquad \mathcal{H}_i^{(1)} = -2\partial_j \pi^j{}_i.}$$

Spatial re-embedding is generated by the candidate momentum constraint and becomes null only on $\mathcal{H}_i^{(1)} \approx 0$.

For a normal deformation N, the linearized metric-sector action is

$$\boxed{\delta_N \gamma_{ij} = 0, \qquad \delta_N \pi^{ij} = \partial^i\partial^j N - \delta^{ij}\partial^2 N.}$$

Then

$$\begin{aligned}
\Omega_{\text{metric}}(\delta\Phi, \delta_N\Phi) &= -\int d^3x \, (\partial^i\partial^j N - \delta^{ij}\partial^2 N)\delta\gamma_{ij} \\
&= \delta \int d^3x \, N(-\partial^i\partial^j\gamma_{ij} + \partial^2\gamma) + \text{boundary}.
\end{aligned}$$

Therefore

$$\boxed{H[N] = \int d^3x \, N\mathcal{H}^{(1)}, \qquad \mathcal{H}^{(1)} = -\partial^i\partial^j\gamma_{ij} + \partial^2\gamma.}$$

Normal re-embedding is generated by the candidate Hamiltonian constraint and becomes null only on $\mathcal{H}^{(1)} \approx 0$.

The record-phase sector uses the fixed-background convention

$$\boxed{\theta = \nu_0\tau + \vartheta, \qquad \nu_0\tau \text{ fixed by the input freeze,} \qquad \vartheta \text{ dynamical.}}$$

Under this convention

$$\boxed{\mathcal{L}_N\vartheta = N\partial_\tau\vartheta,}$$

so at the selected background $\vartheta = 0$ there is no background $N\nu_0$ term in the B2 calculation. The alternative convention, where θ is dynamical while τ is held as a fixed reference label, can produce an $N\nu_0$ phase shift; it is not used for the MVP B2 pass.

For spatial shifts,

$$\boxed{\delta_M\vartheta = M^i\partial_i\vartheta}$$

has no first-order background contribution because $\partial_i\theta_{\rm bg} = 0$. Its later contribution to the full momentum generator has the schematic form

$$\boxed{D_\vartheta[M] = \int d^3x\, \pi_\vartheta M^i\partial_i\vartheta.}$$

For normal deformations the corresponding phase contribution is written schematically as

$$\boxed{H_\vartheta[N] = \int d^3x\, N\mathcal{H}_\vartheta,}$$

with the actual status of $\mathcal{H}_\vartheta$ deferred. B2 therefore passes the record-phase embedding check only conditionally under the fixed-background convention. The fifth-force issue is not solved: Root C and B6 must still show that shift symmetry and the record/matter bridge prevent species-private scalar sourcing.

The B2 result is:

$$\boxed{\begin{array}{c}\text{spatial metric embedding: conditional pass,}\\ \text{normal metric embedding: conditional pass,}\\ \text{off-shell null direction: not required and not claimed,}\\ \text{null on constraint surface: conditional on B3,}\\ \vartheta\text{ embedding: conditional safe at B2,}\\ \text{Root B proof: not passed.}\end{array}}$$

Boundary and remainder blocks are classified as follows: $\Omega_{\rm boundary}$ vanishes for compactly supported variations and may become an ADM charge for non-compact variations; $\Omega_{\rm internal}$ contains the gapped η, q_{ij} sectors with $m_\eta = \ell_*^{-1}$ and $m_Q = 2\ell_*^{-1}$; and $\Omega_{\Delta\text{no}-\text{dot}}$ must hide no lapse/shift kinetic block. Any preferred-frame bulk term, dot-lapse or dot-shift symplectic block, extra generator not proportional to H or D, or species-private metric/clock direction turns the B2 result into a failure or a B4/B5 anomaly.

The next Root B step is B3:

$$\boxed{\dot{\pi}_n \approx 0 \Rightarrow H \approx 0, \qquad \dot{\pi}_i \approx 0 \Rightarrow H_i \approx 0.}$$

If B3 fails, the B2 candidate generators do not become constraints and the route stops before ADM closure.

Root B B3 constraint generators. B3 uses the B1 primary lapse/shift constraints and the B2 candidate generators to run the first Dirac consistency step. The central question is whether

$$\boxed{\pi_n \approx 0, \qquad \pi_i \approx 0}$$

generate the secondary constraints

$$\boxed{\mathcal{H} \approx 0, \qquad \mathcal{H}_i \approx 0}$$

rather than turning lapse and shift into physical or auxiliary variables. The B3 verdict is conditional: the secondary constraints are obtained under the no-algebraic-lapse/shift rule, but first-class status and ADM closure are not yet proven.

B3 imports from B1

$$\boxed{\pi_n \approx 0, \qquad \pi_i \approx 0}$$

and

$$\boxed{\Omega_{\mathcal{R}} = \int d^3x \left(\delta\pi^{ij} \wedge \delta\gamma_{ij} + \delta\pi_\vartheta \wedge \delta\vartheta\right) + \text{classified remainders.}}$$

It imports from B2

$$\boxed{\mathcal{H}_i^{(1)} = -2\partial_j \pi^j{}_i, \qquad \mathcal{H}^{(1)} = -\partial^i\partial^j\gamma_{ij} + \partial^2\gamma,}$$

and

$$\boxed{D[M^i] = \int d^3x\, M^i \mathcal{H}_i^{(1)}, \qquad H[N] = \int d^3x\, N\mathcal{H}^{(1)}.}$$

All of these imports remain conditional on $S_{\mathcal{R}}^{\text{bridge}}$ and on the B1/B2 no-dot and no-hidden-lapse rules.

The B3 total Hamiltonian is

$$\boxed{H_T = \int d^3x \left[n\mathcal{H}^{(1)} + n^i\mathcal{H}_i^{(1)} + H^{(2)}_{\text{phys}} + \Delta^{(2)}_{\text{no-dot}} + u\pi_n + u^i\pi_i\right],}$$

where u, u^i are arbitrary multipliers for the primary constraints. The no-dot remainder is split as

$$\boxed{\Delta^{(2)}_{\text{no-dot}} = \Delta^{(2)}_{\text{safe}} + \Delta^{(2)}_{\text{alg-lapse}} + \Delta^{(2)}_{\text{alg-shift}} + \Delta^{(2)}_{\text{alg-mixed}}.}$$

B3 passes only if

$$\boxed{\Delta^{(2)}_{\text{alg-lapse}} = \Delta^{(2)}_{\text{alg-shift}} = \Delta^{(2)}_{\text{alg-mixed}} = 0}$$

up to pure boundary or gauge degeneracies. This is stricter than B1: B1 forbids $\dot{n}$ and $\dot{n}^i$, whereas B3 forbids algebraic terms such as n^2, n_in^i, nn^i, or n times a local field, whenever they solve for lapse/shift instead of producing constraints.

For example, if

$$\boxed{\Delta^{(2)}_{\text{no-dot}} \supset \frac{m_n^2}{2}n^2,}$$

then preservation of π_n gives

$$\boxed{\mathcal{H}^{(1)} + m_n^2 n \approx 0,}$$

which solves for n rather than producing the Hamiltonian constraint. This is a B3 failure. Physically, an algebraic lapse mass is a local massive/non-ADM gravity sector; B3 forbids it because the local MVP target is a massless ADM core.

The circularity is explicit. B3 is internally consistent if the no-algebraic-lapse/shift rules hold, but it does not prove that the microscopic medium produces a Hamiltonian obeying those rules. Therefore the B3 result is

> If $S_{\mathcal{R}}^{\mathrm{bridge}}$ has the conditional Hamiltonian form above, and if $\Delta^{(2)}_{\mathrm{no-dot}}$ has no algebraic lapse/shift pieces, then the primary constraints generate the secondary ADM candidates.

The B3 canonical brackets are

$$\boxed{\{\gamma_{ij}(x), \pi^{kl}(y)\} = \delta^{k}_{(i}\delta^{l}_{j)}\delta^3(x-y), \qquad \{\vartheta(x), \pi_\vartheta(y)\} = \delta^3(x-y),}$$

and

$$\boxed{\{n(x), \pi_n(y)\} = \delta^3(x-y), \qquad \{n^i(x), \pi_j(y)\} = \delta^i{}_j\delta^3(x-y).}$$

The sign convention is the standard Hamilton equation $\dot{p} = -\partial H/\partial q$:

$$\boxed{\dot{\pi}_n = \{\pi_n, H_T\} = -\frac{\delta H_T}{\delta n}, \qquad \dot{\pi}_i = \{\pi_i, H_T\} = -\frac{\delta H_T}{\delta n^i}.}$$

For the lapse primary constraint,

$$\begin{aligned}\dot{\pi}_n = \{\pi_n, H_T\} &= -\frac{\delta H_T}{\delta n}\\ &= -\mathcal{H}^{(1)} - \frac{\delta \Delta^{(2)}_{\mathrm{no-dot}}}{\delta n}.\end{aligned}$$

Under the B3 no-algebraic-lapse rule,

$$\boxed{\frac{\delta \Delta^{(2)}_{\mathrm{no-dot}}}{\delta n} = 0}$$

up to boundary/gauge terms, so

$$\boxed{\dot{\pi}_n \approx 0 \quad \Rightarrow \quad \mathcal{H}^{(1)} \approx 0.}$$

Thus

$$\boxed{\mathcal{H}^{(1)} = -\partial^i\partial^j\gamma_{ij} + \partial^2\gamma \approx 0.}$$

For compactly supported lapse variations, boundary terms vanish. For lapse choices nonzero at spatial infinity, the boundary piece belongs to later ADM-charge bookkeeping, not to the local MVP constraint test.

For the shift primary constraint,

$$\begin{aligned}\dot{\pi}_i = \{\pi_i, H_T\} &= -\frac{\delta H_T}{\delta n^i}\\ &= -\mathcal{H}^{(1)}_i - \frac{\delta \Delta^{(2)}_{\mathrm{no-dot}}}{\delta n^i}.\end{aligned}$$

Under the B3 no-algebraic-shift rule,

$$\boxed{\frac{\delta\Delta^{(2)}_{\mathrm{no-dot}}}{\delta n^i}=0}$$

up to boundary/gauge terms, so

$$\boxed{\dot{\pi}_i\approx 0 \quad\Rightarrow\quad \mathcal{H}_i^{(1)}\approx 0.}$$

Thus

$$\boxed{\mathcal{H}_i^{(1)}=-2\partial_j\pi^j{}_i\approx 0.}$$

For compactly supported shift variations, boundary terms vanish. For shifts nonzero at spatial infinity, the boundary piece belongs to later ADM momentum bookkeeping.

After B3 the conditional constraint set is

$$\boxed{\pi_n\approx 0,\quad \pi_i\approx 0,\quad \mathcal{H}^{(1)}\approx 0,\quad \mathcal{H}_i^{(1)}\approx 0.}$$

This is 4 primary plus 4 secondary constraint functions. B3 has not proven that all eight are first class. That requires B4:

$$\boxed{\{D,D\},\qquad \{H,D\},\qquad \{H,H\}.}$$

If B4 proves all eight constraints first class, the local phase-space count is

$$\boxed{22\text{ starting variables}-8\times 2=6\text{ remaining phase-space variables.}}$$

The interpretation is

$$\boxed{4\text{ variables}=2\text{ TT metric modes with momenta,}\qquad 2\text{ variables}=(\vartheta,\pi_\vartheta).}$$

Therefore, if the final MVP target is exactly two radiative tensor modes, the record phase must later be shown to be gauge, decoupled, or non-radiative in the gravitational sector. Otherwise the local sector contains two TT modes plus one long-range record-phase mode.

At B3, ϑ remains a conditional canonical pair:

$$\boxed{\vartheta,\pi_\vartheta.}$$

It does not enter the lapse/shift preservation equations unless $\Delta^{(2)}_{\mathrm{no-dot}}$ contains an algebraic phase coupling. The B3 safety check is

$$\boxed{\Delta^{(2)}_{\mathrm{no-dot}}\text{ contains no }n\vartheta\text{ and no }n^i\partial_i\vartheta\text{ algebraic coupling.}}$$

The protected route is the shift symmetry

$$\boxed{\theta\mapsto\theta+\text{constant},}$$

which forbids explicit θ-charge in the local Hamiltonian. Derivative couplings are not automatically forbidden; they must be classified as gauge, decoupled, suppressed, or part of the allowed record structure. B3 therefore does not prove that ϑ is harmless. That remains a later scalar-leakage and matter-coupling question for B6 and Root C.

The B3 pass/fail ledger is:

B1 primary constraints	:	conditional pass,
B2 generators $\mathcal{H}, \mathcal{H}_i$	:	conditional pass,
H_T with primary multipliers	:	conditional pass,
$\dot{\pi}_n \to \mathcal{H}^{(1)} \approx 0$	:	conditional pass,
$\dot{\pi}_i \to \mathcal{H}_i^{(1)} \approx 0$	:	conditional pass,
first-class status	:	open until B4,
ϑ safety	:	open,
TT origin	:	open/fail-risk until B6.

B3 proves only that, within the conditional Hamiltonian and the no-algebraic-lapse/shift rule, preserving the primary lapse/shift constraints produces the secondary candidates. It does not prove first-class closure, the hypersurface-deformation algebra, $\lambda = 1/2$, the EH fixed point, two TT modes, matter preservation of constraints, or derivation of the bridge from the microscopic input.

The forwarding map is:

first-class constraints	$\to$	B4 bracket calculation,
hypersurface-deformation closure	$\to$	B4 anomaly test $A[N, M] = 0$,
nonlinear $\{H, H\}$ structure	$\to$	B4 quadratic consistency,
$\lambda = 1/2$	$\to$	B5 EH fixed-point filter,
two TT modes	$\to$	B6 hyperbolicity certificate,
ϑ under matter coupling	$\to$	B6 and Root C,
$S_{\mathcal{R}}^{\text{bridge}}$ from medium	$\to$	deferred medium-to-bridge derivation.

B3 must stop, or be marked failed, if the conditional Hamiltonian contains

- n^2 or $n_i n^i$ terms that solve for lapse/shift;
- nn^i terms that make lapse/shift auxiliary physical variables;
- $n\vartheta$ or $n^i \partial_i \vartheta$ terms that create a scalar fifth-force constraint failure;
- algebraic preferred-frame terms that make the embedding generator background-dependent;
- a remainder variation that changes $\mathcal{H}^{(1)}$ or $\mathcal{H}_i^{(1)}$ by an unsuppressed local operator.

Here "unsuppressed" means that the operator contributes at the local MVP test scale. Acceptable suppression must be explicit, for example

$$\boxed{\begin{array}{c} (E/\Lambda_{\text{UV}})^2 < 10^{-5} \quad \text{for Solar-System scale checks,} \\ m > 1/r_{\text{test}} \quad \text{for the relevant propagation scale.} \end{array}}$$

The B3 reviewer stance is to accept the conditional pass with the no-algebraic-lapse/shift rule explicit. B3 is a canonical consistency calculation, not the derivation of $\Delta^{(2)}_{\text{no-dot}}$ from the medium. The stricter alternative is to stop B3 until that derivation is complete; the MVP instead keeps the conditional bridge visible.

The cumulative bridge audit after B3 is:

Rule	Source	Statement	Status
R-B1-1	B1	$\Delta^{(2)}_{\mathcal{M}}$ has no $\dot{n}X$	conditional
R-B1-2	B1	$\Delta^{(2)}_{\mathcal{M}}$ has no $\dot{n}^i X$	conditional
R-B1-3	B1	$\Delta^{(2)}_{\mathcal{M}}$ has no $\dot{n}\,\dot{\gamma}$	conditional
R-B2-1	B2	re-embedding is on-constraint null	proven mod constraints
R-B2-2	B2	ϑ uses fixed-background convention	locked
R-B3-1	B3	$\Delta_{\text{no-dot}}$ has no algebraic $n^2, n_i n^i$	conditional
R-B3-2	B3	$\Delta_{\text{no-dot}}$ has no algebraic nn^i	conditional
R-B3-3	B3	$\Delta_{\text{no-dot}}$ has no $n\vartheta$	conditional under shift
R-B3-4	B3	$\Delta_{\text{no-dot}}$ has no preferred-frame operator	conditional.

The B1 and B3 conditional rules must be verified by the deferred medium-to-bridge derivation. R-B2-1 is tested by B4 brackets. R-B2-2 remains the locked MVP convention. If any rule fails when the microscopic input is derived concretely, the corresponding B step must be revisited and its conditional pass cannot be counted.

The next Root B artifact is B4:

$$\boxed{\text{compute } \{D,D\},\ \{H,D\},\ \{H,H\} \text{ from the B1–B3 canonical structure.}}$$

If B4 fails, the B3 secondary constraints do not form the ADM first-class system and Root B fails in the local MVP.

Root B B4 bracket closure. B4 computes the constraint brackets from the B1–B3 canonical structure. Its job is to separate a real linear checkpoint from the full hypersurface-deformation algebra. The B4 verdict is:

> The primary constraints and the linear secondary constraints close at abelian linear order. This is a conditional pass, but it is not ADM closure. The full ADM/HDA test requires the quadratic supplement B4q.

B4 imports the canonical brackets

$$\boxed{\{\gamma_{ij}(x), \pi^{kl}(y)\} = \delta^k_{(i}\delta^l_{j)}\delta^3(x-y), \qquad \{\vartheta(x), \pi_\vartheta(y)\} = \delta^3(x-y),}$$

and

$$\boxed{\{n(x), \pi_n(y)\} = \delta^3(x-y), \qquad \{n^i(x), \pi_j(y)\} = \delta^i{}_j\delta^3(x-y).}$$

Here

$$\boxed{\delta^k_{(i}\delta^l_{j)} = \frac{1}{2}(\delta_i{}^k\delta_j{}^l + \delta_i{}^l\delta_j{}^k)}$$

is the symmetric Kronecker projector. B4 also imports

$$\boxed{\mathcal{H}^{(1)}_i = -2\partial_j\pi^j{}_i, \qquad \mathcal{H}^{(1)} = -\partial^i\partial^j\gamma_{ij} + \partial^2\gamma,}$$

with

$$\boxed{D[M^i] = \int d^3x\, M^i\mathcal{H}^{(1)}_i, \qquad H[N] = \int d^3x\, N\mathcal{H}^{(1)}.}$$

All smearing functions are compactly supported unless a boundary term is being classified. Non-compact smearing belongs to ADM-charge bookkeeping, not to the local MVP bracket test. B4 inherits the B1–B3 conditional rules: no $\dot{n}$, no $\dot{n}^i$, no algebraic lapse/shift mass, no algebraic $n\vartheta$, and no preferred-frame algebraic remainder. If these fail in a later medium derivation, B4 is revoked with the affected earlier step.

For the momentum generator,

$$\boxed{D[M] = \int d^3x\, M^i(-2\partial_j \pi^j{}_i) = \int d^3x\, \pi^{ij}(\partial_i M_j + \partial_j M_i)}$$

up to boundary. Therefore

$$\boxed{= \frac{\delta D[M]}{\delta \pi^{ij}} = \partial_i M_j + \partial_j M_i, \qquad \frac{\delta D[M]}{\delta \gamma_{ij}} = 0}$$

at linear metric order. For the Hamiltonian generator,

$$\boxed{H[N] = \int d^3x\, N(-\partial^i\partial^j\gamma_{ij} + \partial^2\gamma)}$$

and, after integrating derivatives onto N,

$$\boxed{H[N] = \int d^3x\, \gamma_{ij}(-\partial^i\partial^j N + \delta^{ij}\partial^2 N)}$$

up to boundary. Thus

$$\boxed{= \frac{\delta H[N]}{\delta \gamma_{ij}} = -\partial^i\partial^j N + \delta^{ij}\partial^2 N, \qquad \frac{\delta H[N]}{\delta \pi^{ij}} = 0.}$$

This matches the B2 normal deformation:

$$\boxed{\delta_N \pi^{ij} = -\frac{\delta H[N]}{\delta \gamma_{ij}} = \partial^i\partial^j N - \delta^{ij}\partial^2 N.}$$

The primary brackets close because the linear secondary constraints are independent of the lapse/shift variables and their momenta:

$$\boxed{\{\pi_n, \mathcal{H}^{(1)}\} = 0, \quad \{\pi_n, \mathcal{H}_i^{(1)}\} = 0, \quad \{\pi_i, \mathcal{H}^{(1)}\} = 0, \quad \{\pi_i, \mathcal{H}_j^{(1)}\} = 0, \quad \{\pi_n, \pi_i\} = 0.}$$

The reason is purely canonical: $\mathcal{H}^{(1)}$ depends only on γ_{ij}, $\mathcal{H}_i^{(1)}$ depends only on π^{ij}, and the primary lapse/shift momenta have no canonical bracket with those variables. This pass is conditional on the B3 no-algebraic-lapse/shift rule.

For the linear momentum bracket, $D^{(1)}[M]$ is π-linear:

$$\boxed{D^{(1)}[M] = 2\int d^3x\, (\partial_j M_i)\pi^{ij}.}$$

Since

$$\boxed{\{\pi^{ij}(x), \pi^{kl}(y)\} = 0,}$$

the bracket of two purely π-linear generators vanishes:

$$\boxed{\{D[L], D[M]\}_{\text{lin}} = 0.}$$

This is the abelianized linear result. It is not the full spatial diffeomorphism algebra

$$\boxed{\{D[L], D[M]\} = D[[L, M]], \qquad [L, M]^i = L^j \partial_j M^i - M^j \partial_j L^i.}$$

The non-abelian term requires nonlinear corrections such as

$$\boxed{D^{(2)}[M] \sim \int \gamma\, \pi\, \partial M,}$$

whose bracket with $D^{(1)}$ produces the commutator.

For the mixed bracket,

$$\begin{aligned}
\{H[N], D[M]\} &= \int d^3x \, (-\partial^i \partial^j N + \delta^{ij} \partial^2 N)(\partial_i M_j + \partial_j M_i) \\
&= -2 \int d^3x \, (\partial^i \partial^j N)(\partial_i M_j) + 2 \int d^3x \, (\partial^2 N)(\partial_i M^i).
\end{aligned}$$

With compact support, integration by parts gives

$$\boxed{\begin{aligned}
-2 \int (\partial^i \partial^j N)(\partial_i M_j) &\to 2 \int M_j \, \partial_i \partial^i \partial^j N, \\
2 \int (\partial^2 N)(\partial_i M^i) &\to -2 \int M^i \, \partial_i \partial^2 N.
\end{aligned}}$$

The sum is

$$\boxed{2 \int d^3x \, M_j \left(\partial_i \partial^i \partial^j N - \partial^j \partial^2 N \right) = 0,}$$

because spatial derivatives commute on smooth fields. Therefore

$$\boxed{\{H[N], D[M]\}_{\text{lin}} = 0.}$$

This means that $\mathcal{H}^{(1)}$ is invariant under the linearized spatial gauge transformation. The full ADM expectation,

$$\boxed{\{H[N], D[M]\} = H[M^i \partial_i N],}$$

requires nonlinear or density-weight corrections.

For the Hamiltonian bracket, at linear order both smeared Hamiltonian constraints depend only on γ_{ij}:

$$\boxed{\frac{\delta H[N]}{\delta \pi^{ij}} = 0, \qquad \frac{\delta H[M]}{\delta \pi^{ij}} = 0.}$$

Hence

$$\boxed{\{H[N], H[M]\}_{\text{lin}} = 0.}$$

This result is structurally trivial:

$$\boxed{\{H^{(1)}[N], H^{(1)}[M]\} = 0}$$

because $H^{(1)}$ is configuration-only. Any linearized theory with a configuration-only linear Hamiltonian constraint has this property. It does not distinguish the ADM/EH fixed point from nearby non-ADM theories.

The full target is instead

$$\boxed{\{H[N], H[M]\} = D[h^{ij}(N\partial_j M - M\partial_j N)].}$$

At the first nontrivial order one must write

$$\boxed{H = H^{(1)} + H^{(2)} + \cdots, \qquad D = D^{(1)} + D^{(2)} + \cdots}$$

and test

$$\boxed{\begin{aligned} A^{(1)}[N, M] = \{H^{(1)}[N], H^{(2)}[M]\} + \{H^{(2)}[N], H^{(1)}[M]\} \\ - D^{(1)}[\delta^{ij}(N\partial_j M - M\partial_j N)] = 0 \end{aligned}}$$

up to boundary/gauge/constraint terms. This B4q condition is the first real ADM-vs-non-ADM discriminator. It must produce the specific structure function

$$\boxed{h^{ij}(N\partial_j M - M\partial_j N)}$$

from the same record-geometry variable h_{ij}. A different coefficient or a private background tensor is a non-ADM result.

Thus the B4 linear bracket result is

$$\boxed{\{D, D\}_{\text{lin}} = 0, \qquad \{H, D\}_{\text{lin}} = 0, \qquad \{H, H\}_{\text{lin}} = 0.}$$

Together with the primary brackets, the B1–B3 constraint set is first-class at abelian linear order. This is a real checkpoint because it excludes an immediate linear local anomaly, lapse/shift auxiliary equation, or preferred-frame obstruction. But it is not Root B closure.

The record-phase sector remains open. B4 has used the pure metric linear constraints, and ϑ is harmless at this level only if the Hamiltonian has no explicit ϑ-charge and no algebraic lapse/phase coupling. The dangerous schematic term is

$$\boxed{\{H[N], H[M]\}_\vartheta \supset \int \frac{\delta H[N]}{\delta\vartheta}\frac{\delta H[M]}{\delta\pi_\vartheta} - (N \leftrightarrow M).}$$

Under the B3 safety rule and the shift symmetry

$$\boxed{\theta \mapsto \theta + \text{constant},}$$

there is no explicit ϑ-charge in the B4 local metric bracket. Pure derivative phase couplings must still be classified as boundary, gauge, decoupled or suppressed. If they survive as long-range scalar source terms, B4 passes only in the metric sector and the conservative route fails later at B6/Root C.

The B4 anomaly ledger is:

$$\boxed{\begin{array}{lcl} \text{primary lapse/shift anomaly} & : & \text{absent under B3 rules,} \\ \{D, D\}_{\text{lin}} \text{ anomaly} & : & \text{absent by } \pi\text{-linearity,} \\ \{H, D\}_{\text{lin}} \text{ anomaly} & : & \text{absent by IBP,} \\ \{H, H\}_{\text{lin}} \text{ anomaly} & : & \text{absent trivially,} \\ \text{missing nonlinear structure function} & : & \text{open until B4q,} \\ \vartheta \text{ scalar bracket leak} & : & \text{open until B6/Root C,} \\ \text{TT-origin anomaly} & : & \text{open/fail-risk until B6.} \end{array}}$$

No anomaly may be called suppressed unless the scale is explicit. The working MVP thresholds are:

$$\boxed{\begin{gathered}\text{Solar-System/PPN local anomaly: below } 10^{-5},\\ \text{GW cone-changing anomaly: below } 10^{-15} \text{ or absent by structure},\\ \text{binary-pulsar extra loss: below } 10^{-3} \text{ at MVP level},\\ \text{gapped internal sector: } m > 1/r_{\text{test}} \text{ or explicit EFT decoupling.}\end{gathered}}$$

These are working thresholds, not final data-analysis bounds. They prevent the word "suppressed" from being used without a scale.

B4 therefore proves only:

- primary constraints commute with the linear secondary constraints;
- the linear momentum constraints close abelianly;
- the linear Hamiltonian is invariant under linear spatial gauge;
- linear Hamiltonian constraints commute with each other;
- there is no immediate linear bracket anomaly.

B4 does not prove the full hypersurface-deformation algebra, the structure function $h^{ij}(N\partial_j M - M\partial_j N)$, $A[N,M]=0$ beyond the abelian linear test, $\lambda = 1/2$, the EH fixed point, two TT modes, matter-coupled preservation, or microscopic derivation of the bridge.

The honest next artifact is the quadratic supplement:

$$\boxed{\texttt{Root\_B\_B4q\_Quadratic\_HDA\_Closure.md}.}$$

The naming convention is

$$\boxed{B4 = \text{linear abelian bracket check}, \qquad B4q = \text{quadratic ADM/HDA discriminator.}}$$

Together, B4 and B4q complete the MVP bracket-closure calculation; B4 alone does not. B4q must derive or explicitly condition $H^{(2)}$ and $D^{(2)}$, then compute $A^{(1)}[N,M]$. The expected outcomes are:

$$\boxed{\begin{array}{lcl} A^{(1)} = 0 & : & \text{conditional ADM closure at MVP order,}\\ \delta A \text{ below bound} & : & \text{conditional non-ADM correction,}\\ A^{(1)} = O(1) \text{ or preferred tensor} & : & \text{conservative route fails,}\\ \text{no derivable } H^{(2)} & : & \text{bridge incomplete.}\end{array}}$$

For local Solar-System style checks the first compatibility target is

$$\boxed{|\delta A/A_{\text{ADM}}| < 10^{-5}.}$$

Any deviation that changes the GW cone must meet the stronger 10^{-15} cone/speed target or be absent by structure.

Root B B4q quadratic HDA closure. B4q is the quadratic supplement to B4. B4 showed only the abelian linear closure; B4q tests the first order where $\{H, H\}$ must reproduce the ADM momentum generator with the correct structure function. Its status is:

$$\boxed{\text{B4q conditionally passes the ADM-candidate kinetic sector iff } \lambda = 1/2 \text{ and } \Delta^{(2)} \text{ contributes no local HDA anomaly. It does not prove that the medium derives } H^{(2)}.}$$

B4q imports

$$\boxed{H^{(1)}[N] = \int d^3x\, N(-\partial^i\partial^j\gamma_{ij} + \partial^2\gamma), \qquad D^{(1)}[V] = \int d^3x\, V^i(-2\partial_j\pi^j{}_i).}$$

Equivalently, after integration by parts,

$$\boxed{H^{(1)}[N] = \int d^3x\, \gamma_{ij}(-\partial^i\partial^j N + \delta^{ij}\partial^2 N), \qquad D^{(1)}[V] = 2\int d^3x\, (\partial_j V_i)\pi^{ij}.}$$

It tests

$$\boxed{H = H^{(1)} + H^{(2)} + \cdots, \qquad D = D^{(1)} + D^{(2)} + \cdots,}$$

through

$$\boxed{\begin{aligned} A^{(1)}[N,M] = \{H^{(1)}[N], H^{(2)}[M]\} + \{H^{(2)}[N], H^{(1)}[M]\} \\ - D^{(1)}[\delta^{ij}(N\partial_j M - M\partial_j N)]. \end{aligned}}$$

The pass condition is $A^{(1)}[N,M] = 0$, up to boundary/gauge/constraint terms.

B4q uses the quadratic audit ansatz

$$\boxed{\mathcal{H}^{(2)} = \mathcal{H}^{(2)}_{\mathrm{kin}} + \mathcal{H}^{(2)}_{\mathrm{curv}} + \Delta^{(2)}.}$$

The kinetic part is

$$\boxed{\mathcal{H}^{(2)}_{\mathrm{kin}} = \pi^{ij}\pi_{ij} - \lambda\pi^2, \qquad \pi = \delta_{ij}\pi^{ij}.}$$

This is an ADM-candidate audit form, not yet derived from $S^{\mathrm{bridge}}_{\mathcal{R}}$. Indices are raised and lowered with δ_{ij} at B4q order; γ-corrections to index raising enter at the next perturbative order.

The curvature term $\mathcal{H}^{(2)}_{\mathrm{curv}}[\gamma,\gamma]$ is configuration-only, so it does not contribute to the leading B4q bracket with $H^{(1)}$:

$$\boxed{\{H^{(1)}, H^{(2)}_{\mathrm{curv}}\} = 0}$$

at this order. B5 must still check that this curvature sector reduces to the EH quadratic curvature operator, not to an unsuppressed $f(R)$, $R_{ij}R^{ij}$, Weyl-type, or preferred-frame operator.

From B4,

$$\boxed{\frac{\delta H^{(1)}[N]}{\delta\gamma_{ij}} = -\partial^i\partial^j N + \delta^{ij}\partial^2 N, \qquad \frac{\delta H^{(1)}[N]}{\delta\pi^{ij}} = 0.}$$

For the kinetic ansatz,

$$\boxed{\frac{\delta H^{(2)}_{\mathrm{kin}}[M]}{\delta\pi^{ij}} = 2M(\pi_{ij} - \lambda\delta_{ij}\pi), \qquad \frac{\delta H^{(2)}_{\mathrm{kin}}[M]}{\delta\gamma_{ij}} = 0.}$$

Therefore

$$\boxed{\{H^{(1)}[N], H^{(2)}_{\mathrm{kin}}[M]\} = 2\int d^3x\, M(-\partial^i\partial^j N + \delta^{ij}\partial^2 N)(\pi_{ij} - \lambda\delta_{ij}\pi).}$$

The antisymmetrized bracket is

$$\boxed{B_\lambda[N,M] = 2\int d^3x\, [ML^{ij}_N - NL^{ij}_M](\pi_{ij} - \lambda\delta_{ij}\pi),}$$

where

$$\boxed{L_N^{ij} = -\partial^i\partial^j N + \delta^{ij}\partial^2 N.}$$

Terms symmetric under $N \leftrightarrow M$ cancel. Expanding gives

$$\boxed{\begin{aligned} B_\lambda[N,M] = -2\int d^3x\,(M\partial^i\partial^j N - N\partial^i\partial^j M)\pi_{ij} \\ + 2(1-2\lambda)\int d^3x\,(M\partial^2 N - N\partial^2 M)\pi. \end{aligned}}$$

The first term is the ADM momentum generator. The second term is the trace anomaly.

The flat-background ADM structure function is

$$\boxed{V^i = \delta^{ij}(N\partial_j M - M\partial_j N).}$$

With the B2–B4 sign convention,

$$\boxed{D^{(1)}[V] = \int d^3x\,V^i(-2\partial_j\pi^j{}_i) = 2\int d^3x\,(\partial_j V_i)\pi^{ji}.}$$

Since

$$\partial_j V_i = \partial_j N\,\partial_i M + N\partial_j\partial_i M - \partial_j M\,\partial_i N - M\partial_j\partial_i N,$$

the first and third terms drop out after contraction with symmetric π^{ij}. Thus

$$\boxed{D^{(1)}[V] = -2\int d^3x\,(M\partial_i\partial_j N - N\partial_i\partial_j M)\pi^{ij}.}$$

Comparing with $B_\lambda[N,M]$, the anomaly is

$$\boxed{A_\lambda^{(1)}[N,M] = 2(1-2\lambda)\int d^3x\,(M\partial^2 N - N\partial^2 M)\pi.}$$

Hence B4q closes at this order only if

$$\boxed{1-2\lambda = 0, \qquad \lambda = \frac{1}{2}.}$$

If $\lambda \neq 1/2$, the bracket leaves a scalar trace anomaly. In the strict local MVP core the target is

$$\boxed{A_\lambda^{(1)}[N,M] = 0.}$$

Residual quadratic corrections are collected as $\Delta^{(2)}$. Then

$$\boxed{A^{(1)}[N,M] = A_\lambda^{(1)}[N,M] + A_\Delta^{(1)}[N,M].}$$

B4q passes only if

$$\boxed{A_\Delta^{(1)}[N,M] = 0}$$

or if the term is boundary/gauge/constraint-proportional with explicit classification. Failing examples include a preferred tensor $K^{ij}\partial_i N\partial_j M$, a trace kinetic correction shifting λ, a $\delta\vartheta$-driven lapse coupling at quadratic order, a second metric structure function, or a species/private record deformation vector. Working thresholds are inherited from the observation table:

$$\boxed{\begin{array}{c}\text{local PPN/Cassini anomaly: below } 10^{-5},\\ \text{GW cone-changing anomaly: below } 10^{-15} \text{ or absent by structure,}\\ \text{binary-pulsar scalar/vector channel: below } 10^{-3} \text{ at MVP level,}\\ \text{gapped/internal sector: } m > 1/r_{\text{test}} \text{ or explicit decoupling.}\end{array}}$$

The status of $D^{(2)}$ remains open. B4q uses only $D^{(1)}$ because the leading structure function is $\delta^{ij}(N\partial_j M - M\partial_j N)$. The next spatial-metric correction

$$\boxed{h^{ij} = \delta^{ij} - \gamma^{ij} + O(\gamma^2)}$$

requires the corresponding $D^{(2)}$ and higher metric-dependence terms. B4q's conditional pass can be revised later if the actual $D^{(2)}$ calculation cannot extend $D^{(1)}$ into the spatial-diffeomorphism generator needed by the HDA structure function.

The B4q pass/fail ledger is:

$$\boxed{\begin{array}{lcl}\text{B4 linear closure} & : & \text{conditional import,}\\ H^{(2)}_{\text{kin}} \text{ ADM-candidate form} & : & \text{conditional,}\\ \{H^{(1)}, H^{(2)}_{\text{kin}}\} & : & \text{computed,}\\ \text{ADM structure function} & : & \text{matched iff } \lambda = 1/2,\\ \text{trace anomaly} & : & 2(1-2\lambda),\\ H^{(2)}_{\text{curv}} & : & \text{open for B5,}\\ \Delta^{(2)} & : & \text{open and must not contribute } A^{(1)}_{\Delta},\\ \text{medium derivation of } H^{(2)} & : & \text{open,}\\ \text{full nonlinear HDA} & : & \text{open.}\end{array}}$$

B4q therefore proves only that the ADM-candidate kinetic completion passes the first nontrivial bracket discriminator iff $\lambda = 1/2$. It does not prove that the microscopic medium derives $H^{(2)}$, that $\lambda = 1/2$ emerges without tuning, that the curvature operator is exactly EH, that $\alpha = 1$, that all $\Delta^{(2)}$ terms vanish, that the full nonlinear HDA closes, or that two TT modes and matter-coupled preservation are secured.

For the B4q conditional pass to become a medium-derived pass, the bridge must produce the quadratic Hamiltonian structure. The medium-derivation targets are:

$$\boxed{\begin{array}{lcl}\text{kinetic sector} & : & S^{\text{bridge}}_{\mathcal{R}} \to \pi^{ij}\pi_{ij} - \frac{1}{2}\pi^2 \text{ automatically,}\\ \text{curvature sector} & : & S^{\text{bridge}}_{\mathcal{R}} \to \text{EH-compatible } H^{(2)}_{\text{curv}},\\ \Delta \text{ sector} & : & \Delta^{(2)} \text{ classified with no local HDA anomaly.}\end{array}}$$

If the medium gives $\lambda \neq 1/2$, this is a Root B failure and may not be repaired by rescaling or manual tuning. If the curvature sector gives a leading $f(R)$, $R_{ij}R^{ij}$, Weyl-type, or preferred-frame operator at the same local order, B5 must mark it as a fixed-point failure unless it is explicitly suppressed or outside the local tested regime. The word "suppressed" is not a pass without a scale.

The next artifact is B5:

$$\boxed{\texttt{Root\_B\_B5\_EH\_Fixed\_Point\_Filter.md}.}$$

B5 imports only

$$\boxed{\lambda = \frac{1}{2}, \qquad A_\lambda^{(1)} = 0}$$

as conditional bracket-closure results for the ADM-candidate kinetic sector. It must keep

$$\boxed{\Delta^{(2)} \text{ open}, \qquad H^{(2)} \text{ not yet medium-derived.}}$$

If B5 cannot show $\alpha = 1$ and safe $\Delta_{\mathcal{M}}$, Root B remains open even though B4q passed conditionally.

Root B B5 EH fixed-point filter. B5 uses the B4q result as a conditional input and asks whether the local two-derivative record Hamiltonian is the EH fixed point rather than a nearby non-EH, preferred-frame, scalar/vector, massive-tensor, or second-metric theory. Its status is deliberately a filter status:

> B5 imports $\lambda = 1/2$ only as a conditional bracket-closure result. It does not import $\alpha = 1$, the EH curvature operator, or $\Delta_{\mathcal{M}} = 0$. B5 defines the fixed-point filter; it does not yet pass that filter.

The central B5 Hamiltonian form to audit is

$$\boxed{\mathcal{H}_{\mathrm{local}} = \frac{1}{\sqrt{h}}\left(\pi^{ij}\pi_{ij} - \frac{1}{2}\pi^2\right) - \sqrt{h}\,(\alpha R - 2\Lambda) + \Delta_{\mathcal{M}}.}$$

Here $\lambda = 1/2$ is inherited from B4q. B5 tests α, the curvature operator, the remainder $\Delta_{\mathcal{M}}$, and the equality of the gravitational normalization used in static, near-zone, radiative, and matter-coupling contexts.

B5 imports only:

$$\boxed{\begin{array}{lcl} \lambda = 1/2 & : & \text{conditional bracket-closure result,} \\ A_\lambda^{(1)}[N, M] = 0 & : & \text{ADM-candidate kinetic result,} \\ \Delta^{(2)} & : & \text{open and to be classified,} \\ H^{(2)} \text{ from the medium} & : & \text{not derived,} \\ D^{(2)} \text{ continuation} & : & \text{open.} \end{array}}$$

Forbidden imports are:

> Do not import Einstein equations; do not import $\alpha = 1$; do not import EH curvature as a premise; do not import $\Delta_{\mathcal{M}} = 0$ without classification; do not use MOND, cosmology, quantum, or strong-field branches to repair a local fixed-point failure; and do not pass a local core that requires a second metric.

B5 alpha normalization gate. The coefficient α is read from the local curvature response. The extraction rule is:

$$\boxed{\begin{array}{ll} 1. & \mathcal{H}_{\mathrm{local}} = \mathcal{H}_{\mathrm{kin}} + \mathcal{H}_{\mathrm{curv}} + \Delta_{\mathcal{M}}, \\ 2. & \text{isolate the coefficient of } -\sqrt{h}\,R \text{ in } \mathcal{H}_{\mathrm{curv}}, \\ 3. & \text{call that coefficient } \alpha_{\mathrm{raw}}, \\ 4. & \text{divide by the single measured-}G_{\mathrm{record}} \text{ normalization,} \\ 5. & \text{call the result } \alpha_{\mathrm{norm}}, \\ 6. & \text{pass only if } |\alpha_{\mathrm{norm}} - 1| < 10^{-5}. \end{array}}$$

A single universal normalization is allowed only if

$$\boxed{G_{\text{static}} = G_{\text{dynamic near}} = G_{\text{radiation}} = G_{\text{matter}} = G_{\text{record}}.}$$

Separate G-values for static, orbital, radiative, or matter coupling sectors are a B5 failure. PSR-style orbital decay tests are especially sensitive to $G_{\text{radiation}}/G_{\text{dynamic near}}$.

B5 curvature gate. The local two-derivative curvature target is

$$\boxed{\mathcal{H}_{\text{curv}} \to -\sqrt{h}\, R}$$

with the B1–B4q sign convention $H^{(1)} = -R^{(1)}$ and

$$\boxed{R^{(1)} = \partial^i \partial^j \gamma_{ij} - \partial^2 \gamma.}$$

At quadratic order the EH target, up to boundary terms and integration by parts, is

$$\boxed{\begin{aligned} \mathcal{H}^{(2)}_{\text{curv},EH} = {} & \frac{1}{4}(\partial_k \gamma_{ij})(\partial^k \gamma^{ij}) - \frac{1}{4}(\partial_k \gamma)(\partial^k \gamma) \\ & - \frac{1}{2}(\partial_i \gamma^{ij})(\partial^k \gamma_{kj}) + \frac{1}{2}(\partial_i \gamma^{ij})(\partial_j \gamma). \end{aligned}}$$

The execution file may use an integration-by-parts equivalent expression, but it must state the equivalence. A leading $f(R)$, $R_{ij}R^{ij}$, Weyl-type, preferred-frame, nonlocal, or second-metric curvature term is not the EH fixed point unless it is boundary/gauge/constraint-proportional or explicitly suppressed outside the local tested regime.

B5 $\Delta_{\mathcal{M}}$ classifier. Every extra local term must be assigned to exactly one class:

absent	:	passes,
boundary	:	local pass; charge audit later,
gauge	:	pass only if bracket-verified,
constraint-proportional	:	pass only if preserved,
internal/gapped	:	pass only with a gap and range,
high-energy suppressed	:	pass only with scale,
large-scale-only	:	not a local-core repair,
anomaly	:	fail.

The words "small", "irrelevant", "negligible", and "suppressed" are not passes without a mechanism and a scale. Working thresholds are:

PPN/local curvature anomaly	:	$< 10^{-5}$ or absent,
GW cone-changing operator	:	$< 10^{-15}$ or absent,
binary-pulsar scalar/vector leak	:	$< 10^{-3}$ at MVP,
gapped/internal mode	:	$m > 1/r_{\text{test}}$ or decoupled,
high-energy operator	:	$(E/\Lambda_{\text{UV}})^p$ with p and Λ_{UV}.

Useful mass-range bookkeeping examples are:

$$\boxed{\begin{array}{lll} r_{\text{test}} = 1\,\text{mm} & \Rightarrow & m_{\text{gap}} \gtrsim 2 \times 10^{-4}\,\text{eV}, \\ r_{\text{test}} = 1\,\text{AU} & \Rightarrow & m_{\text{gap}} \gtrsim 10^{-18}\,\text{eV}, \\ r_{\text{test}} = 10\,\text{km} & \Rightarrow & m_{\text{gap}} \gtrsim 2 \times 10^{-11}\,\text{eV}, \\ r_{\text{test}} = 10^9\,\text{m} & \Rightarrow & m_{\text{gap}} \gtrsim 2 \times 10^{-16}\,\text{eV}. \end{array}}$$

The actual execution file must state the actual r_{test} used for each term.

The main $\Delta_{\mathcal{M}}$ sources to check are:

$\eta R,\ \eta\gamma,\ \eta\partial^2\gamma$	:	amplitude integration; gapped/internal or absent,
$q_{ij}R^{ij},\ q_{ij}\gamma^{ij}$	:	strain integration; gapped/internal or TT-route revision,
$\partial_\mu\vartheta\,\partial_\nu\vartheta\,\gamma^{\mu\nu}$	:	gauge/clock/decoupled or Root C audit,
$u^iu^jR_{ij}$	:	preferred-frame anomaly unless gauge/internal,
$R^2,\ R_{ij}R^{ij},\ C^2$	:	high-energy suppressed or fail,
$R\,F(\Box)R$	:	requires explicit expansion scale,
$R[g_2]$	:	second metric; fail in the local MVP,
$m_g^2(\gamma_{ij}\gamma^{ij}-\gamma^2)$	:	fail unless cosmological-scale and local PPN protected.

A Fierz–Pauli-like mass term is not harmless merely because its wavelength is large. It must also avoid local PPN discontinuities and preserve the local two-TT-mode target; any screening mechanism must itself be proven, not named.

As a worked B5 example, a term

$$\boxed{c_u\,u^iu^jR_{ij}}$$

is classified as an anomaly unless u^i is gauge/internal and non-observable. Its scale is c_u plus any explicit suppression scale; its windows are local preferred-frame and GW-cone tests; its failure consequence is preferred-frame local curvature and possible cone splitting. It passes only if $c_u = 0$, if the whole term is boundary/gauge/internal by derivation, or if it is suppressed below the relevant local threshold.

B5 observation links. The B5 filter blocks local structures that would make the observation gates impossible:

Newtonian	:	$\Delta G/G < 10^{-5}$	one G_{record},
Shapiro/light bending	:	$\gamma_{\text{PPN}} - 1 < 2.3\times10^{-5}$	EH curvature + one cone,
perihelion/LLR	:	$\beta_{\text{PPN}} - 1 < 8\times10^{-5}$	nonlinear EH weak field,
frame dragging	:	preferred-frame PPN $< 10^{-4}$	no vector/tensor leak,
GW speed	:	$\lvert c_{\text{GW}} - c\rvert/c < 10^{-15}$	no cone-changing term,
binary pulsars	:	dipole excess $< 10^{-3}$ at MVP	no scalar/vector leak,
compact binaries	:	no leading non-tensor channel	correct local tensor baseline.

B5 does not prove these observations. It rejects local cores that would make matching them impossible.

B5 execution algorithm. The executable B5 procedure is:

1. read the candidate $\mathcal{H}_{\text{local}}$,
2. $\mathcal{H}_{\text{local}} = \mathcal{H}_{\text{kin}} + \mathcal{H}_{\text{curv}} + \Delta_{\mathcal{M}}$,
3. read λ from $\mathcal{H}_{\text{kin}} = \pi^{ij}\pi_{ij} - \lambda\pi^2 + \cdots$,
4. read α_{norm} from $\mathcal{H}_{\text{curv}} = -\sqrt{h}\,\alpha R + \cdots$,
5. list every non-EH curvature operator,
6. classify every $\Delta_{\mathcal{M}}$ term,
7. test $G_{\text{static}} = G_{\text{dynamic near}} = G_{\text{radiation}} = G_{\text{matter}}$,
8. output the B5 decision table.

The decision table must report λ, α, the curvature status, the complete $\Delta_{\mathcal{M}}$ table, $G_{\rm record}$ consistency, and B6 permission. B6 permission is yes iff

$$\boxed{\begin{array}{lcl} \alpha=1 & : & \text{conditional pass,} \\ \mathcal{H}_{\rm curv}=EH & : & \text{conditional pass,} \\ \Delta_{\mathcal{M}} & : & \text{no anomaly row,} \\ G_{\rm record} & : & \text{one universal value passes.} \end{array}}$$

Otherwise B6 may proceed only as a conditional template. Any B5 pass is twice conditional: B4q's $\lambda=1/2$ result must survive the $D^{(2)}$ continuation, and $S_{\mathcal{R}}^{\rm bridge}$ must produce $H^{(2)}$ automatically without retuning the candidate Hamiltonian by hand.

B5 pass/fail ledger. At the current stage:

$$\boxed{\begin{array}{lcl} \text{B5-1 import } \lambda=1/2 & : & \text{conditional pass from B4q,} \\ \text{B5-2 candidate Hamiltonian} & : & \text{open; medium derivation pending,} \\ \text{B5-3 } \alpha=1 & : & \text{open,} \\ \text{B5-4 EH curvature operator} & : & \text{open,} \\ \text{B5-5 } \Delta_{\mathcal{M}} \text{ classified} & : & \text{open,} \\ \text{B5-6 one record metric in all sectors} & : & \text{open,} \\ \text{B5-7 no window repair} & : & \text{active rule,} \\ \text{B5 fixed point passed} & : & \text{no.} \end{array}}$$

The next paragraph executes this filter on the minimal conditional candidate.

Root B B5 execution on the minimal EH candidate. B5 execution tests the filter machinery on

$$\boxed{\mathcal{H}_{\rm local}^{\rm min}=\frac{1}{\sqrt{h}}\left(\pi^{ij}\pi_{ij}-\frac{1}{2}\pi^2\right)-\sqrt{h}\,(R-2\Lambda)+\Delta_{\mathcal{M}}^{\rm local}.}$$

For the first local MVP,

$$\boxed{\Lambda=0, \qquad \Delta_{\mathcal{M}}^{\rm local}=0,}$$

so

$$\boxed{\mathcal{H}_{\rm local}^{\rm min}=\frac{1}{\sqrt{h}}\left(\pi^{ij}\pi_{ij}-\frac{1}{2}\pi^2\right)-\sqrt{h}\,R.}$$

Its short status is:

The minimal candidate passes the B5 filter, but the medium has not yet been shown to produce the minimal candidate. This is a candidate-level consistency check, not a medium-level derivation.

B5 execution on the minimal candidate is necessary but tautological: it confirms that the filter works on the EH candidate. The discriminating test begins only when a medium-derived $\mathcal{H}_{\rm local}$ replaces $\mathcal{H}_{\rm local}^{\rm min}$.

The operator decomposition is

$$\boxed{\begin{array}{lcl} \pi^{ij}\pi_{ij} & : & +1, \\ \pi^2 & : & -1/2, \\ -\sqrt{h}\,R & : & +1 \text{ relative to the EH target,} \\ 2\Lambda\sqrt{h} & : & 0 \text{ in the local MVP,} \\ \Delta_{\mathcal{M}}^{\rm local} & : & 0. \end{array}}$$

Thus no unclassified operator remains inside the minimal candidate. If $S_{\mathcal{R}}^{\text{bridge}}$ later produces an additional local two-derivative operator, this table must be reopened and the term must be classified.

The kinetic read-off is

$$\boxed{\mathcal{H}_{\text{kin}} = \pi^{ij}\pi_{ij} - \lambda\pi^2, \qquad \lambda = \frac{1}{2},}$$

which is a conditional pass because it uses the B4q-required value. It is not a new derivation of $\lambda = 1/2$.

The curvature read-off is

$$\boxed{\mathcal{H}_{\text{curv}} = -\sqrt{h}\,R = -\sqrt{h}\,\alpha R, \qquad \alpha_{\text{raw}} = 1, \qquad \alpha_{\text{norm}} = 1,}$$

with

$$\boxed{|\alpha_{\text{norm}} - 1| = 0 < 10^{-5}.}$$

This is a conditional pass for the minimal candidate. It does not prove that $S_{\mathcal{R}}^{\text{bridge}}$ produces $\alpha_{\text{raw}} = 1$ automatically.

The EH curvature check passes by candidate construction:

$$\boxed{\mathcal{H}_{\text{curv}}^{\min} = -\sqrt{h}\,R,}$$

whose weak-field expansion is the EH quadratic operator listed in the B5 filter above. The sign convention matches

$$R^{(1)} = \partial^i\partial^j\gamma_{ij} - \partial^2\gamma;$$

after integration by parts, the standard linearized EH curvature term gives that quadratic density. The medium still has to derive this operator rather than another two-derivative curvature operator.

In the minimal candidate all non-EH local curvature operators are absent:

$$\boxed{\begin{array}{lcl} f(R)\text{ beyond EH} & : & 0, \\ R^2,\ R_{ij}R^{ij},\ C^2 & : & 0, \\ u^iu^jR_{ij} & : & 0, \\ R\,F(\Box)R & : & 0, \\ R[g_2]\text{ or mixed second-metric curvature} & : & 0, \\ m_g^2(\gamma_{ij}\gamma^{ij} - \gamma^2) & : & 0. \end{array}}$$

If the medium derivation later produces any of these operators, B5 must be rerun with nonzero coefficients.

For the minimal candidate,

$$\boxed{\Delta_{\mathcal{M}}^{\text{local}} = 0.}$$

Hence the amplitude, strain, phase-gradient, preferred-frame, higher-curvature, nonlocal, second-metric, and mass-term sources are absent in the candidate. This does not mean derived absent from the medium. Promotion requires $S_{\mathcal{R}}^{\text{bridge}}$ to show that the coefficients are zero, boundary/gauge, constraint-proportional, gapped/internal with a scale, or high-energy suppressed with a scale.

The preferred-frame worked example is

$$\boxed{c_u\,u^iu^jR_{ij}.}$$

In the minimal candidate $c_u = 0$, so the term is absent. If $c_u \neq 0$, it must be bounded:

$$\boxed{\begin{array}{lll} \text{GW cone safety} & : & |c_u|/|c_{EH}| < 10^{-15}, \\ \text{preferred-frame PPN} & : & |c_u|/|c_{EH}| < 10^{-7}, \\ \text{strongest local pass} & : & c_u = 0 \text{ by derivation.} \end{array}}$$

A nonzero and unbounded c_u is a B5 failure. Calling it small without a bound is also a failure.

B5 also imposes the one-G normalization

$$\boxed{G_{\text{static}} = G_{\text{dynamic near}} = G_{\text{radiation}} = G_{\text{matter}} = G_{\text{record}}.}$$

This is a conditional pass in the minimal candidate, but it is a major fail-risk. The equality must be revisited in the matter-coupling, tensor normalization, and GW-emission calculations; binary-pulsar timing already constrains $G_{\text{radiation}}/G_{\text{dynamic near}}$ at roughly the 10^{-3} MVP level.

The B5 observation snapshot for the minimal candidate is:

$$\boxed{\begin{array}{llll} \text{Newtonian} & : & \Delta G/G < 10^{-5} & \text{conditional,} \\ \text{Shapiro/light bending} & : & \gamma_{\text{PPN}} - 1 < 2.3 \times 10^{-5} & \text{needs matter/light coupling,} \\ \text{perihelion/LLR} & : & \beta_{\text{PPN}} - 1 < 8 \times 10^{-5} & \text{needs nonlinear expansion,} \\ \text{frame dragging} & : & \alpha_1 < 10^{-4},\ \alpha_2 < 10^{-7} & \text{conditional,} \\ \text{GW speed} & : & |c_{\text{GW}} - c|/c < 10^{-15} & \text{B6 still required,} \\ \text{binary pulsars} & : & \dot{P}_b \text{ within } 10^{-3} \text{ at MVP} & \text{matter/emission still required.} \end{array}}$$

B5 blocks local fixed-point failures; it does not by itself match these observations.

The B5 decision table is:

row	result	conditional type	reason
λ	pass	derived conditional	B4q value; $D^{(2)}$ open
α	pass	candidate-imposed	$\alpha_{\text{raw}} = \alpha_{\text{norm}} = 1$
$\mathcal{H}_{\text{curv}}$	pass	candidate-imposed	$-\sqrt{h}\,R$
non-EH curvature	pass	candidate stability	all coefficients zero
$\Delta_{\mathcal{M}}$	pass	candidate-imposed	$\Delta_{\mathcal{M}}^{\text{local}} = 0$
G_{record}	pass	imposed normalization	Root C/GW must preserve it
B5 filter	pass	candidate-level only	medium derivation open.

This conditional pass does not equal a Root B proof.

B6 permission is granted only as a conditional template:

$$\boxed{\begin{array}{lll} \alpha = 1 & : & \text{conditional pass,} \\ \mathcal{H}_{\text{curv}} = EH & : & \text{conditional pass,} \\ \Delta_{\mathcal{M}} & : & \text{no anomaly row in the candidate,} \\ G_{\text{record}} & : & \text{one imposed normalization.} \end{array}}$$

B6 may not treat this as a medium-derived hyperbolicity proof. The active conditional stack entering B6 is:

conditional	source	type	test
K1 kinetic completion	Root A temporal	conditional	derive from medium,
$S_{\mathcal{R}}^{\text{bridge}}$	Root B bridge	candidate	construct from medium,
$H^{(2)}$ ADM form	B4q	candidate-imposed	medium reproduces it,
$\lambda = 1/2$	B4q	derived conditional	$D^{(2)}$ and HDA continuation,
$\alpha = 1$	B5 execution	candidate-imposed	derive normalization,
$\mathcal{H}_{\text{curv}} = -\sqrt{h}\,R$	B5 execution	candidate-imposed	derive EH operator,
$\Delta_{\mathcal{M}} = 0$	B5 execution	candidate-imposed	integrate out remainders,
G_{record} single	B5 execution	imposed	matter + emission,
two TT modes	B6 entry	fail-risk	rank/sign/cone.

This is the anti-amnesia ledger for B6. No row may be upgraded to established unless B6 or a later derivation performs the listed test. The next artifact is

$$\boxed{\texttt{Root\_B\_B6\_Hyperbolicity\_Certificate.md}.}$$

Its task is to test whether the conditional local fixed-point baseline supports exactly two positive TT modes on the K1 record cone, with no scalar/vector/massive spin-2 or second-metric leakage.

Root B B6 hyperbolicity certificate. B6 tests whether the conditional B5 local EH-candidate baseline supports the right tensor sector. Its status is:

> The minimal TT candidate passes the B6 hyperbolicity certificate, but the medium has not yet been shown to produce this TT candidate. B6 is a candidate-level certificate, not a medium-derived proof of gravitational tensor modes.

B6 imports the B5 stack only as open debt. It may not upgrade any previous conditional row to established, may not relabel the conservative gapped q_{ij} sector as TT gravity, and may not call the TT candidate medium-derived.

The minimal TT action is

$$\boxed{S_{TT}^{(2)} = \frac{1}{2}\int dt\, d^3x\, \left[\dot{\gamma}_{ij}^{TT}\dot{\gamma}_{ij}^{TT} - (\partial_k \gamma_{ij}^{TT})(\partial_k \gamma_{ij}^{TT})\right],}$$

with

$$\boxed{\partial^i \gamma_{ij}^{TT} = 0, \qquad \delta^{ij}\gamma_{ij}^{TT} = 0, \qquad \gamma_{ij}^{TT} = \gamma_{ji}^{TT}.}$$

This is the standard minimal local tensor candidate consistent with the B5 EH baseline. It is not derived from $S_{\mathcal{R}}^{\text{bridge}}$ here.

For $k_i \neq 0$,

$$\boxed{P_{ij}(k) = \delta_{ij} - \frac{k_i k_j}{k^2}, \qquad \Lambda_{ij,kl}^{TT} = \frac{1}{2}(P_{ik}P_{jl} + P_{il}P_{jk}) - \frac{1}{2}P_{ij}P_{kl}.}$$

It satisfies

$$\boxed{k^i \Lambda_{ij,kl}^{TT} = 0, \qquad \delta^{ij}\Lambda_{ij,kl}^{TT} = 0, \qquad \Lambda^{TT}\Lambda^{TT} = \Lambda^{TT}.}$$

The rank count is

$$\boxed{\text{symmetric } \gamma_{ij} : 6 \quad \to \quad \text{TT components at } k \neq 0 : 2, \qquad \text{rank}\, K_{TT} = 2.}$$

The projector exists for the conditional record-geometry variable γ_{ij}^{TT}. It has not been derived as a projection of the conservative medium variables; in the conservative route q_{ij} is gapped and may not be declared to be the two TT gravitons.

Varying the action gives

$$\boxed{(\partial_t^2 - \partial^2)\gamma_{ij}^{TT} = 0.}$$

For $\gamma_{ij}^{TT} = \epsilon_{ij} e^{-i\omega t + ik\cdot x}$,

$$\boxed{\omega^2 = k^2, \qquad P_{TT}(\omega, k) = -\omega^2 + k^2.}$$

Thus

$$\boxed{P_{TT} = P_{\text{record}}, \qquad c_{TT} = c_{\text{record}} = 1}$$

in canonical MVP units. The cone-origin caution is important: in the current Root A spectrum the only medium mode with $\omega^2 = k^2$ is the gapless phase scalar ϑ. The candidate γ_{ij}^{TT} is required to share that cone, but it is not identical to ϑ. A later derivation must decide whether these are distinct fields or coupled/composite at higher order.

The canonical momentum and Hamiltonian density are

$$\boxed{\Pi_{TT}^{ij} = \dot{\gamma}_{ij}^{TT}, \qquad \mathcal{H}_{TT}^{(2)} = \frac{1}{2}\Pi_{TT}^{ij}\Pi_{TT}^{ij} + \frac{1}{2}(\partial_k \gamma_{ij}^{TT})(\partial_k \gamma_{ij}^{TT}).}$$

Both terms are positive squares. The sign convention is compatible with the B5 EH quadratic curvature target: the apparent coefficient shift between the B5 curvature density and the B6 TT Hamiltonian comes from the Legendre transform and restriction to the TT sector. Variation gives $\omega^2 = k^2$, fixing the sign.

The B6 leakage audit for the candidate is:

scalar trace γ	:	projected out,
longitudinal scalar σ	:	projected out,
vector V_i^T	:	projected out,
lapse n	:	no kinetic sector; inherited conditional pass,
shift n^i	:	no kinetic sector; inherited conditional pass,
massive spin-2 branch	:	absent,
second metric tensor	:	absent,
preferred-frame tensor	:	absent,
phase scalar ϑ	:	not in TT action; matter-coupling risk.

Projected out in the candidate does not mean absent in the medium-derived action. If the medium produces scalar/vector/tensor leakage before the TT projection, B6 must be rerun. The ϑ row is especially sensitive because ϑ is gapless; scalar matter coupling must remain below laboratory fifth-force, Solar-System PPN, and binary-pulsar dipole thresholds.

Forbidden local tensor terms include

$m_2^2 \gamma_{ij}^{TT}\gamma_{ij}^{TT}$	:	massive spin-2,
$\gamma_{ij}^{(2)}$	:	second TT tensor,
bimetric kinetic mixing	:	second metric leak,
private tensor cone	:	species/private propagation,
K^{ij} in the principal symbol	:	preferred tensor.

All have coefficient zero in the minimal candidate. A nonzero coefficient from the medium is a B6 failure unless it is outside the tested local sector with an explicit mechanism and bound.

TT-origin ledger. The possible TT-origin routes are:

γ_{ij}^{TT} as conditional record geometry	:	current candidate; allows template pass,
q_{ij}^{TT} from conservative Q	:	blocked because Q is gapped,
gapless Q route	:	deferred; input freeze and Root A rerun,
phase/strain composite	:	speculative; transversality not automatic,
second metric	:	forbidden in the local MVP.

The gapless Q route would require

$$\boxed{\begin{array}{l}\omega_Q^2 = k^2 + 4 \;\to\; \omega_Q^2 = k^2, \\ \text{remove or neutralize the } Q\text{-sector mass,} \\ \text{rerun the input freeze and Root A spectrum.}\end{array}}$$

It must also verify that η and ϑ remain stable and that no extra non-TT Q components become observable long-range modes. A tentative phase-composite route such as

$$\boxed{\gamma_{ij}^{TT} \sim \left(\partial_i\partial_j - \frac{1}{3}\delta_{ij}\partial^2\right)\vartheta}$$

is traceless but not automatically transverse and may be constrained/gauge rather than a propagating spin-2 field.

The actionable TT-origin resolution map is:

route	mechanism	revision	status
A	$\gamma_{ij}^{TT} \in \Phi_{\mathcal{M}}$	medium input revision	not attempted
B	composite from ϑ	Root A re-derivation	speculative
C	gapless Q	input freeze + Root A rerun	not attempted
D	higher-derivative emergent tensor	higher-order kinetic audit	not attempted.

Route A gives a direct spin-2 carrier but changes the field inventory. Route B uses the existing gapless cone but risks scalar/fifth-force leakage. Route C uses the tensor Q structure but may create extra long-range tensor components. Route D may preserve the conservative spectrum but risks higher-derivative ghosts. Until one route is executed, TT origin remains the leading Root B fail-risk.

The B6 observation snapshot is:

GW speed	:	$\lvert c_{\rm GW} - c\rvert/c < 10^{-15}$	conditional pass,
GW polarizations	:	non-tensor fraction below bounds	conditional,
no breathing/longitudinal	:	scalar mode below detector bounds	open with matter,
binary-pulsar damping	:	$\dot{P}_b$ within 10^{-3} at MVP	emission still required,
preferred-frame	:	$\alpha_1 < 10^{-4},\ \alpha_2 < 10^{-7}$	conditional.

B6 supplies the propagation and mode-count side of the tensor channel. It does not prove GW emission.

The B6 pass/fail ledger is:

TT rank	pass	candidate-imposed	low priority,
kinetic sign	pass	candidate-imposed	low,
gradient sign	pass	candidate-imposed	low,
principal cone	pass	K1-conditional	medium,
scalar/vector leakage	pass in candidate	projection-dependent	high,
massive spin-2	pass in candidate	coefficient zero	medium,
second metric	pass in candidate	absent	medium,
$G_{\rm record}$	inherited conditional	imposed normalization	high,
TT origin	open/fail-risk	not derived	high,
B6 certificate	conditional pass	medium derivation open	high.

B6 proves only that the minimal TT candidate has rank two, positive kinetic and gradient signs, the K1 principal cone, no candidate-level leakage, and a baseline for the GW-emission memo. It does not prove that the medium produces γ_{ij}^{TT}, that the gapped q_{ij} is a graviton, that tensor radiation is emitted with the correct strength, or that binary-pulsar damping is correct.

Root B three pillars after B6. Root B closure requires more than the B6 local tensor certificate:

local-core hyperbolicity, B1–B6	:	conditional candidate pass; no medium derivation,
matter-coupling interface	:	open; one $G_{\rm record}$ and no fifth force,
GW emission/radiation reaction	:	open; quadrupole flux and binary-pulsar damping.

B6 closes only the first pillar at candidate level. Root B is not near final until all three pillars pass with the same record metric and normalization.

If B6 is accepted, the next natural artifact is

`Root_B_Conditional_Local_Core_Summary.md`.

The summary must explicitly state that Root B has not passed: it has only a candidate-level local-core hyperbolicity chain through B6; the medium has not produced the TT origin; and matter coupling plus GW emission remain open. It must avoid language such as "near final", "mostly complete", or "Root B essentially passed."

Root B conditional local-core summary. The post-B6 summary is an anti-amnesia import gate for all later Root B, Root C and GW-emission work. Its short verdict is

B1–B6 local-core chain	conditional candidate pass,
medium-derived local gravity mechanism	not proven,
ADM-like canonical setup	conditional,
$\lambda = 1/2$ HDA kinetic value	conditional bracket result,
EH fixed point	minimal-candidate pass only,
two TT modes	minimal-TT-candidate pass only,
TT origin from medium	open; leading fail-risk,
matter-coupling pillar	open,
GW-emission pillar	open,
Root B status	not passed.

Thus Root C and GW artifacts may proceed only as conditional templates.

Position in the program. Root A supplies the conditional record-carrier inputs: K1 cone and spectrum, temporal kinetic completion, and the candidate medium spectrum. Root B uses those inputs in three pillars:

- local-core canonical/hyperbolic chain: conditional candidate pass; seven audited files are done including this summary;
- matter-coupling interface: open; about five to eight artifacts remain;
- GW emission and radiation reaction: open; about three to five artifacts remain.

Root C supplies the total-energy and matter-coupling interface, including one G_{record} and the ϑ/fifth-force audit. The cross-cutting files are:

- `MVP_Input_Freeze.md`;
- `MVP_Local_Core_Proof_Tasks.md`;
- `Observation_Matching_Table.md`.

The conditional stack is roughly four Root A inherited items, six Root B local-core items, and two Root C interface items. Consequently Root B cannot be final until its inherited Root A conditions and its Root C interface conditions are also resolved.

Non-upgrade rule. No later file may use phrases such as "Root B has passed", "Root B is near final", "Root B is mostly complete", "the TT modes are derived", "the EH action is derived", $\Delta_{\mathcal{M}} = 0$ is derived, or one G_{record} is derived. Allowed wording is:

> Root B has a conditional local-core template through B6. The template is suitable for conditional Root C/GW interface work. The medium-derivation debt remains open. TT origin remains the leading fail-risk.

Any stronger statement requires a separate medium-derivation calculation.

B1–B6 chain ledger. The ledger is:

- B1, presymplectic extraction, builds on the bridge candidate: conditional pass; open item is medium derivation of the no-dot rule.
- B2, embedding null test, builds on B1: partial conditional pass; open items are nonlinear embedding and scalar-sector risk.
- B3, constraint generators, builds on B1–B2: conditional pass; open item is deriving the no-algebraic lapse/shift rule from the medium.
- B4, linear bracket closure, builds on B1–B3: conditional pass; open item is the nontrivial HDA structure.
- B4q, quadratic HDA, builds on B4: conditional pass only if $\lambda = 1/2$; open items are $D^{(2)}$, $H^{(2)}$, and full HDA.
- B5, fixed-point filter, builds on B4q: filter defined; open item is execution on the actual medium-derived candidate.

- B5 execution builds on the B5 filter: minimal-EH-candidate pass; open items are medium derivations of α, H_{curv}, and $\Delta_{\mathcal{M}}$.
- B6 hyperbolicity builds on B5 execution: minimal-TT-candidate pass; open item is TT origin from the medium.

B1–B6 form a coherent conditional local-core template. They do not form a medium-derived Root B proof.

What the template achieves. Conditionally, the chain provides an ADM-like phase-space form, non-kinetic lapse and shift, candidate Hamiltonian and momentum constraints, linear bracket consistency, the B4q discriminator $\lambda = 1/2$, a minimal EH fixed-point candidate with $\alpha = 1$ and $\Delta_{\mathcal{M}} = 0$, a minimal TT candidate with two positive luminal tensor modes, and explicit stop conditions for later medium derivation. This is enough to build conditional interface files. It is not enough to claim the gravitational mechanism has been proven.

What the template does not prove. It does not prove that the medium produces $S_{\mathcal{R}}^{\mathrm{bridge}}$, the ADM quadratic Hamiltonian, $\lambda = 1/2$ without candidate tuning, $\alpha = 1$ without candidate tuning, $H_{\mathrm{curv}} = -\sqrt{h}\,R$, $\Delta_{\mathcal{M}} = 0$ after internal-sector integration, one G_{record} in matter and radiation, a medium origin for γ_{ij}^{TT}, harmless ϑ matter coupling, correct binary-pulsar damping, GW emission, or full nonlinear stability.

Cumulative conditional stack. The blocking rows and their resolving artifacts are:

- K1 kinetic completion: `Root_A_K1_Derivation_From_Medium.md`.
- $S_{\mathcal{R}}^{\mathrm{bridge}}$ exists: `Root_B_Bridge_Action_Construction.md`.
- No lapse/shift kinetic sector: `Root_B_NoDot_Rule_Derivation.md`.
- No algebraic lapse/shift mass: `Root_B_NoAlgebraic_LapseShift_Derivation.md`.
- $\lambda = 1/2$: `Root_B_B4q_Extended_to_D2.md`.
- $\alpha = 1$: `Root_B_Alpha_Normalization_Derivation.md`.
- $H_{\mathrm{curv}} = -\sqrt{h}\,R$: `Root_B_Curvature_Operator_Derivation.md`.
- $\Delta_{\mathcal{M}} = 0$: `Root_B_DeltaM_Remainder_Audit.md`.
- One G_{record}: `Root_C_Conditional_Matter_Interface.md` plus the GW bridge.
- ϑ harmlessness: `Root_C_Vartheta_Matter_Coupling_Audit.md`.
- TT origin: `Root_B_TT_Origin_Derivation_Map.md`.

The HDA extension through $D^{(2)}$ is deferrable for the narrow MVP but blocking for a final nonlinear Root B proof.

TT-origin decision point. The leading risk is not the candidate TT action, which passes. The leading risk is the question

$$\boxed{\text{where does } \gamma_{ij}^{TT} \text{ come from in the medium?}}$$

The recommended order of attack is:

1. relational shear projection: keep the conservative spectrum and test whether the common record oscillation plus a local Q/shear response can produce detector-visible and source-compatible TT response;

2. revised gapless Q: reuse the existing tensor variable, but redo the input freeze and Root A spectrum;

3. direct spin-2 field: clean carrier, but it revises the medium input;

4. composite from ϑ: uses the existing gapless cone, but transversality and gauge status are not automatic;

5. higher-derivative emergence: delayed tensor route, but with ghost risk.

The relational shear route is a new conservative pre-test introduced by the TT-origin map. It must be back-propagated into this summary because it was not part of the first four-route enumeration. It does not upgrade the conservative branch to a proof: it only asks whether the missing tensor response might be an emergent detector/source projection of a common record carrier. Until one route is executed, TT origin is unresolved. This is the main reason Root B cannot be called passed.

Import rules and enforcement. Root C and GW files may import the conditional record-metric baseline, conditional ADM-like constraints, conditional EH local-core candidate, conditional TT propagation candidate, the K1 candidate shared cone, and the B6 tensor rank/sign/cone result. They may not import a medium-derived EH action, medium-derived TT origin, medium-derived $\Delta_{\mathcal{M}} = 0$, medium-derived one G_{record}, or a completed Root B proof.

Every downstream file that imports the local-core template must begin its verdict section with:

> Root B status per `Root_B_Conditional_Local_Core_Summary.md`: not passed; local-core template conditional only; all open debt inherited.

If a downstream file violates this import rule, the file is non-compliant, its result may not enter the proof chain, and it must be revised before it is cited. This is a hard constraint, not a style preference.

Summary stop conditions. This summary is invalidated or must be revised if any triggering computation finds a stop condition:

- $S_{\mathcal{R}}^{\text{bridge}}$ cannot be constructed: `Root_B_Bridge_Action_Construction.md`.
- Lapse/shift kinetic terms appear: `Root_B_NoDot_Rule_Derivation.md`.
- Algebraic lapse/shift terms solve for multipliers: `Root_B_NoAlgebraic_LapseShift_Derivation.md`.
- $\lambda \neq 1/2$ after HDA continuation: `Root_B_B4q_Extended_to_D2.md`.
- $\alpha \neq 1$ after one normalization: `Root_B_Alpha_Normalization_Derivation.md`.
- Non-EH curvature survives locally: `Root_B_Curvature_Operator_Derivation.md`.
- $\Delta_{\mathcal{M}}$ has an unsuppressed local anomaly: `Root_B_DeltaM_Remainder_Audit.md`.
- TT origin requires relabeling gapped q_{ij}: `Root_B_TT_Origin_Derivation_Map.md`.

- Matter coupling creates a fifth force or scalar/vector radiation: `Root_C_Vartheta_Matter_Coupling_Audit.md`.
- GW emission needs a non-tensor channel: `GW_Emission_Conditional_Tensor_Bridge.md`.

Decision and next artifact. Later work may proceed only as conditional interface/execution work, not as a derived Root B input, completed gravity mechanism proof, or publication-level local-core proof. The next path depends on priority:

matter observations	:	`Root_C_Conditional_Matter_Interface.md`,
binary-pulsar/GW anchor	:	`GW_Emission_Conditional_Tensor_Bridge.md`,
debt reduction	:	`Root_B_TT_Origin_Derivation_Map.md`.

If undecided, the conservative default is

`Root_B_TT_Origin_Derivation_Map.md`

because it attacks the leading Root B fail-risk before more conditional observation-side structure is added.

Root B TT-origin derivation map. The TT-origin map is a debt-reduction artifact after the conditional local-core summary. Its purpose is to turn the B6 leading fail-risk–the missing medium origin of the two TT tensor modes–into an explicit route-by-route derivation agenda. Its short verdict is:

Root B status	not passed; all B1–B6 debt inherited,
conservative Root A branch	no medium-derived TT origin,
minimal B6 TT action	candidate-level pass only,
gapped Q as graviton	blocked,
ϑ-only tensor route	likely insufficient,
relational shear route	no-direct-readout consistency pass,
direct spin-2 route	possible but revises medium input,
splash/scattering route	next measurement-response route,
gapless-Q route	carrier-revision fallback,
higher-derivative route	last resort; ghost-risk.

The map is not a proof of TT origin. It is a routing document: it defines what must be attempted next and what cannot be silently called a graviton.

Minimal TT-origin requirements. A successful TT-origin derivation must provide either a medium-side carrier T_{ij}^{TT} or a medium-side shear projection Π_{shear} that produces a detector-visible and source-compatible TT response. The required checks are:

spin/rank	:	exactly two observable TT polarizations,
principal cone	:	$\omega^2 = k^2$ on the K1 record cone,
mass	:	no unsuppressed local mass term,
sign	:	positive kinetic and gradient signs,
constraint status	:	scalar/vector parts constrained, gauge, or gapped,
metric role	:	same record geometry for rods, clocks, matter, constraints,
source compatibility	:	coupling to total energy/momentum,
radiation compatibility	:	binary systems lose tensor flux, not scalar/vector flux.

B6 already passes rank, sign and cone at the candidate-action level. The missing step is identifying the actual medium carrier or proving that a common carrier plus a shear projector is physically equivalent for propagation, detection and radiation reaction.

Conservative branch audit. The frozen conservative spectrum is:

$$\boxed{\omega_\eta^2 = k^2 + 1, \qquad \omega_\vartheta^2 = k^2, \qquad \omega_Q^2 = k^2 + 4.}$$

Thus the conservative branch has one gapless scalar and no gapless tensor. The immediate TT-origin audit is:

$$\boxed{\begin{array}{lcl} \eta & : & \text{scalar and gapped,} \\ \vartheta & : & \text{scalar; one field does not generically give two TT polarizations,} \\ q_{ij} & : & \text{tensor-like but gapped,} \\ q_{ij} \text{ as local projector} & : & \text{possible pre-test, not a carrier,} \\ \gamma_{ij}^{TT} & : & \text{correct candidate variable, not medium-derived.} \end{array}}$$

Therefore conservative Root A plus the B6 minimal TT candidate gives a useful template, but not a medium-derived TT origin. This blocks a Root B pass.

Route matrix. The ordered route map is:

- P, relational shear projection: common record oscillation plus local Q/shear response. Minimal execution says the common carrier cannot be read out directly. This strengthens the no-direct-measurement core.
- S, splash/scattering response: the common wave strikes the local matter/gravity field and creates a measurable disturbance at the same frequency. This is now the next measurement-response route.
- C, revised gapless Q: input freeze and Root A rerun. This remains the first carrier-revision fallback if the splash route fails.
- A, direct spin-2 field: medium input revision. First test: can the new field avoid a second metric?
- B, composite from ϑ: Root A/B rerun. First test: can one scalar supply two polarizations?
- D, higher-derivative tensor emergence: higher-order kinetic audit. First test: ghosts or extra modes.

The original order was $P \to C \to A \to B \to D$. The minimal Route P execution has now shown that passive direct readout of the common carrier does not work. In RMG this is a core-consistency result: common medium pressure or rhythm changes should not be locally visible directly, because rods, clocks, masses and detectors co-vary with the same medium state. The next route is therefore S, the active splash/scattering response. Route C remains the first carrier-revision fallback. Route D is last because higher derivatives can hide ghosts and extra polarizations.

Route P: relational shear projection. Route P formalizes the intuition that local oscillations may start as distinguishable sectors, then mix into one collective record oscillation at large distance, while the detector sees only the differential shear projection of that common oscillation. Its first-audit carrier specification is:

$$\boxed{\text{carrier}_P = \vartheta, \qquad \omega_\vartheta^2 = k^2.}$$

The Q_{ij} sector is not a propagating graviton in this route. It is a local projector or susceptibility:

$$\boxed{\text{common record oscillation} + \text{local } Q/\text{shear susceptibility} \longrightarrow \text{detector-visible differential shear.}}$$

This inherits the central scalar-carrier challenge: a scalar common carrier must, through local projection, reproduce the observational data normally carried by two independent TT polarizations for each wave-vector k. If the carrier is not ϑ but a new collective record mode, Root A must identify that mode explicitly before Route P can pass.

Route P must distinguish:

$$\boxed{\begin{array}{lcl} \text{common carrier} & : & \text{long-range record oscillation on K1,} \\ \text{local projector} & : & \text{anisotropic/shear response of records or rods,} \\ \text{observable TT signal} & : & \text{two detector-visible shear patterns.} \end{array}}$$

It passes only if all of the following are true:

$$\boxed{\begin{array}{lcl} \text{projection rank} & : & \text{exactly two independent shear patterns,} \\ \text{common-mode removal} & : & \text{breathing mode cancels or is below bounds,} \\ \text{detector independence} & : & \text{not an L-shaped-interferometer artifact,} \\ \text{source side} & : & \text{binary energy loss maps to the same tensor channel,} \\ \text{network polarizations} & : & \text{non-tensor contamination below MVP bounds,} \\ \text{rod/clock universality} & : & \text{not apparatus-private.} \end{array}}$$

Detector independence means testing at least an L-shaped Michelson interferometer, a triangular three-arm interferometer, and a ring/spherical or equivalent omnidirectional response model. If only the L-shaped geometry makes the scalar carrier look TT-like, Route P fails. The working polarization-network anchor is that non-tensor contamination must be below the MVP $O(10^{-2})$ amplitude level until the observation table supplies the exact data-source row.

Route P may explain detector-visible shear only if the source-side test also passes. A detector-only partial pass is not allowed: detector strain and orbital energy loss must use the same channel.

Minimal Route P no-direct-readout result. The minimal relational ansatz is

$$\boxed{S_{ij}(k) = \chi_{ij}(k)\,\vartheta(k), \qquad r_A(k) = D_A^{ij}(x_A)\chi_{ij}(k)\vartheta(k) = F_A^P(k, x_A)\vartheta(k).}$$

Here $\chi_{ij}(k)$ is the universal record-shear susceptibility and $D_A^{ij}(x_A)$ is the apparatus tensor. If χ_{ij} is apparatus-private, Route P fails universality. If it is universal, the rank-one obstruction remains because $\vartheta(k)$ is the only carrier amplitude.

The detector-network correlation matrix makes the obstruction explicit:

$$\boxed{C_{AB}^P = \langle r_A r_B^* \rangle = F_A^P F_B^{P*}\,\langle|\vartheta|^2\rangle.}$$

This is an outer product of one response vector with itself, so

$$\boxed{\operatorname{rank} C^P \leq 1.}$$

By contrast, a two-polarization tensor signal has

$$\boxed{C_{AB}^{TT} = F_A^+ F_B^{+*}\langle|h_+|^2\rangle + F_A^\times F_B^{\times *}\langle|h_\times|^2\rangle + \text{cross terms,}}$$

which can have rank two when h_+ and $h_\times$ are independent. Changing detector geometry changes F^P_A; it does not create a second carrier amplitude. This is not a failure of the RMG core. It is what the core expects: a common medium change cannot be read out directly by a local apparatus whose time, length and mass standards are made from the same medium. A projector can produce a real measurement only if there is an active local response–a splash–or if the projector itself carries independent two-state radiative data.

The source-side scaling warns against treating the carrier as an ordinary directly sourced scalar charge. A massless scalar with non-universal scalar charge generically has dipole power relative to the GR tensor quadrupole of order

$$\boxed{\frac{P_{\text{scalar,dipole}}}{P_{\text{GR,tensor}}} \sim K\,(\Delta\alpha)^2 \left(\frac{c}{v}\right)^2,}$$

where $K = O(1)$ and $\Delta\alpha$ is the difference in scalar charge per unit mass. For binary-pulsar scales, $v/c \sim 10^{-3}$, hence

$$\boxed{\left(\frac{c}{v}\right)^2 \sim 10^6.}$$

The MVP binary-pulsar gate requires roughly

$$\boxed{\frac{P_{\text{extra}}}{P_{\text{GR,tensor}}} < 10^{-3}.}$$

Thus a natural non-universal scalar charge, $\Delta\alpha = O(1)$, misses the target by about

$$\boxed{10^6/10^{-3} = 10^9.}$$

Equivalently, the dipole channel would require

$$\boxed{(\Delta\alpha)^2 < 10^{-9}}$$

or exact cancellation at dipole order. Such a cancellation would have to be proved in Root C. More importantly, the splash route does not identify the observed record with direct scalar charge radiation.

Therefore the minimal Route P result is:

detector intuition	:	useful,
direct carrier readout	:	not locally visible,
core implication	:	no-direct-measurement principle strengthened,
ordinary scalar-charge warning	:	generic 10^9 mismatch,
downstream import	:	forbidden as direct TT input,
next route	:	Route S, splash/scattering response.

Route S: splash/scattering response. Route S is the corrected continuation of Route P. It does not claim that the common wave is directly measured. Instead:

$$\boxed{\text{incoming common medium wave} + \text{local matter/gravity field} \longrightarrow \text{measurable splash disturbance.}}$$

The measured disturbance should inherit the driving frequency of the incoming wave, just as a vibration can inherit the frequency of a rotating engine even when the rotation itself is not measured directly. The difference between Michelson–Morley and LIGO is then structural: Michelson–Morley searched for a direct local readout of the common medium state, while LIGO-like detection is modeled as an indirect response to a wave interacting with local matter/gravity.

Route S must prove:

no direct pressure readout	:	common state remains locally hidden,
local splash	:	matter/gravity field produces measurable disturbance,
same frequency	:	splash inherits the wave frequency,
two shear channels	:	response has the observed differential patterns,
source/detector unity	:	radiation and detection use one mechanism.

This route must be calculated before RMG commits to a carrier-revision route.

Route C: revised gapless Q. Route C changes the Q sector from

$$\boxed{\omega_Q^2 = k^2 + 4} \qquad \text{to} \qquad \boxed{\omega_{Q,TT}^2 = k^2}$$

for the TT projection only. This is not merely setting a mass to zero. A symmetric traceless Q_{ij} has five components:

$$\boxed{Q_{ij} = Q_{ij}^{TT} + (\partial_i V_j^T + \partial_j V_i^T) + \left(\partial_i\partial_j - \frac{1}{3}\delta_{ij}\partial^2\right)\phi_Q,}$$

with component count $2+2+1=5$. Route C must separately remove or suppress V_i^T and ϕ_Q. The permitted mechanisms are a derived gauge symmetry, consistent Lagrange-multiplier constraints, or a derived mass hierarchy for the non-TT pieces. If all five Q components become long-range, Route C fails by scalar/vector leakage.

Route A, B and D. Route A adds a primary spin-2-capable medium variable S_{ij}. It can pass only if the low-energy projection of S_{ij} is the record metric TT sector, not a private second metric. Possible mechanisms are direct identification with the record geometry, kinetic mixing whose massless eigenstate is γ_{ij}^{TT}, or a derived constraint. Each mechanism revises the medium input.

Route B tries a phase composite such as

$$\boxed{\gamma_{ij}^{\text{comp}} = \left(\partial_i\partial_j - \frac{1}{3}\delta_{ij}\partial^2\right)\vartheta.}$$

This is traceless but not automatically transverse. A single scalar Fourier amplitude normally cannot produce two independent tensor polarizations. Route B is therefore useful mostly as a no-go audit unless it is supplemented by a non-private tensor structure.

Route D tries higher-derivative tensor emergence. It is allowed only if all extra poles are auxiliary, constrained or positive-energy. The principal cone must remain K1, and no extra scalar, vector or tensor branch may survive. Otherwise it fails by Ostrogradsky or extra-mode leakage.

Decision tree. The execution order is:

1. Route P minimal form has been tested as a pre-check and gives the expected no-direct-readout result.

2. Test Route S: can the incoming common wave, by interacting with local matter/gravity, produce a measurable splash with the observed differential channels and the same frequency?

3. If Route S fails, test Route C by rerunning the input freeze and Root A spectrum with a protected gapless Q_{ij}^{TT} sector.
4. If Route C fails, test Route A. If Route A fails, run Route B as a no-go/composite audit. Route D is last.

The preferred next artifact is

$$\boxed{\texttt{Root\_B\_TT\_Origin\_Scattering\_Splash\_Response.md}.}$$

Observation-table impact. If Route P is attempted, `Observation_Matching_Table.md` must separate detector and source information. The required columns are:

detector tensor amplitude,
detector scalar/vector amplitude bound,
source tensor flux, especially binary pulsars,
source scalar/vector flux bound,
network polarization test result.

This prevents an interferometer-response success from being mistaken for a complete gravitational-radiation success.

Pass/fail status of the map. The current status is:

conservative branch TT origin	fail	no gapless tensor,
gapped Q as graviton	fail	$\omega_Q^2 = k^2 + 4$,
ϑ-only tensor	fail-risk	one scalar,
Route P	core-consistency pass	no direct local readout,
Route S	open	splash/scattering response,
minimal B6 TT action	conditional pass	candidate rank/sign/cone,
Route C	open	carrier-revision fallback,
Route A	open	larger medium revision,
Route B	open/likely no-go	scalar composite audit,
Route D	open/last resort	higher-derivative ghost risk,
Root B	not passed	TT origin unresolved.

Stop conditions. Stop a TT-origin route if it requires any of the following:

declaring gapped q_{ij} to be a massless graviton; claiming detector shear while binary energy loss is scalar; treating an L-shaped interferometer cancellation as a universal polarization proof; five long-range Q polarizations; a private tensor cone; a second metric; an unsuppressed scalar breathing mode; an unsuppressed vector mode; a higher-derivative ghost; matter coupling to a tensor other than the record metric; or using Root C/GW windows to hide the missing TT origin.

If all routes fail. If Routes P, C, A, B, D all fail, the current medium structure

$$\boxed{\Phi_{\mathcal{M}} = (\eta, \vartheta, Q_{ij})}$$

is insufficient to support local gravitational radiation in RMG. The allowed resolution is medium augmentation or an input revision. The forbidden resolution is hiding the failure in matter coupling, GW emission, MOND, cosmology or strong-field windows. At that point the program must either revise the medium input with a viable tensor/shear structure or mark the Root B tensor origin as failed.

Physical significance of Route P. In standard GR, gravitational radiation is carried by a propagating spin-2 mode, and detectors see TT polarizations because the propagating field itself has TT components. In most modified-gravity alternatives, the spin-2 mode is kept and scalar/vector modes are added around it.

Route P is different. In the conservative branch it assumes no propagating spin-2 carrier in the medium. The common long-range carrier is scalar-like, provisionally ϑ, and the detector registers TT-like polarizations through differential record/shear projection filtered by local Q-shear susceptibility. If this route passes, RMG is not merely a medium derivation of GR; it is a distinct gravity theory whose detector tensor patterns are projection effects rather than intrinsic medium polarizations.

The passive direct-readout route does not pass, and that strengthens the theory rather than weakening it. If common medium pressure were directly locally measurable, the core claim that pressure determines time, length, mass and detector response would be violated. Its rank is at most one in the passive detector-network amplitude space. Treating the common carrier as an ordinary non-universal scalar charge also gives a generic binary-pulsar mismatch of about 10^9. Therefore the next RMG-consistent task is not to pretend the common wave is directly seen, but to calculate the local splash/scattering response it produces when it meets matter's gravitational field.

Reviewer questions for the TT-origin map.

1. Is Route P a legitimate TT-origin route, or only a detector-response route?
2. Should Route C remain the first carrier-revision route if Route P fails?
3. Is the scalar-composite route treated strongly enough as a likely no-go?
4. Are the stop conditions strict enough to prevent relabeling gapped Q?
5. Does Route P need a worked detector-response model before it is counted?

The recommended answer is that Route P should be tested first because it preserves the conservative medium, but it cannot pass by detector response alone. It must also reproduce tensor polarization-network behavior and binary energy loss through the same record-shear projection.

Root C execution tasks. Root C begins with an inheritance table saying exactly which Root A/B outputs are used and with what status. It must verify universal matter action $S_m[g_{\rm rec}, \Psi_A]$, define the lapse/compression bridge with u varied at fixed h_{ij}, N^i and matter variables, and mark $N = e^{-u}$ or its replacement as derived from Root B or conditional in MVP-0. It must compute

$$\boxed{(N\sqrt{h})^{-1}\frac{\delta S_m}{\delta u} = \rho_E = T_{\mu\nu}n^\mu n^\nu}$$

for the included matter sectors; derive the far-tail monopole $Q_A^{(0)} = \eta E_A/c^2$; enumerate and eliminate species-private force terms; audit photons, binding energy, internal energy, composites and compact objects; define $\alpha_A = q_A/(E_A/c^2)$ and prove leading-order $\alpha_A = \alpha_B$; and perform the Nordtvedt/self-binding audit.

GW-emission execution tasks. The primary MVP anchor is PSR B1913+16. PSR J0737–3039 and PSR J1738+0333 are extension anchors. The emission task must derive the radiative TT action, the retarded wave-zone solution, the multipole hierarchy, the tensor flux P_{TT}, radiation reaction

$$\boxed{\frac{dE_{\mathrm{orbit}}}{dt} = -P_{TT}}$$

and angular-momentum flux where needed. It must build source masses from $m_A = E_A/c^2$, recheck the Root C no-dipole audit in the binary context, and produce a quantitative $\dot{P}_b$ anchor row. A correct $c_{\mathrm{GW}} = c$ propagation result does not by itself pass this emission gate.

Observation execution tasks. The MVP observation checklist contains only local anchors supported by the completed gates: Newtonian $1/r$ with $M = E_{\mathrm{total}}/c^2$; redshift/lapse universality; light bending and Shapiro with one cone and $\gamma = 1$; perihelion/ranging with $\lambda = 1/2 \to$ ADM closure $\to$ EH $\to \beta = 1$; no preferred-frame local flow; GW propagation with $c_{\mathrm{GW}} = c$ and two TT modes; no scalar dipole from Root C; and binary-pulsar tensor $\dot{P}_b$ from the GW-emission gate.

MVP pass, fail and soft-fail rules. The MVP passes only if all six blocks are derived at the stated MVP order, all six artifacts are completed, and each artifact receives an external reviewer check before the next artifact is treated as settled. The MVP fails if it leaves an uncontrolled private cone, species-private principal metric, non-closing ADM bracket, extra scalar/vector/massive tensor channel, lapse/shift kinetic leak, trace-only or species-label source, unequal G_{static} and $G_{\mathrm{radiation}}$, or any MOND/cosmology/strong-field patch for a local-core failure.

Borderline cases are soft failures until numerically controlled: small but nonzero cone splitting, tiny species-dependent tail charge or small extra radiative channel require an explicit bound and an observation scale. They may not be dismissed by phrases such as "approximately universal" or "effectively suppressed" without a calculation.

Execution order and fast track. The recommended order is

$$\boxed{\begin{array}{c}\text{MVP\_Input\_Freeze}\\ \Downarrow \\ \text{Root\_A\_MVP\_Spectrum}\\ \Downarrow \\ \text{Root\_B\_Linearized\_ADM}\\ \Downarrow \\ \text{Root\_C\_Source}\\ \Downarrow \\ \text{PSR\_B1913\_Emission}\\ \Downarrow \\ \text{MVP\_Observation\_Checklist}\end{array}}$$

The first physically blocking calculation is the Root A spectrum/one-cone test. A PSR B1913+16 emission template may be drafted earlier only after Root A passes, and only as a conditional numerical template; it is not a publishable result before Root B and Root C have delivered the tensor channel and total-energy source. Every suppression claim in every artifact must name a scale or bound.

17 Open Items

RMG-1 is not yet a closed theory. The open items group into the following programs.

Microscopic foundation. Derive $S_{\rm eff}^{\rm rec}[\mathcal{M}_0]$ from the stable base-medium phase; specify the input package $\mathcal{D}_{\mathcal{M}} = (\Phi_{\mathcal{M}}, S_{\mathcal{M}}, \mathcal{M}_0, \ell_*, \Omega_{\mathcal{M}}, \mathcal{G}_{\rm int}, \mathcal{O}_{\rm rec})$; test the candidate order parameter $\Phi_{\mathcal{M}} = \sqrt{n}e^{i\theta}\chi$ against Roots A–C; compute the stable-phase spectrum; integrate out the internal sector and verify the absence of background leaks; derive ℓ_* from the base-medium record dynamics; verify that the stable phase contains no negative-energy physical branch; carry at least one working toy instantiation through its quadratic Hessian, spectrum, projectors and record/internal split.

Root A. Derive one common external record carrier from the stable phase; compute $\mathrm{Spec}(\mathcal{L}_{\mathcal{M}_0})$ and prove one gapless record branch; derive phase locking of stable tails to ν_0; prove the phase/acoustic branch gives only the common cone, not a scalar graviton; prove every long-range stable record family uses the coefficients $C_t : C_0 : C_x = 1 : \rho : \rho^2$.

Root B. Derive record-embedding first-class ADM closure; construct $\Gamma_{\mathcal{R}}$ and the presymplectic form $\Omega_{\mathcal{R}}$; derive the embedding vector fields δ_ξ and the local generators D, H; derive the Noether split $J_\xi = C_\xi + dQ_\xi$; verify B1–B5 and prove the anomaly gate $A[N, M] = 0$; derive the construction bridge $\mathcal{D}_{\mathcal{M}} \to S_{\rm eff}^{\rm rec} \to S_{\mathcal{R}}$; exclude unsuppressed lapse/shift scalar-vector kinetic terms as a derived Dirac result rather than as a gauge assumption; compute brackets from $\Omega_{\mathcal{R}}$ with Dirac brackets where needed; derive the hypersurface-deformation algebra; derive the DeWitt coefficient $\lambda = 1/2$; prove $\Delta_{\mathcal{M}} = 0$ or suppressed with explicit structural or scale tests; give a linearized hyperbolicity certificate, match the principal symbol to the record cone, and exclude ghost/gradient instabilities; verify constraint preservation under evolution and matter-coupled Dirac closure.

Root C. Verify C1–C5 in the base-medium matter sector; prove $S_u = \rho_E$ for every stable record matter family with $\rho_E = T_{\mu\nu}n^\mu n^\nu$ in the ADM normal frame; prove metric-exhaustion of local u-dependence; prove the absence of explicit species-private u-coupling; prove $\sigma_{\rm species}^{\rm long,local} = 0$; derive the universal gravitational tail projection $q_A \propto E_A/c^2$ and the far-tail monopole $Q_A^{(0)} = \eta E_A/c^2$ in the regime $r \gg R_{\rm source}$ and $r \gg \ell_*$; verify that photons and traceless fields source through energy density rather than trace; handle quantum matter Hamiltonian expectation values and vacuum subtractions consistently with the cosmological term; audit compact-object sensitivities using $\alpha_A = q_A/(E_A/c^2)$ so that no leading scalar dipole loss appears in binary-pulsar systems; include self-gravitational binding energy in the strong-equivalence/Nordtvedt audit.

Factorization and protection. Derive $Q_{AB}^{\mu\nu} = G^{\mu\nu}K_{AB}$ from the record kinetic kernel; prove the stable record phase has only one observable principal tensor; derive non-common mode elimination; prove the observable tensor basis gate; eliminate unsuppressed $U^\mu U^\nu$ and flow/vorticity tensors; derive local flow decoupling and the preferred-frame PPN null target; prove no physical low-energy birefringence; derive one-cone clock universality; prove the failure matrix $B_{AB}^{\mu\nu}$ vanishes on physical principal modes.

Composite metric and tensor sector. Derive the composite map $g_{\rm rec}^{\mu\nu} = \mathcal{G}^{\mu\nu}[\Phi_{\mathcal{M}}]$; prove $\delta g_{\rm rec}$ contains a protected h_{ij}^{TT} sector; close the tensor kinetic-rank test (exactly two positive luminal TT modes and no extra tensor/scalar/vector radiation); prove amplitude and internal modes are gapped, constrained, gauge, confined or suppressed.

Radiation and detection. Separate propagation from emission. For propagation, derive scalar/vector constraints and tensor/shear radiation; derive lapse/shift no-kinetic structure from $\mathcal{M}$; prove first-class constraint closure removes scalar/vector radiation; prove pure common-mode breathing is locally self-calibrated and detector-null; derive the detector-visible TT/shear or defect projection for GW signals. For emission, derive tensor quadrupole energy loss, radiation reaction and compact-binary waveform phase from the same record metric and total-energy source; prove $G_{\mathrm{Newton}} = G_{\mathrm{radiation}} = G_{\mathrm{record}}$; derive P_{TT}, $dE_{\mathrm{orbit}}/dt = -P_{TT}$ and the angular-momentum flux for eccentric systems; build the leading chirp mass from $m_A = E_A/c^2$; run the PSR B1913+16-style binary-pulsar timing MVP; and exclude massive vector modes, massive spin-2 leakage, a second metric tensor mode and any hidden scalar/vector monopole or dipole energy-loss channel in the tested local sector.

Observational audit package. Map each local observable to a gate: Newtonian/orbital limit; gravitational redshift; light bending and Shapiro delay; perihelion, ranging and PPN β; frame dragging and preferred-frame null tests; binary-pulsar orbital decay; GW propagation; compact-binary waveform phase; and finite strong-field tests such as ISCO, shadow, ringdown, strong lensing and late-inspiral behavior. Separate GR-like local-core recovery from RMG-distinctive deviation-window observables. The execution sequence is: first build $MVP\_Local\_Core\_Proof\_Tasks$, then execute the Root A toy spectrum/one-cone test, the Root B linearized ADM closure audit, the Root C total-energy/no-dipole audit, and finally the PSR B1913+16-style numerical emission comparison.

Foundational and extension programs. Derive Operational Relational Covariance from microscopic $\mathcal{M}$; derive the record/internal projection $\Pi_{\mathcal{R}}$ from $\mathcal{M}$; derive the flow/vorticity gate and the MOND reentry function $\mathcal{F}$; derive the compact-object branch and the strong-field projection gate; derive the cosmological background state and the matter sector from $\mathcal{M}$; state the quantum-extension boundary, namely which quantum claims are outside the local-gravity MVP and which must be derived later from the base-medium record action.

18 Pass/Fail Gates

Gate 8 (Pass condition)**.** *The framework passes if $\mathcal{M}$ derives $S_{\mathrm{EH}} + S_{\mathrm{matter}}$ as the low-energy fixed point, with positive physical Hamiltonian, universal metric coupling, and exactly two vacuum tensor gravitational degrees of freedom.*

Gate 9 (Fail conditions)**.** *The framework fails in the tested local sector if any of the following occurs:*

- *observable preferred frame, species-dependent metric coupling, extra scalar/vector radiation, wrong gravitational-wave speed, negative-energy physical modes, or non-closing ADM constraints;*
- *factorization failure $Q^{\mu\nu}_{AB} \neq G^{\mu\nu} K_{AB}$, with the failure matrix $B^{\mu\nu}_{AB}$ producing species-dependent cones, a preferred-frame tensor, or a geometry cone distinct from the matter cone;*
- *composition-dependent gravitational projection $q_A/(E_A/c^2)$, or long-range tails requiring distinct spacetime carriers;*
- *independent $u^{\mu}_{\mathcal{M}}$ in local operators, or low-energy kinetic terms for lapse or shift;*
- *residue $\Delta_{\mathcal{M}}$ leaving an unsuppressed preferred-frame term, extra operational tensor, non-ADM constraint bracket, or scalar/vector propagating mode;*
- *observed interferometer signal explained only by pure common-mode breathing that should be self-calibrated, or unsuppressed breathing/scalar polarization in local polarization tests;*

- *record dynamics that do not yield exactly two positive, luminal, common-cone TT modes, or that leave an unsuppressed scalar/vector, massive spin-2, negative-energy or extra shear channel;*

- *binary-pulsar timing or compact-binary waveforms requiring an energy-loss law different from the protected tensor channel, or requiring an unsuppressed scalar dipole/monopole radiation channel;*

- *GW emission with $G_{\text{radiation}} \neq G_{\text{Newton}}$, outgoing tensor flux not balanced by $dE_{\text{orbit}}/dt = -P_{TT}$, missing angular momentum flux for eccentric timing systems, or a chirp mass not built from the Root C total-energy masses $m_A = E_A/c^2$;*

- *unsuppressed MOND-type correction in solar-system, binary-pulsar or local GW tests, or modification of $\gamma = 1$, $\beta = 1$, $c_{\text{GW}} = c$, universal free fall, or the two-tensor-mode vacuum sector;*

- *compact-object sector hiding energy from gravity, creating an unsuppressed breathing/scalar radiation channel, defining a species-private external cone, altering $c_{\text{GW}} = c$, or forcing shadow, ringdown, binary-pulsar or strong-field lensing deviations already excluded by observation.*

Current Status

$$\boxed{\text{RMG-1 has a conditional leading-order local GR core, not yet a closed theory.}}$$

Principal theorem to be derived:

$$\boxed{\begin{aligned} \mathcal{M} &\Rightarrow \text{exact ORC/no-extra-tensor} \\ &\Rightarrow Q^{\mu\nu}_{AB} = G^{\mu\nu} K_{AB} \\ &\Rightarrow \Delta_{\mathcal{M}} = 0 \\ &\Rightarrow \text{ADM/Einstein-Hilbert low-energy fixed point.} \end{aligned}}$$